

Topological Energy Condensate Theory (TECT): 16

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I. HONEST STATUS OF THE PDE

The final target must be the PDE derived from the full complete TECT action.

However, the current derivation has not yet reached that final unconditional stage. What is currently available is the strongest explicit working microscopic branch

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi],$$

which is not yet claimed to be the final complete TECT action.

What is already closed is the reduced infrared Dirac form:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

What remains open is the actual coefficient extraction from the final complete microscopic TECT operator:

$$L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

II. MINIMAL FINAL-COMPLETE TECT ACTION ANSATZ CONSISTENT WITH ALL CLOSED CONSTRAINTS

III. GOAL

The reduced Dirac theorem, the S2 transport structure, the Branch A / Branch B longitudinal audit, and the remote-sector completion logic are already closed at the formal level. What is still missing is not another reduction theorem, but a complete microscopic action that contains every carrier slot required by those closed conditions.

Therefore we now write the *minimal final-complete TECT action ansatz* that is consistent with all already-established constraints.

IV. CLOSED CONSTRAINTS THAT THE FINAL ACTION MUST SATISFY

The complete microscopic action must simultaneously satisfy the following facts already established in the notes:

1. the plain scalar Bragg route is insufficient;
2. the scalar identity-channel Bragg block is not cancelled inside the currently safe scalar library;
3. the relevant infrared carrier must therefore be a grading-odd first-order $2 + 2$ block;
4. the actual local projector-defect geometry yields

$$V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi));$$

5. the natural reduced branch is S2:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2);$$

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6. the transverse sector is controlled by (Λ_E, Λ_O) ;
7. the longitudinal sector must realize either

$$\Lambda_{\parallel}^{(0)} \neq 0 \quad (\text{Branch A}),$$

or

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0 \quad (\text{Branch B}).$$

Therefore the final complete action must contain enough microscopic carrier structure to generate:

$$h_{\parallel}(\theta), \quad h_E(\theta), \quad h_O(\theta),$$

with at least one surviving admissible longitudinal seed and at least one nonzero transverse doublet seed.

V. MINIMAL COMPLETE FIELD CONTENT

The minimal complete microscopic field content compatible with the closed constraints is:

$$\boxed{\Psi(x, t) \in \mathbb{C}^3, \quad P(x, t) = z(x, t)z(x, t)^{\dagger}, \quad z^{\dagger}z = 1, \quad X_i^a(x, t), \quad \sigma(x, t).} \quad (1)$$

Their roles are:

- Ψ : condensate triplet field;
- P : projector / orientational sector encoding the CP^2 defect geometry;
- X_i^a : enriched Class II carrier field generating derivative witness transport;
- σ : heavy volumetric / amplitude sector needed for remote-gap structure.

VI. MINIMAL FINAL-COMPLETE ACTION ANSATZ

The minimal complete action consistent with all closed logic is

$$\boxed{S_{\text{TECT}}^{\text{comp}} = \int dt d^3x \left(\mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \mathcal{L}_X^{\text{II}} + \mathcal{L}_{\text{mix}}^{\text{comp}} \right).} \quad (2)$$

We now specify each piece.

A. Condensate core

The condensate core must contain at least the locked Brazovskii-type shell-forming piece:

$$\boxed{\mathcal{L}_{\Psi}^{\text{core}} = |\partial_t \Psi|^2 - \mu^2 \Psi^{\dagger} \Psi + Z \partial_i \Psi^{\dagger} \partial_i \Psi - Y (\nabla^2 \Psi)^{\dagger} (\nabla^2 \Psi) - u (\Psi^{\dagger} \Psi)^2 + \mathcal{L}_{\Psi}^{\text{ref}}.} \quad (3)$$

The role of $\mathcal{L}_{\Psi}^{\text{ref}}$ is crucial: it contains the still-undetermined complete microscopic refined derivative terms that may generate a genuine constant-leading longitudinal Branch A coefficient

$$\Lambda_{\parallel}^{(0)} \neq 0.$$

B. Projector / geometry sector

The projector sector must encode the actual local CP^2 defect geometry:

$$\mathcal{L}_P^{\text{geom}} = \frac{\kappa_P}{2} (\partial_\mu P \partial^\mu P) + \sum_{n \geq 2} a_n \mathcal{I}_n[P, \partial P, \partial^2 P, \dots]. \quad (4)$$

The first term gives the basic projector stiffness. The higher invariants $\mathcal{I}_n$ represent the still-open even/odd microscopic geometric operators that may contribute to the actual transport kernel

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

C. Volumetric sector

The heavy volumetric mode is included as

$$\mathcal{L}_\sigma^{\text{vol}} = \frac{1}{2} (\partial_t \sigma)^2 - \frac{1}{2} M_\sigma^2 \sigma^2 - \frac{1}{2} c_\sigma^2 (\nabla \sigma)^2. \quad (5)$$

This sector is needed for the remote spectral benchmark and for the wrong-internal splitting scale.

D. Class II carrier sector

The enriched Class II carrier sector is

$$\mathcal{L}_X^{\text{II}} = \frac{1}{2} M_X^2 X_i^a X_i^a + \mathcal{L}_{X,\text{kin}}^{\text{ref}}. \quad (6)$$

The term $\mathcal{L}_{X,\text{kin}}^{\text{ref}}$ collects any still-allowed kinetic or derivative refinements of the heavy Class II field that are absent in the minimal explicit branch but may occur in the full complete TECT action.

E. Complete microscopic mixing sector

The full complete mixing sector must contain, at minimum, the currently explicit Class II carrier couplings:

$$\mathcal{L}_{\text{mix}}^{\text{comp}} = \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \mathcal{L}_{\text{mix}}^{\text{extra}}. \quad (7)$$

Here

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T^a \Psi + \Psi^\dagger T^a (\partial_i \Psi), \quad (8)$$

while $K_i^a[P, \psi]$ denotes the projector-coupled enriched derivative current.

The extra mixing term $\mathcal{L}_{\text{mix}}^{\text{extra}}$ is not optional in principle: it is the placeholder for all still-open full-complete TECT microscopic couplings that are invisible in the strongest explicit branch but may be needed to realize the actual coefficient point of the complete theory.

VII. FIELD EQUATIONS OF THE MINIMAL COMPLETE ANSATZ

Varying (2) gives the full microscopic PDE system:

a. Condensate PDE

$$\frac{\delta S_{\text{TECT}}^{\text{comp}}}{\delta \Psi^\dagger} = 0 \implies \partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - 2u(\Psi^\dagger \Psi) \Psi + \mathcal{F}_\Psi^{\text{ref}} + \alpha_X \frac{\delta(X_i^a J_i^a)}{\delta \Psi^\dagger} + \beta_X \frac{\delta(X_i^a K_i^a)}{\delta \Psi^\dagger} + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta \Psi^\dagger}. \quad (9)$$

b. *Projector PDE*

$$\boxed{\frac{\delta S_{\text{TECT}}^{\text{comp}}}{\delta P} = 0 \implies \kappa_P \partial_\mu \partial^\mu P + \mathcal{F}_P^{\text{geom}} + \beta_X \frac{\delta(X_i^a K_i^a)}{\delta P} + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta P} = 0.} \quad (10)$$

c. *Volumetric PDE*

$$\boxed{\frac{\delta S_{\text{TECT}}^{\text{comp}}}{\delta \sigma} = 0 \implies \partial_t^2 \sigma - c_\sigma^2 \nabla^2 \sigma + M_\sigma^2 \sigma + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta \sigma} = 0.} \quad (11)$$

d. *Class II carrier equation*

$$\boxed{\frac{\delta S_{\text{TECT}}^{\text{comp}}}{\delta X_i^a} = 0 \implies M_X^2 X_i^a + \frac{\delta \mathcal{L}_{X,\text{kin}}^{\text{ref}}}{\delta X_i^a} + \alpha_X J_i^a[\Psi] + \beta_X K_i^a[P, \psi] + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta X_i^a} = 0.} \quad (12)$$

VIII. BLOCH-REDUCED EFFECTIVE PDE

Linearizing around the locked periodic background and passing to Bloch momentum k , one obtains

$$\boxed{\mathcal{H}_{\text{Bloch}}^{\text{comp}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi P}(k) & H_{\Psi\sigma}(k) & H_{\Psi X}(k) \\ H_{P\Psi}(k) & H_{PP}(k) & H_{P\sigma}(k) & H_{PX}(k) \\ H_{\sigma\Psi}(k) & H_{\sigma P}(k) & H_{\sigma\sigma}(k) & H_{\sigma X}(k) \\ H_{X\Psi}(k) & H_{XP}(k) & H_{X\sigma}(k) & H_{XX}(k) \end{pmatrix}.} \quad (13)$$

This is the actual complete microscopic Hessian whose shell reduction must be analyzed.

After integrating out the heavy remote sectors (P, σ, X) or, more precisely, after taking the Schur complement relative to the isolated low-energy shell carrier, the effective shell operator is

$$\boxed{H_{\text{eff}}^{\text{comp}}(k) = H_{\text{low,low}}(k) - H_{\text{low,rem}}(k) H_{\text{rem,rem}}(k)^{-1} H_{\text{rem,low}}(k).} \quad (14)$$

IX. FINAL REDUCED INFRARED PDE

The shell-projected infrared principal symbol must still take the closed S2 form

$$\boxed{A_1^{\text{comp}}(p) = \lambda_{\parallel}^{\text{comp}} p_{\parallel} \sigma_3 + \alpha^{\text{comp}}(p_1 \sigma_1 + p_2 \sigma_2) + \beta^{\text{comp}}(-p_2 \sigma_1 + p_1 \sigma_2).} \quad (15)$$

The complete microscopic coefficient point is therefore

$$\boxed{(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O) \quad \text{or equivalently} \quad (\lambda_{\parallel}^{\text{comp}}, \alpha^{\text{comp}}, \beta^{\text{comp}}).} \quad (16)$$

X. FINAL EXACT TASKS

The remaining complete microscopic closure problem is now exactly:

1. compute the full Hessian $\mathcal{H}_{\text{Bloch}}^{\text{comp}}(k)$;
2. extract the actual transport kernel $V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi))$;
3. compute

$$\Lambda_{\parallel}^{(0)} = \frac{1}{2}[\sigma_3 n_\mu \mathcal{M}^\mu(0)];$$

4. if $\Lambda_{\parallel}^{(0)} = 0$, compute the primitive coefficient λ_c ;
5. compute the transverse coefficients (Λ_E, Λ_O) ;
6. verify the final nondegeneracy condition

$$[\Lambda_{\parallel}^{(0)} \neq 0 \text{ or } (\Lambda_{\parallel}^{(0)} = 0, \lambda_c \neq 0)] \quad \text{and} \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

XI. HONEST INTERPRETATION

This ansatz is not yet the proved final complete TECT action. It is the minimal complete microscopic action form that contains every slot required by the already-closed Dirac reduction logic.

Therefore the frontier is no longer structural. It is the explicit evaluation of the coefficient point of the complete action.

XII. FINAL SUMMARY

The final target is not the strongest explicit working branch alone, but the PDE derived from a complete TECT action.

The minimal complete action ansatz consistent with all closed constraints is

$$S_{\text{TECT}}^{\text{comp}} = \int dt d^3x \left(\mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \mathcal{L}_X^{\text{II}} + \mathcal{L}_{\text{mix}}^{\text{comp}} \right),$$

with field content

$$\Psi, \quad P = zz^{\dagger}, \quad \sigma, \quad X_i^a.$$

Its microscopic PDE system consists of the coupled Euler-Lagrange equations for

$$\Psi, \quad P, \quad \sigma, \quad X_i^a.$$

After Bloch linearization around the locked periodic background, the full microscopic Hessian is

$$\mathcal{H}_{\text{Bloch}}^{\text{comp}}(k),$$

and the effective low-energy shell operator is obtained by the Schur complement

$$H_{\text{eff}}^{\text{comp}}(k) = H_{\text{low,low}}(k) - H_{\text{low,rem}}(k) H_{\text{rem,rem}}(k)^{-1} H_{\text{rem,low}}(k).$$

The reduced infrared principal symbol must still be

$$A_1^{\text{comp}}(p) = \lambda_{\parallel}^{\text{comp}} p_{\parallel} \sigma_3 + \alpha^{\text{comp}} (p_1 \sigma_1 + p_2 \sigma_2) + \beta^{\text{comp}} (-p_2 \sigma_1 + p_1 \sigma_2).$$

Thus the final complete microscopic closure problem is reduced to extracting the coefficient point

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O),$$

equivalently

$$(\lambda_{\parallel}^{\text{comp}}, \alpha^{\text{comp}}, \beta^{\text{comp}}),$$

from the complete background-dependent TECT Hessian.

So the true remaining task is not to invent a new reduced PDE form, but to insert the complete action into the already-closed extraction formalism and compute the actual coefficient point.

XIII. MINIMAL INDEPENDENT INVARIANT SLOTS FOR THE COMPLETE MIXING SECTOR

XIV. GOAL

We now determine the minimal independent invariant content that must be allowed inside

$$\mathcal{L}_{\text{mix}}^{\text{extra}}$$

so that the final complete TECT action can realize the already-closed reduced S2 symbol

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2).$$

The point is not to enumerate every imaginable coupling. The point is to identify the smallest set of independent microscopic carrier slots required by the already-established extraction formalism.

XV. WHAT THE EXTRA SECTOR MUST GENERATE

The final microscopic operator must produce a local transport kernel of the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

with invariant decomposition

$$\mathcal{M}_{\text{inv}}^\mu(\theta) = h_0(\theta) n^\mu \mathbf{1}_2 + h_\parallel(\theta) n^\mu \sigma_3 + h_E(\theta) E^\mu{}_a \sigma_a + h_O(\theta) J^\mu{}_a \sigma_a. \quad (17)$$

Therefore the complete mixing sector must supply, at minimum, three nontrivial carrier channels:

1. one longitudinal σ_3 -type channel;
2. one transverse even doublet channel;
3. one transverse odd doublet channel.

Equivalently, the extra sector must be able to generate the shell tensors

$$\hat{n}_i \delta^{a3}, \quad e_{1i} \delta^{a1} + e_{2i} \delta^{a2}, \quad -e_{1i} \delta^{a2} + e_{2i} \delta^{a1}. \quad (18)$$

XVI. RESTRICTION FROM THE GRADED PRINCIPAL SYMBOL

Only terms linear in the fluctuation and containing exactly one worldvolume derivative contribute to the graded principal symbol. Thus the first variations must have the normal form

$$J_i^{a(1)}[\eta] = \mathcal{J}_i^{a(0)}(\xi) \eta + \mathcal{J}_i^{a\mu}(\xi) (-i \partial_{y^\mu}) \eta + \cdots, \quad (19)$$

$$K_i^{a(1)}[\eta] = \mathcal{K}_i^{a(0)}(\xi) \eta + \mathcal{K}_i^{a\mu}(\xi) (-i \partial_{y^\mu}) \eta + \cdots. \quad (20)$$

Hence $\mathcal{L}_{\text{mix}}^{\text{extra}}$ only needs to enlarge the coefficient content of the (i, a) -carrier sector. It does not need to introduce qualitatively new derivative order.

XVII. MINIMAL INDEPENDENT INVARIANT SLOTS

A minimal complete choice is to add exactly three independent invariant families:

A. Slot L: longitudinal even carrier

The first independent slot must be able to generate a longitudinal σ_3 -type seed. The minimal invariant representative is

$$\boxed{\mathcal{I}_L = X_i^a \hat{n}^i \mathcal{S}_a[P, \Psi, \sigma]}, \quad (21)$$

where $\mathcal{S}_a$ is an internal adjoint scalar built from the background projector/condensate data. Its role is to generate either:

- a constant-leading Branch A term $h_\parallel(0) \neq 0$, or
- the primitive Branch B commutator-square seed.

Thus Slot L is not yet tied to Branch A or Branch B specifically. It is the minimal longitudinal carrier slot that must exist in the complete action.

B. Slot E: transverse even doublet carrier

The second independent slot must generate the even transverse structure

$$E^\mu{}_a \sigma_a \iff p_1 \sigma_1 + p_2 \sigma_2.$$

The minimal invariant representative is

$$\boxed{\mathcal{I}_E = X_i^a \mathcal{E}^{ia}[P, \Psi, \sigma]}, \quad (22)$$

where $\mathcal{E}^{ia}$ transforms as the even shell doublet tensor.

Its reduced projection must have nonzero overlap with

$$e_{1i} \delta^{a1} + e_{2i} \delta^{a2}. \quad (23)$$

C. Slot O: transverse odd doublet carrier

The third independent slot must generate the odd transverse structure

$$J^\mu{}_a \sigma_a \iff -p_2 \sigma_1 + p_1 \sigma_2.$$

The minimal invariant representative is

$$\boxed{\mathcal{I}_O = X_i^a \mathcal{O}^{ia}[P, \Psi, \sigma]}, \quad (24)$$

where $\mathcal{O}^{ia}$ transforms as the odd shell doublet tensor.

Its reduced projection must have nonzero overlap with

$$-e_{1i} \delta^{a2} + e_{2i} \delta^{a1}. \quad (25)$$

XVIII. MINIMAL COMPLETE MIXING ANSATZ

Therefore the minimal complete extra mixing sector can be written as

$$\boxed{\mathcal{L}_{\text{mix}}^{\text{extra}} = \gamma_L \mathcal{I}_L + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O + \text{h.c.}} \quad (26)$$

with three independent couplings

$$\gamma_L, \quad \gamma_E, \quad \gamma_O.$$

Then the full complete mixing sector becomes

$$\boxed{\mathcal{L}_{\text{mix}}^{\text{comp}} = \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_L \mathcal{I}_L + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O + \text{h.c.}} \quad (27)$$

XIX. WHY EXACTLY THREE SLOTS ARE MINIMAL

The reason this is minimal is:

1. the reduced S2 symbol has exactly one longitudinal traceless channel;
2. it has exactly one even transverse doublet channel;
3. it has exactly one odd transverse doublet channel.

No smaller set can span all three independent reduced witness directions. Any larger set is redundant at the level of first-order principal-symbol closure.

So the complete action does not need an arbitrary zoo of extra couplings. It needs, minimally, one generator for each of these three carrier directions.

XX. CORRESPONDENCE WITH THE COEFFICIENT POINT

After Bloch reduction and shell projection, the three extra slots contribute only through the already-closed coefficient point:

$$\boxed{(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O).} \quad (28)$$

More explicitly:

- $\mathcal{I}_L$ contributes to the longitudinal branch decision:

$$\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0 \quad \text{or} \quad \lambda_c \stackrel{?}{\neq} 0;$$

- $\mathcal{I}_E$ contributes to Λ_E ;
- $\mathcal{I}_O$ contributes to Λ_O .

Therefore the microscopic completion problem is now compressed to the question:

$$\boxed{\text{which actual complete TECT invariants realize the three slots } \mathcal{I}_L, \mathcal{I}_E, \mathcal{I}_O?} \quad (29)$$

XXI. MINIMAL FINAL-COMPLETE PDE STATEMENT

With these extra slots included, the complete microscopic action is now organized as

$$S_{\text{TECT}}^{\text{comp}} = \int dt d^3x \left(\mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \mathcal{L}_X^{\text{II}} + \mathcal{L}_{\text{mix}}^{\text{comp}} \right), \quad (30)$$

where $\mathcal{L}_{\text{mix}}^{\text{comp}}$ is given by (27).

The corresponding complete microscopic PDE is the coupled Euler–Lagrange system for

$$\Psi, \quad P, \quad \sigma, \quad X_i^a.$$

After periodic-background linearization and shell projection, the infrared principal symbol must be

$$\boxed{A_1^{\text{comp}}(p) = \lambda_{\parallel}^{\text{comp}} p_{\parallel} \sigma_3 + \alpha^{\text{comp}}(p_1 \sigma_1 + p_2 \sigma_2) + \beta^{\text{comp}}(-p_2 \sigma_1 + p_1 \sigma_2).} \quad (31)$$

Thus the complete PDE search is no longer underdetermined in structure. It is underdetermined only in the microscopic realization of the three invariant slots

$$\mathcal{I}_L, \quad \mathcal{I}_E, \quad \mathcal{I}_O.$$

XXII. FINAL SUMMARY

The complete TECT action does not need an arbitrary number of extra couplings. To realize the already-closed reduced S2 Dirac structure, the extra mixing sector needs only three independent invariant slots:

$$\boxed{\mathcal{I}_L \quad (\text{longitudinal carrier}),}$$

$$\boxed{\mathcal{I}_E \quad (\text{transverse even carrier}),}$$

$$\boxed{\mathcal{I}_O \quad (\text{transverse odd carrier}).}$$

Hence the minimal complete extra mixing ansatz is

$$\mathcal{L}_{\text{mix}}^{\text{extra}} = \gamma_L \mathcal{I}_L + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O + \text{h.c.}$$

The full complete mixing sector is therefore

$$\mathcal{L}_{\text{mix}}^{\text{comp}} = \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_L \mathcal{I}_L + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O + \text{h.c.}$$

After reduction, these slots contribute only through the coefficient point

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O).$$

So the complete PDE search is now compressed to one final microscopic question:

$$\text{which actual complete TECT invariants realize } \mathcal{I}_L, \mathcal{I}_E, \mathcal{I}_O?$$

In other words, the structure of the final PDE is no longer the problem. The remaining problem is the microscopic realization of the three minimal carrier slots.

XXIII. LOWEST-ORDER INVARIANT CANDIDATES FOR THE THREE OPEN CARRIER SLOTS

XXIV. SETUP

Let

$$P = zz^\dagger, \quad z^\dagger z = 1, \quad \Pi_Q = \mathbf{1}_3 - P, \quad \mathcal{T}_\mu := i[P, \partial_\mu P].$$

On the defect background, the shell-compatible odd transport kernel must take the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

with invariant decomposition

$$\mathcal{M}^\mu = h_0 n^\mu \mathbf{1}_2 + h_{\parallel} n^\mu \sigma_3 + h_E E^\mu{}_a \sigma_a + h_O J^\mu{}_a \sigma_a.$$

Hence we need one microscopic invariant for each of:

$$n^\mu \sigma_3, \quad E^\mu{}_a \sigma_a, \quad J^\mu{}_a \sigma_a.$$

XXV. SLOT L: LOWEST-ORDER LONGITUDINAL CANDIDATE

The already-closed primitive longitudinal source is the traceless commutator-square channel

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Therefore the lowest-order longitudinal invariant candidate is

$$\mathcal{I}_L^{(2)} = X_i^a n^i \left[T^a (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} \right]. \quad (32)$$

On the defect background,

$$\mathcal{I}_L^{(2)} \sim X_{\parallel}^3 \theta'(\xi)^2.$$

After solving the heavy-field equation for X_i^a , the induced odd transport piece becomes

$$V_{(L)}^\mu(\xi) \propto \theta'(\xi)^3 n^\mu \sigma_3,$$

so this realizes the primitive Branch B slot

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}.$$

XXVI. SLOT E: LOWEST-ORDER TRANSVERSE-EVEN CANDIDATE

The minimal transverse-even carrier must project to

$$E^\mu{}_a \sigma_a \iff p_1 \sigma_1 + p_2 \sigma_2.$$

The lowest-order projector/condensate candidate is obtained from one projector derivative and one tangent derivative:

$$\mathcal{I}_E^{(1,1)} = X_i^a (T^a \mathcal{T}_i) \Re(z^\dagger D_\parallel \Psi). \quad (33)$$

More invariantly, after restriction to the tangent bundle, this is equivalent to taking the even shell-doublet tensor

$$\mathcal{I}_E^{(1,1)} = X_i^a \mathcal{E}^{ia}, \quad \mathcal{E}^{ia} := (T^a \mathcal{T}_j) E^{ji}. \quad (34)$$

On the defect background,

$$\mathcal{T}_j \sim \theta'(\xi) \times (\text{tangent adjoint}),$$

hence

$$\mathcal{I}_E^{(1,1)} \sim X_i^a \theta'(\xi) E^{ia}.$$

After eliminating X_i^a ,

$$V_{(E)}^\mu(\xi) \propto \theta'(\xi) E^\mu{}_a \sigma_a,$$

so this contributes to

$$\alpha = I_\theta \Lambda_E.$$

XXVII. SLOT O: LOWEST-ORDER TRANSVERSE-ODD CANDIDATE

The minimal transverse-odd carrier must project to

$$J^\mu{}_a \sigma_a \iff -p_2 \sigma_1 + p_1 \sigma_2.$$

The lowest-order candidate is the parity-odd partner of the previous tensor:

$$\mathcal{I}_O^{(1,1)} = X_i^a \mathcal{O}^{ia}, \quad \mathcal{O}^{ia} := (T^a \mathcal{T}_j) J^{ji}. \quad (35)$$

On the defect background,

$$\mathcal{I}_O^{(1,1)} \sim X_i^a \theta'(\xi) J^{ia}.$$

After eliminating X_i^a ,

$$V_{(O)}^\mu(\xi) \propto \theta'(\xi) J^\mu{}_a \sigma_a,$$

so this contributes to

$$\beta = I_\theta \Lambda_O.$$

XXVIII. COMPLETE LOWEST-ORDER EXTRA MIXING SECTOR

Therefore the lowest-order extra mixing sector consistent with the already-closed S2 carrier structure is

$$\mathcal{L}_{\text{mix}}^{\text{extra, LO}} = \gamma_L X_i^a n^i \left[T^a (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} \right] + \gamma_E X_i^a \mathcal{E}^{ia} + \gamma_O X_i^a \mathcal{O}^{ia} + \text{h.c.} \quad (36)$$

Equivalently, in the shell-adapted seed basis,

$$m^a = m \delta^{a3}, \quad C_{ij}^{33} = c_\parallel \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \quad (37)$$

Then

$$g_c \propto c_\parallel, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

XXIX. INDUCED REDUCED SYMBOL

After integrating out X_i^a , the induced local transport kernel is

$$V_{\text{LO}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right]. \quad (38)$$

Therefore the reduced principal symbol is

$$A_1^{\text{LO}}(p) = \frac{\lambda_c}{2} I_{\theta^3} p_\parallel \sigma_3 + I_\theta \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_\theta \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (39)$$

So the lowest-order complete closure condition is

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (40)$$

XXX. MINIMAL FINAL-COMPLETE PDE

The corresponding minimal final-complete microscopic action is

$$S_{\text{TECT}}^{\text{comp, LO}} = \int dt d^3x \left(\mathcal{L}_\Psi^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_\sigma^{\text{vol}} + \mathcal{L}_X^{\text{II}} + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \mathcal{L}_{\text{mix}}^{\text{extra, LO}} \right). \quad (41)$$

Its Euler–Lagrange equations for (Ψ, P, σ, X_i^a) define the minimal final-complete PDE system compatible with all already-closed reduced constraints.

XXXI. FINAL SUMMARY

The lowest-order open carrier slots are realized by exactly three invariant candidates:

$$\mathcal{I}_L^{(2)} = X_i^a n^i \left[T^a (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} \right],$$

which generates the primitive longitudinal seed

$$V_{(L)}^\mu(\xi) \propto \theta'(\xi)^3 n^\mu \sigma_3;$$

$$\mathcal{I}_E^{(1,1)} = X_i^a \mathcal{E}^{ia},$$

which generates the transverse-even seed

$$V_{(E)}^\mu(\xi) \propto \theta'(\xi) E^\mu{}_a \sigma_a;$$

$$\mathcal{I}_O^{(1,1)} = X_i^a \mathcal{O}^{ia},$$

which generates the transverse-odd seed

$$V_{(O)}^\mu(\xi) \propto \theta'(\xi) J^\mu{}_a \sigma_a.$$

Thus the lowest-order extra mixing sector is

$$\mathcal{L}_{\text{mix}}^{\text{extra, LO}} = \gamma_L \mathcal{I}_L^{(2)} + \gamma_E \mathcal{I}_E^{(1,1)} + \gamma_O \mathcal{I}_O^{(1,1)} + \text{h.c.}$$

After integrating out the heavy mediator X_i^a , the induced local transport kernel is

$$V_{\text{LO}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) [\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a].$$

Therefore the reduced principal symbol is

$$A_1^{\text{LO}}(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_\parallel \sigma_3 + I_\theta \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_\theta \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2),$$

and the lowest-order nondegeneracy condition is

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).}$$

So the minimal final-complete PDE search is now reduced to evaluating the three couplings

$$\gamma_L, \quad \gamma_E, \quad \gamma_O,$$

equivalently the induced coefficient point

$$(\lambda_c, \Lambda_E, \Lambda_O).$$

XXXII. EXPLICIT ELIMINATION OF THE HEAVY CLASS II CARRIER

XXXIII. LOWEST-ORDER COMPLETE MICROSCOPIC ACTION

Take the lowest-order complete microscopic action

$$S_{\text{TECT}}^{\text{comp,LO}} = \int dt d^3x \left(\mathcal{L}_\Psi^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_\sigma^{\text{vol}} + \mathcal{L}_X^{\text{II}} + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \mathcal{L}_{\text{mix}}^{\text{extra,LO}} \right), \quad (42)$$

with

$$\mathcal{L}_X^{\text{II}} = \frac{1}{2} M_X^2 X_i^a X_i^a, \quad (43)$$

and

$$\mathcal{L}_{\text{mix}}^{\text{extra,LO}} = \gamma_L \mathcal{I}_L^{(2)} + \gamma_E \mathcal{I}_E^{(1,1)} + \gamma_O \mathcal{I}_O^{(1,1)} + \text{h.c.} \quad (44)$$

Introduce the total X -source

$$\boxed{\mathcal{J}_i^{a,\text{tot}} := \alpha_X J_i^a[\Psi] + \beta_X K_i^a[P, \psi] + \gamma_L \mathfrak{L}_i^a + \gamma_E \mathfrak{E}_i^a + \gamma_O \mathfrak{O}_i^a}, \quad (45)$$

where

$$\mathfrak{L}_i^a, \quad \mathfrak{E}_i^a, \quad \mathfrak{O}_i^a$$

denote the local operator densities corresponding to $\mathcal{I}_L^{(2)}, \mathcal{I}_E^{(1,1)}, \mathcal{I}_O^{(1,1)}$.

Then the X -dependent part of the action is simply

$$\boxed{\mathcal{L}_X^{\text{tot}} = \frac{1}{2} M_X^2 X_i^a X_i^a + X_i^a \mathcal{J}_i^{a,\text{tot}}}. \quad (46)$$

XXXIV. EXACT HEAVY-FIELD ELIMINATION

The Euler–Lagrange equation for X_i^a is algebraic:

$$\frac{\delta S}{\delta X_i^a} = 0 \quad \implies \quad M_X^2 X_i^a + \mathcal{J}_i^{a,\text{tot}} = 0. \quad (47)$$

Therefore

$$\boxed{X_i^a = -\frac{1}{M_X^2} \mathcal{J}_i^{a,\text{tot}}.} \quad (48)$$

Substituting (48) into (46) gives the exact tree-level effective interaction

$$\boxed{\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \mathcal{J}_i^{a,\text{tot}} \mathcal{J}_i^{a,\text{tot}}.} \quad (49)$$

Thus the complete lowest-order effective action is

$$\boxed{S_{\text{eff}}^{\text{LO}} = \int dt d^3x \left(\mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} - \frac{1}{2M_X^2} \mathcal{J}_i^{a,\text{tot}} \mathcal{J}_i^{a,\text{tot}} \right).} \quad (50)$$

XXXV. EXPANSION OF THE INDUCED EFFECTIVE INTERACTION

Expanding (49),

$$\begin{aligned} \mathcal{L}_{\text{eff}}^{(X)} = & -\frac{\alpha_X^2}{2M_X^2} J_i^a J_i^a - \frac{\beta_X^2}{2M_X^2} K_i^a K_i^a - \frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a \\ & - \frac{\alpha_X \gamma_L}{M_X^2} J_i^a \mathfrak{L}_i^a - \frac{\alpha_X \gamma_E}{M_X^2} J_i^a \mathfrak{E}_i^a - \frac{\alpha_X \gamma_O}{M_X^2} J_i^a \mathfrak{D}_i^a \\ & - \frac{\beta_X \gamma_L}{M_X^2} K_i^a \mathfrak{L}_i^a - \frac{\beta_X \gamma_E}{M_X^2} K_i^a \mathfrak{E}_i^a - \frac{\beta_X \gamma_O}{M_X^2} K_i^a \mathfrak{D}_i^a \\ & - \frac{1}{2M_X^2} \left(\gamma_L^2 \mathfrak{L}_i^a \mathfrak{L}_i^a + \gamma_E^2 \mathfrak{E}_i^a \mathfrak{E}_i^a + \gamma_O^2 \mathfrak{D}_i^a \mathfrak{D}_i^a \right) \\ & - \frac{\gamma_L \gamma_E}{M_X^2} \mathfrak{L}_i^a \mathfrak{E}_i^a - \frac{\gamma_L \gamma_O}{M_X^2} \mathfrak{L}_i^a \mathfrak{D}_i^a - \frac{\gamma_E \gamma_O}{M_X^2} \mathfrak{E}_i^a \mathfrak{D}_i^a. \end{aligned} \quad (51)$$

XXXVI. LOAD-BEARING FIRST-ORDER TRANSPORT SECTOR

The graded principal symbol keeps only terms that are:

1. quadratic in the light fluctuations;
2. odd under $\xi \mapsto -\xi$;
3. linear in one worldvolume derivative ∂_{y^μ} .

Therefore, among (51), the load-bearing piece is the projection of

$$\boxed{\mathcal{L}_{\text{eff,lb}}^{(X)} = -\frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a - \frac{\alpha_X \gamma_E}{M_X^2} J_i^a \mathfrak{E}_i^a - \frac{\alpha_X \gamma_O}{M_X^2} J_i^a \mathfrak{D}_i^a - \frac{\beta_X \gamma_L}{M_X^2} K_i^a \mathfrak{L}_i^a + \dots} \quad (52)$$

to the shell-/defect-adapted tangent sector.

The dots indicate terms of the same schematic type that survive the oddness filter and contain exactly one world-volume derivative.

XXXVII. QUADRATIC SHELL TEMPLATE

At the shell-relevant quadratic level, the complete X -source has the form

$$\boxed{\mathcal{J}_i^{a,\text{tot}} = \lambda m^a \partial_i \phi_0 + g_X C_{ij}^{ab} (D_j \psi) \xi^b + \ell L_i^a + e E_i^a + o O_i^a.} \quad (53)$$

Here:

- $m^a \partial_i \phi_0$ is the shell scalar-gradient source;
- $C_{ij}^{ab}(D_j \psi) \xi^b$ is the enriched tensor source;
- L_i^a, E_i^a, O_i^a are the shell-adapted representatives of the three extra slots.

Substituting (53) into (49), the cross term between $m^a \partial_i \phi_0$ and the enriched tensor source reproduces

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}}, \quad (54)$$

which is the already-closed Class II induced enriched tensor.

Likewise, the cross terms with the extra slots give

$$\delta N_{ij}^{b,(L)} \propto -\frac{\lambda \ell}{M_X^2} m^a L_{ij}^{ab}, \quad \delta N_{ij}^{b,(E)} \propto -\frac{\lambda e}{M_X^2} m^a E_{ij}^{ab}, \quad \delta N_{ij}^{b,(O)} \propto -\frac{\lambda o}{M_X^2} m^a O_{ij}^{ab}. \quad (55)$$

XXXVIII. SEED-BASIS EVALUATION

Take the shell-adapted seed family

$$m^a = m \delta^{a3}, \quad (56)$$

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \quad (57)$$

Then

$$\boxed{\Gamma := \frac{\lambda g_X m}{M_X^2}, \quad \delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J.} \quad (58)$$

Hence

$$\boxed{g_c = \nu_{\parallel} \Gamma c_{\parallel}, \quad g_E = \nu_{\perp} \Gamma c_I, \quad g_O = \nu_{\perp} \Gamma c_J,} \quad (59)$$

up to the nonzero shell/defect overlap normalizations $\nu_{\parallel}, \nu_{\perp}$. This is the strongest honest normalization extraction currently available. :contentReference[oaicite:3]index=3

XXXIX. LOCAL TRANSPORT KERNEL

In the local defect language, the induced mediator transport kernel is therefore

$$\boxed{V_{\text{med}}^{\mu}(\xi) = \theta'(\xi) \left[\Lambda_{\parallel}^{(0)} n^{\mu} \Sigma_3 + \Lambda_E E^{\mu}{}_a \Sigma_a + \Lambda_O J^{\mu}{}_a \Sigma_a \right] + \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \Sigma_3 + \dots} \quad (60)$$

and in the exact concrete Class II branch

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.} \quad (61)$$

Therefore the reduced principal symbol is

$$\boxed{A_1(p) = \frac{\lambda_c}{2} I_{\theta^3} p_{\parallel} \sigma_3 + I_{\theta} \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta} \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (62)$$

XL. MINIMAL NONDEGENERACY CONDITION

Thus the lowest-order complete closure condition is

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0),} \quad (63)$$

equivalently

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).} \quad (64)$$

XLI. RESULTING EFFECTIVE PDE

The effective lowest-order microscopic PDE is obtained by varying (50) with respect to the light fields only:

$$\boxed{\frac{\delta S_{\text{eff}}^{\text{LO}}}{\delta \Psi^\dagger} = 0, \quad \frac{\delta S_{\text{eff}}^{\text{LO}}}{\delta P} = 0, \quad \frac{\delta S_{\text{eff}}^{\text{LO}}}{\delta \sigma} = 0.} \quad (65)$$

At the shell-/defect-reduced level, its principal part is exactly (62). So the heavy-field elimination has reduced the complete microscopic PDE search to the coefficient extraction problem for the mediator-induced transport triplet

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

XLII. FINAL SUMMARY

Integrating out the heavy Class II carrier X_i^a is algebraic:

$$M_X^2 X_i^a + \mathcal{J}_i^{a, \text{tot}} = 0 \quad \implies \quad X_i^a = -\frac{1}{M_X^2} \mathcal{J}_i^{a, \text{tot}}.$$

Hence the exact tree-level induced interaction is

$$\boxed{\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \mathcal{J}_i^{a, \text{tot}} \mathcal{J}_i^{a, \text{tot}}.}$$

At the shell-relevant quadratic level,

$$\mathcal{J}_i^{a, \text{tot}} = \lambda m^a \partial_i \phi_0 + g_X C_{ij}^{ab} (D_j \psi) \xi^b + \dots,$$

so the cross term reproduces the already-closed induced enriched tensor

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} .}$$

For the shell-adapted seed family,

$$\boxed{g_c = \nu_{\parallel} \Gamma c_{\parallel}, \quad g_E = \nu_{\perp} \Gamma c_I, \quad g_O = \nu_{\perp} \Gamma c_J, \quad \Gamma = \frac{\lambda g_X m}{M_X^2} .}$$

In the local defect language, the induced transport kernel is

$$V_{\text{med}}^{\mu}(\xi) = \theta'(\xi) \left[\Lambda_{\parallel}^{(0)} n^{\mu} \Sigma_3 + \Lambda_E E^{\mu}{}_a \Sigma_a + \Lambda_O J^{\mu}{}_a \Sigma_a \right] + \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \Sigma_3 + \dots$$

and in the exact concrete Class II branch

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.$$

Therefore the reduced principal symbol is

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta^3} p_{\parallel} \sigma_3 + I_{\theta} \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta} \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

So after eliminating X_i^a , the final remaining problem is just the coefficient extraction of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O,$$

equivalently

$$g_c, \quad g_E, \quad g_O.$$

XLIII. COMPLETE TECT REALIZATION OF THE EFFECTIVE CLASS II SOURCE ON THE DEFECT BACKGROUND

XLIV. LOCKED DEFECT BACKGROUND

Take

$$\Psi_B(\xi, x_{\parallel}) = \phi_0(x_{\parallel}) z_B(\xi), \quad P_B(\xi) = z_B(\xi) z_B(\xi)^{\dagger}, \quad z_B^{\dagger} z_B = 1,$$

with explicit representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \partial_{\xi} z_B = \theta'(\xi) e_1(\xi).$$

The tangent fluctuation is

$$\delta P = \eta z_B^{\dagger} + z_B \eta^{\dagger}, \quad \eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^{\perp}.$$

Write the low tangent mode as

$$\eta(\xi, y) = \sum_{a=1}^2 \chi_a(y) u_a(\xi), \quad u_a(\xi) = f(\xi) e_a(\xi),$$

and the shell/test mode as $\psi(y)$.

XLV. MICROSCOPIC CURRENTS

Use the minimal complete TECT currents

$$J_i^a[\Psi] = (\partial_i \Psi^{\dagger}) T_a \Psi + \Psi^{\dagger} T_a (\partial_i \Psi), \quad (66)$$

$$K_i^a[P, \psi] = (T_a \partial_i P) \mathcal{O}[\psi]. \quad (67)$$

On the defect background,

$$\partial_i \Psi_B = (\partial_i \phi_0) z_B + \phi_0 \partial_i z_B.$$

Hence the shell-gradient piece of J_i^a is

$$\begin{aligned} J_i^a[\Psi_B] &= (\partial_i \phi_0) (z_B^{\dagger} T_a z_B + z_B^{\dagger} T_a z_B) |\phi_0| + \phi_0^2 ((\partial_i z_B^{\dagger}) T_a z_B + z_B^{\dagger} T_a (\partial_i z_B)) \\ &= 2(\partial_i \phi_0) \phi_0 m_a(\theta) + \phi_0^2 \mathcal{A}_i^a(\theta, \theta'), \end{aligned} \quad (68)$$

where

$$m_a(\theta) := z_B^\dagger T_a z_B, \quad \mathcal{A}_i^a := (\partial_i z_B^\dagger) T_a z_B + z_B^\dagger T_a (\partial_i z_B).$$

Since $\partial_i z_B \neq 0$ only in the normal direction $i = \xi$, the shell-gradient carrier is

$$\boxed{J_i^a[\Psi_B] = 2(\partial_i \phi_0) \phi_0 m_a(\theta) + \delta_{i\xi} \phi_0^2 \mathcal{A}_\xi^a(\theta, \theta').} \quad (69)$$

Now

$$\partial_i P_B = \delta_{i\xi} \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger),$$

so

$$\boxed{K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \mathcal{O}[\psi], \quad k_a(\theta) := \left[T_a (e_1 z_B^\dagger + z_B e_1^\dagger) \right].} \quad (70)$$

XLVI. EFFECTIVE MIXED SOURCE AFTER ELIMINATING X_i^a

Integrating out X_i^a gives

$$\mathcal{L}_{\text{eff}}^{(II)} = -\frac{\alpha\beta}{M_X^2} J_i^a K_i^a. \quad (71)$$

Substituting (69) and (70),

$$\begin{aligned} \mathcal{L}_{\text{eff}}^{(II)} &= -\frac{\alpha\beta}{M_X^2} \left[2(\partial_i \phi_0) \phi_0 m_a(\theta) + \delta_{i\xi} \phi_0^2 \mathcal{A}_\xi^a(\theta, \theta') \right] \delta_{i\xi} \theta'(\xi) k_a(\theta) \mathcal{O}[\psi] \\ &= -\frac{\alpha\beta}{M_X^2} \left[2(\partial_\xi \phi_0) \phi_0 m_a(\theta) k_a(\theta) \theta'(\xi) + \phi_0^2 \mathcal{A}_\xi^a(\theta, \theta') k_a(\theta) \theta'(\xi) \right] \mathcal{O}[\psi]. \end{aligned} \quad (72)$$

Thus the complete TECT realization automatically splits into:

$$\boxed{\mathcal{L}_{\text{eff}}^{(II)} = \mathcal{L}_\perp^{(1)} + \mathcal{L}_\parallel^{\text{prim}},} \quad (73)$$

with

$$\mathcal{L}_\perp^{(1)} = -\frac{2\alpha\beta}{M_X^2} (\partial_\xi \phi_0) \phi_0 \theta'(\xi) [m_a(\theta) k_a(\theta)] \mathcal{O}[\psi], \quad (74)$$

and

$$\mathcal{L}_\parallel^{\text{prim}} = -\frac{\alpha\beta}{M_X^2} \phi_0^2 \theta'(\xi) [\mathcal{A}_\xi^a(\theta, \theta') k_a(\theta)] \mathcal{O}[\psi]. \quad (75)$$

XLVII. TRANSVERSE CHANNEL EXTRACTION

The first piece is linear in $\theta'(\xi)$. After tangent-bundle reduction,

$$m_a(\theta) k_a(\theta) \mapsto \Lambda_E E^\mu{}_b \Sigma_b + \Lambda_O J^\mu{}_a \Sigma_a.$$

Therefore

$$\boxed{V_\perp^\mu(\xi) = \theta'(\xi) \left[\Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right].} \quad (76)$$

Hence

$$\boxed{\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.} \quad (77)$$

XLVIII. PRIMITIVE LONGITUDINAL EXTRACTION

Now evaluate $\mathcal{A}_\xi^a$:

$$\mathcal{A}_\xi^a = (\partial_\xi z_B^\dagger) T_a z_B + z_B^\dagger T_a (\partial_\xi z_B) = \theta'(\xi) \left(e_1^\dagger T_a z_B + z_B^\dagger T_a e_1 \right).$$

Therefore

$$\mathcal{A}_\xi^a(\theta, \theta') k_a(\theta) \theta'(\xi) = \theta'(\xi)^2 Q(\theta),$$

for some scalar $Q(\theta)$ valued in the adjoint-singlet contraction.

After tangent projection, the traceless quadratic commutator channel is

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3$$

and therefore the primitive longitudinal transport is

$$\boxed{V_\parallel^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3.} \quad (78)$$

Hence

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (79)$$

XLIX. FINAL REDUCED PRINCIPAL SYMBOL

Combining (76) and (78),

$$\boxed{V^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \left[\Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right].} \quad (80)$$

Projecting onto the graded low modes gives

$$\boxed{A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_\parallel \sigma_3 + I_\theta \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_\theta \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (81)$$

Thus the complete TECT realization of the minimal Class II carrier yields the final reduced PDE data

$$\boxed{\lambda_c, \quad \Lambda_E, \quad \Lambda_O.} \quad (82)$$

L. CLOSURE CONDITION

The final nondegeneracy condition is

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (83)$$

Equivalently, in the reduced Class II coordinates,

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).} \quad (84)$$

LI. FINAL SUMMARY

Using the minimal complete TECT currents

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \text{h.c.}, \quad K_i^a[P, \psi] = (T_a \partial_i P) \mathcal{O}[\psi],$$

and eliminating the heavy Class II field X_i^a , one gets

$$\mathcal{L}_{\text{eff}}^{(II)} = -\frac{\alpha\beta}{M_X^2} J_i^a K_i^a.$$

On the defect background,

$$J_i^a[\Psi_B] = 2(\partial_i \phi_0) \phi_0 m_a(\theta) + \delta_{i\xi} \phi_0^2 \mathcal{A}_\xi^a(\theta, \theta'),$$

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \mathcal{O}[\psi].$$

Therefore the effective source splits into:

$$\mathcal{L}_\perp^{(1)} \sim (\partial_\xi \phi_0) \phi_0 \theta'(\xi) [m_a k_a] \mathcal{O}[\psi],$$

which generates

$$V_\perp^\mu(\xi) = \theta'(\xi) [\Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a],$$

and hence

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O;$$

and

$$\mathcal{L}_\parallel^{\text{prim}} \sim \phi_0^2 \theta'(\xi) [\mathcal{A}_\xi^a k_a] \mathcal{O}[\psi],$$

which reduces to the primitive commutator-square channel

$$V_\parallel^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3,$$

and hence

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}.$$

Thus the complete TECT realization of the minimal Class II carrier yields the final reduced symbol

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_\parallel \sigma_3 + I_\theta \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_\theta \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

So the final closure condition is

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).}$$

LII. EXPLICIT SHELL/TEST-MODE REALIZATION OF $\mathcal{O}[\psi]$

LIII. MINIMAL SHELL/TEST MODE

Let the shell/test mode be the complex transverse field

$$\boxed{\psi(y) = u_1(y) + i u_2(y),} \tag{85}$$

where (u_1, u_2) are the two transverse components on the polarization plane

$$\hat{k}^\perp = \text{span}\{e_1, e_2\}.$$

The natural first-order test-mode operator is the worldvolume covariant derivative

$$\boxed{\mathcal{O}_j[\psi] = D_j \psi.} \quad (86)$$

This is the unique lowest-order choice compatible with the already-closed shell target form

$$(\partial_i \phi_0)(\mathcal{D}_j \psi) \xi_a.$$

LIV. MINIMAL COMPLETE PROJECTOR CURRENT

We therefore choose the complete lowest-order projector/test-mode current in the form

$$\boxed{K_i^a[P, \psi] = (T^a \partial_i P) \mathcal{T}_j D_j \psi,} \quad (87)$$

where $\mathcal{T}_j$ is a shell tensor acting on the transverse worldvolume derivative index.

To realize the S2 branch, $\mathcal{T}_j$ must split into even and odd transverse parts. Thus the minimal decomposition is

$$\boxed{\mathcal{T}_j = \Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j,} \quad (88)$$

where E_j and J_j are the shell-adapted even/odd transverse tensors.

Equivalently, in index form,

$$\boxed{K_i^a[P, \psi] = (T^a \partial_i P) \left(\Lambda_E^{(0)} E_j{}^b + \Lambda_O^{(0)} J_j{}^b \right) D_j \psi_b.} \quad (89)$$

LV. DEFECT-BACKGROUND INSERTION

On the defect background,

$$\partial_i P_B = \delta_{i\xi} \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger),$$

hence

$$(T^a \partial_i P_B) = \delta_{i\xi} \theta'(\xi) k_a(\theta), \quad k_a(\theta) := \left[T^a (e_1 z_B^\dagger + z_B e_1^\dagger) \right]. \quad (90)$$

Therefore the explicit defect realization becomes

$$\boxed{K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \left(\Lambda_E^{(0)} E_j{}^b + \Lambda_O^{(0)} J_j{}^b \right) D_j \psi_b.} \quad (91)$$

LVI. INDUCED MIXED TERM

After eliminating the heavy mediator,

$$\mathcal{L}_{\text{eff}}^{(II)} = -\frac{\alpha\beta}{M_X^2} J_i^a K_i^a.$$

Using

$$J_i^a[\Psi_B] = 2(\partial_i \phi_0) \phi_0 m_a(\theta) + \delta_{i\xi} \phi_0^2 \mathcal{A}_\xi^a(\theta, \theta'),$$

with

$$m_a(\theta) = z_B^\dagger T_a z_B,$$

the shell-gradient contribution to the mixed term is

$$\mathcal{L}_{\text{eff},\perp}^{(II)} = -\frac{2\alpha\beta}{M_X^2} (\partial_\xi \phi_0) \phi_0 \theta'(\xi) m_a(\theta) k_a(\theta) \left(\Lambda_E^{(0)} E_j{}^b + \Lambda_O^{(0)} J_j{}^b \right) D_j \psi_b. \quad (92)$$

Thus the induced transverse transport kernel is

$$V_\perp^\mu(\xi) = \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right], \quad (93)$$

where

$$\Lambda_E = -\frac{2\alpha\beta}{M_X^2} \mathcal{N}_E, \quad \Lambda_O = -\frac{2\alpha\beta}{M_X^2} \mathcal{N}_O, \quad (94)$$

with microscopic overlap coefficients

$$\mathcal{N}_E := (\partial_\xi \phi_0) \phi_0 m_a(\theta) k_a(\theta) \Lambda_E^{(0)}, \quad \mathcal{N}_O := (\partial_\xi \phi_0) \phi_0 m_a(\theta) k_a(\theta) \Lambda_O^{(0)}. \quad (95)$$

After projection onto the low modes $u_a^\pm(\xi)$, this gives

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (96)$$

LVII. LONGITUDINAL PRIMITIVE BRANCH

The remaining term

$$\delta_{i\xi} \phi_0^2 \mathcal{A}_\xi^a(\theta, \theta')$$

contracts with the same K_i^a and produces the primitive commutator-square channel. Indeed,

$$\mathcal{A}_\xi^a(\theta, \theta') k_a(\theta) \theta'(\xi) \sim \theta'(\xi)^2,$$

and after tangent projection,

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3.$$

Hence

$$V_\parallel^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3, \quad (97)$$

and therefore

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}. \quad (98)$$

LVIII. FINAL REDUCED SYMBOL

Combining the two sectors,

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_\parallel \sigma_3 + I_\theta \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_\theta \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (99)$$

So the complete shell/test-mode realization is obtained by the minimal choice

$$\mathcal{O}_j[\psi] = D_j \psi, \quad K_i^a[P, \psi] = (T^a \partial_i P) (\Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j) D_j \psi.$$

The final nondegeneracy condition is

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (100)$$

LIX. FINAL SUMMARY

The minimal complete shell/test-mode choice is

$$\boxed{\mathcal{O}_j[\psi] = D_j \psi.}$$

Accordingly, the lowest-order projector/test-mode current is

$$\boxed{K_i^a[P, \psi] = (T^a \partial_i P) (\Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j) D_j \psi.}$$

On the defect background,

$$\partial_i P_B = \delta_{i\xi} \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger),$$

so

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) (\Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j) D_j \psi.$$

After eliminating the heavy mediator, the mixed term

$$-\frac{\alpha\beta}{M_X^2} J_i^a K_i^a$$

splits automatically into:

$$\boxed{V_\perp^\mu(\xi) = \theta'(\xi) [\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a],}$$

hence

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O,$$

and

$$\boxed{V_\parallel^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3,}$$

hence

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}.$$

Therefore the final reduced principal symbol is

$$\boxed{A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_\parallel \sigma_3 + I_\theta \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_\theta \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).}$$

So the remaining microscopic task is only to determine the actual signs and nonvanishing of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O$$

inside the full complete TECT action. This is exactly the currently honest frontier.

LX. NONVANISHING CRITERIA FOR THE FINAL REDUCED COEFFICIENTS

LXI. COMPLETE REDUCED SYMBOL

The reduced principal symbol is

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta} \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta} \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (101)$$

Since the explicit defect profile already gives

$$I_{\theta} \neq 0, \quad I_{\theta'^3} \neq 0,$$

the exact nondegeneracy condition is

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (102)$$

LXII. COEFFICIENT EXTRACTION IN THE SHELL-SEED BASIS

Let the microscopic transport seed space be

$$V_{\text{seed}} = \text{span}\{B_{\parallel}, B_I, B_J\},$$

with

$$B_{\parallel} \leftrightarrow \hat{n}_i \delta^{a3}, \quad B_I \leftrightarrow e_{1i} \delta^{a1} + e_{2i} \delta^{a2}, \quad B_J \leftrightarrow -e_{1i} \delta^{a2} + e_{2i} \delta^{a1}.$$

In the canonical Class II basis, the transport-response matrix is diagonal:

$$\mathcal{M}_{\text{seed}} = \text{diag}(\mu_{\parallel}, \mu_I, \mu_J), \quad \mu_{\parallel} \neq 0, \quad \mu_I \neq 0, \quad \mu_J \neq 0. \quad (103)$$

For a general seed combination

$$N = c_{\parallel} B_{\parallel} + c_I B_I + c_J B_J, \quad (104)$$

the reduced coefficients are therefore

$$\lambda_c = \mu_{\parallel} c_{\parallel}, \quad \Lambda_E = \mu_I c_I, \quad \Lambda_O = \mu_J c_J. \quad (105)$$

Hence the sign information is immediate:

$$\text{sgn}(\lambda_c) = \text{sgn}(\mu_{\parallel}) \text{sgn}(c_{\parallel}), \quad \text{sgn}(\Lambda_E) = \text{sgn}(\mu_I) \text{sgn}(c_I), \quad \text{sgn}(\Lambda_O) = \text{sgn}(\mu_J) \text{sgn}(c_J). \quad (106)$$

LXIII. MINIMAL NONVANISHING THEOREM

Let V_{mic} denote the actual microscopic transport space produced by the full complete TECT action. Let

$$P_{V_{\text{mic}}}$$

be the orthogonal projection onto that space.

Then the exact finite completion criterion is:

$$P_{V_{\text{mic}}} B_{\parallel} \neq 0 \implies \lambda_c \neq 0, \quad (107)$$

and

$$P_{V_{\text{mic}}} B_I \neq 0 \text{ or } P_{V_{\text{mic}}} B_J \neq 0 \implies (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (108)$$

Therefore a sufficient condition for the full reduced symbol to be nondegenerate is

$$\boxed{P_{V_{\text{mic}}} B_{\parallel} \neq 0, \quad (P_{V_{\text{mic}}} B_I, P_{V_{\text{mic}}} B_J) \neq (0, 0).} \quad (109)$$

Under (109), one has

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0),} \quad (110)$$

hence by (101),

$$\boxed{A_1(p) \text{ is Dirac-nondegenerate.}} \quad (111)$$

LXIV. DIRECT COEFFICIENT FORMULAS FROM THE CHOSEN COMPLETE REALIZATION

For the chosen lowest-order complete realization

$$K_i^a[P, \psi] = (T^a \partial_i P) (\Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j) D_j \psi,$$

and

$$\mathcal{L}_{\text{eff}}^{(II)} = -\frac{\alpha\beta}{M_X^2} J_i^a K_i^a,$$

the reduced coefficients take the form

$$\boxed{\Lambda_E = -\frac{2\alpha\beta}{M_X^2} \mathcal{N}_E, \quad \Lambda_O = -\frac{2\alpha\beta}{M_X^2} \mathcal{N}_O,} \quad (112)$$

for microscopic overlaps $\mathcal{N}_E, \mathcal{N}_O$, and

$$\boxed{\lambda_c = -\frac{\alpha\beta}{M_X^2} \mathcal{N}_{\parallel},} \quad (113)$$

where $\mathcal{N}_{\parallel}$ is the primitive commutator-square overlap.

Thus the practical sufficient conditions are simply

$$\boxed{\mathcal{N}_{\parallel} \neq 0, \quad (\mathcal{N}_E, \mathcal{N}_O) \neq (0, 0).} \quad (114)$$

If $\alpha\beta \neq 0$, then (114) implies

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (115)$$

LXV. FINAL COMPLETE-PDE CRITERION

Therefore the complete TECT PDE closes at lowest order iff the full microscopic Hessian generates nonzero overlap with the three shell target directions:

$$\boxed{\mathcal{N}_{\parallel} \neq 0, \quad (\mathcal{N}_E, \mathcal{N}_O) \neq (0, 0).} \quad (116)$$

Equivalently, in the seed-language,

$$\boxed{c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0).} \quad (117)$$

Then the final reduced PDE is exactly

$$\boxed{A_1(p) = \frac{\lambda_c}{2} I_{\theta'3} p_{\parallel} \sigma_3 + I_{\theta} \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta} \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (118)$$

LXVI. FINAL SUMMARY

The final reduced PDE is

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'3} p_{\parallel} \sigma_3 + I_{\theta} \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta} \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

Since

$$I_{\theta} \neq 0, \quad I_{\theta'3} \neq 0,$$

its exact nondegeneracy condition is

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).}$$

In the canonical Class II seed basis,

$$\lambda_c = \mu_{\parallel} c_{\parallel}, \quad \Lambda_E = \mu_I c_I, \quad \Lambda_O = \mu_J c_J,$$

with

$$\mu_{\parallel} \neq 0, \quad \mu_I \neq 0, \quad \mu_J \neq 0.$$

Therefore a sufficient microscopic completion criterion is

$$\boxed{P_{V_{\text{mic}}} B_{\parallel} \neq 0, \quad (P_{V_{\text{mic}}} B_I, P_{V_{\text{mic}}} B_J) \neq (0, 0).}$$

Equivalently,

$$\boxed{c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0).}$$

For the chosen complete realization,

$$\Lambda_E = -\frac{2\alpha\beta}{M_X^2} \mathcal{N}_E, \quad \Lambda_O = -\frac{2\alpha\beta}{M_X^2} \mathcal{N}_O, \quad \lambda_c = -\frac{\alpha\beta}{M_X^2} \mathcal{N}_{\parallel}.$$

So the entire final complete-PDE problem is reduced to the three overlap tests

$$\boxed{\mathcal{N}_{\parallel} \neq 0, \quad (\mathcal{N}_E, \mathcal{N}_O) \neq (0, 0).}$$

LXVII. EXPLICIT EVALUATION OF $\mathcal{N}_{\parallel}, \mathcal{N}_E, \mathcal{N}_O$ FOR A CONCRETE GENERATOR CHOICE

LXVIII. CONCRETE EMBEDDED $SU(2)$ CHOICE

Choose the embedded $SU(2)$ acting on the $(1, 3)$ -plane:

$$\boxed{T_1 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad T_2 = \frac{1}{2} \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad T_3 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.} \quad (119)$$

Take the explicit real defect representative

$$\boxed{z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}.} \quad (120)$$

Then

$$z_B^{\dagger} z_B = 1, \quad \partial_{\xi} z_B = \theta'(\xi) e_1(\xi).$$

LXIX. BASIC ADJOINT CONTRACTIONS

Define

$$m_a(\theta) := z_B^\dagger T_a z_B, \quad (121)$$

$$k_a(\theta) := \left[T_a (e_1 z_B^\dagger + z_B e_1^\dagger) \right], \quad (122)$$

$$A_a(\theta) := e_1^\dagger T_a z_B + z_B^\dagger T_a e_1. \quad (123)$$

A direct computation gives

$$\boxed{m_1 = \frac{1}{2} \sin 2\theta, \quad m_2 = 0, \quad m_3 = \frac{1}{2} \cos 2\theta,} \quad (124)$$

$$\boxed{k_1 = \cos 2\theta, \quad k_2 = 0, \quad k_3 = -\sin 2\theta,} \quad (125)$$

and

$$\boxed{A_1 = \cos 2\theta, \quad A_2 = 0, \quad A_3 = -\sin 2\theta.} \quad (126)$$

In particular,

$$\boxed{A_a(\theta) = k_a(\theta).} \quad (127)$$

LXX. TRANSVERSE OVERLAP

The transverse piece of the effective mixed current comes from the $\theta'(\xi)$ -order term

$$m_a(\theta) k_a(\theta).$$

Using (124) and (125),

$$\begin{aligned} m_a k_a &= m_1 k_1 + m_2 k_2 + m_3 k_3 \\ &= \frac{1}{2} \sin 2\theta \cos 2\theta + 0 + \frac{1}{2} \cos 2\theta (-\sin 2\theta) \\ &= 0. \end{aligned} \quad (128)$$

Therefore the minimal bare JK realization produces no transverse overlap:

$$\boxed{\mathcal{N}_E = 0, \quad \mathcal{N}_O = 0.} \quad (129)$$

Hence

$$\boxed{\Lambda_E = 0, \quad \Lambda_O = 0} \quad (130)$$

for this minimal real representative.

LXXI. PRIMITIVE LONGITUDINAL OVERLAP

The longitudinal primitive piece comes from

$$A_a(\theta) k_a(\theta) \theta'(\xi)^2.$$

Using (126) and (125),

$$\begin{aligned} A_a k_a &= A_1 k_1 + A_2 k_2 + A_3 k_3 \\ &= \cos^2 2\theta + 0 + (-\sin 2\theta)(-\sin 2\theta) \\ &= 1. \end{aligned} \tag{131}$$

Thus the primitive longitudinal overlap is strictly nonzero:

$$\boxed{\mathcal{N}_{\parallel} = 1} \tag{132}$$

up to the overall shell/defect normalization conventions.

Therefore

$$\boxed{\lambda_c = -\frac{\alpha\beta}{M_X^2} \mathcal{N}_{\parallel} = -\frac{\alpha\beta}{M_X^2} \neq 0 \quad (\alpha\beta \neq 0).} \tag{133}$$

Consequently,

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'3} = -\frac{\alpha\beta}{2M_X^2} I_{\theta'3}.} \tag{134}$$

LXXII. RESULTING REDUCED SYMBOL

For this minimal real complete realization,

$$\boxed{A_1(p) = -\frac{\alpha\beta}{2M_X^2} I_{\theta'3} p_{\parallel} \sigma_3.} \tag{135}$$

So the reduced symbol is longitudinal-only and fails full Dirac nondegeneracy:

$$\boxed{\lambda_c \neq 0, \quad \Lambda_E = \Lambda_O = 0.} \tag{136}$$

LXXIII. IMMEDIATE CONSEQUENCE

Therefore the complete microscopic conclusion is sharp:

$$\boxed{\text{the minimal bare } J_i^a K_i^a \text{ realization on the real } CP^2 \text{ representative closes the primitive longitudinal channel,}} \tag{137}$$

but

$$\boxed{\text{it does not generate the transverse-even or transverse-odd channels.}} \tag{138}$$

Hence the full complete TECT action must contain additional transverse carrier structure, namely the extra E/O -slots previously identified:

$$\boxed{\mathcal{I}_E, \quad \mathcal{I}_O.} \tag{139}$$

Without them, the final reduced PDE is not fully Dirac-nondegenerate.

LXXIV. FINAL SUMMARY

For the explicit generator choice

$$T_1 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad T_2 = \frac{1}{2} \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad T_3 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

and the real defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix},$$

one finds

$$\begin{aligned} m_1 &= \frac{1}{2} \sin 2\theta, & m_2 &= 0, & m_3 &= \frac{1}{2} \cos 2\theta, \\ k_1 &= \cos 2\theta, & k_2 &= 0, & k_3 &= -\sin 2\theta, \\ A_1 &= \cos 2\theta, & A_2 &= 0, & A_3 &= -\sin 2\theta. \end{aligned}$$

Hence

$$m_a k_a = 0,$$

so the transverse overlaps vanish:

$$\boxed{\mathcal{N}_E = 0, \quad \mathcal{N}_O = 0.}$$

But

$$A_a k_a = 1,$$

so the primitive longitudinal overlap is nonzero:

$$\boxed{\mathcal{N}_{\parallel} = 1.}$$

Therefore

$$\boxed{\lambda_c = -\frac{\alpha\beta}{M_X^2} \neq 0 \quad (\alpha\beta \neq 0), \quad \Lambda_E = \Lambda_O = 0.}$$

Thus the minimal bare $J_i^a K_i^a$ realization on the real CP^2 representative produces only the primitive longitudinal channel and does not generate the transverse Dirac channels.

So the full complete TECT action must contain additional transverse carrier slots $\mathcal{I}_E, \mathcal{I}_O$ if the final reduced PDE is to be fully Dirac-nondegenerate.

LXXV. EXPLICIT TRANSVERSE COMPLETION BY EXTRA E/O SLOTS

LXXVI. DEFECT FRAME

Keep the same real defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (140)$$

Thus

$$Q_B(\xi) = \text{span}\{e_1(\xi), e_2\}.$$

LXXVII. WHY THE PREVIOUS MINIMAL REALIZATION FAILED

With the embedded $SU(2)_{(13)}$ generators

$$T_1, T_2, T_3,$$

all adjoint contractions only saw the $(1, 3)$ -plane. Hence the contraction

$$m_a(\theta)k_a(\theta)$$

cancelled identically, giving

$$\mathcal{N}_E = \mathcal{N}_O = 0,$$

while the primitive longitudinal contraction

$$A_a(\theta)k_a(\theta) = 1$$

survived. Therefore the transverse slots must involve generators that see the second tangent vector e_2 .

LXXVIII. MINIMAL TRANSVERSE GENERATOR PAIR

Choose the lowest-order Hermitian pair mixing the $(1, 2)$ -plane:

$$\boxed{S_E := \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S_O := \frac{1}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.} \quad (141)$$

These are the minimal extra generators needed to couple z_B to e_2 .

LXXIX. EVEN TRANSVERSE CARRIER

Define the even transverse rank-one projector tensor

$$M_E := e_2 z_B^\dagger + z_B e_2^\dagger. \quad (142)$$

Using (140),

$$M_E = \begin{pmatrix} 0 & \cos \theta & 0 \\ \cos \theta & 0 & \sin \theta \\ 0 & \sin \theta & 0 \end{pmatrix}. \quad (143)$$

Now compute

$$q_E(\theta) := (S_E M_E). \quad (144)$$

Since

$$(S_E M_E) = \frac{1}{2}(M_{12} + M_{21}),$$

we get

$$\boxed{q_E(\theta) = \cos \theta.} \quad (145)$$

Therefore the lowest-order even extra slot

$$\boxed{\mathcal{I}_E = X_i^{(E)} (S_E M_E) E_j D_j \psi = X_i^{(E)} \cos \theta(\xi) E_j D_j \psi} \quad (146)$$

is nonzero on the defect background whenever $\cos \theta \neq 0$.

LXXX. ODD TRANSVERSE CARRIER

Define the odd transverse rank-one tensor

$$M_O := i(e_2 z_B^\dagger - z_B e_2^\dagger). \quad (147)$$

Using (140),

$$M_O = \begin{pmatrix} 0 & -i \cos \theta & 0 \\ i \cos \theta & 0 & -i \sin \theta \\ 0 & i \sin \theta & 0 \end{pmatrix}. \quad (148)$$

Now compute

$$q_O(\theta) := (S_O M_O). \quad (149)$$

Since

$$(S_O M_O) = \frac{1}{2}(-i M_{21} + i M_{12}),$$

we obtain

$$\boxed{q_O(\theta) = \cos \theta.} \quad (150)$$

Hence the lowest-order odd extra slot

$$\boxed{\mathcal{I}_O = X_i^{(O)} (S_O M_O) J_j D_j \psi = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi} \quad (151)$$

is also nonzero whenever $\cos \theta \neq 0$.

LXXXI. HEAVY-FIELD ELIMINATION

Include the transverse extra source

$$\mathcal{J}_i^{\text{extra}} = \gamma_E \cos \theta(\xi) E_j D_j \psi + \gamma_O \cos \theta(\xi) J_j D_j \psi. \quad (152)$$

After eliminating the heavy field X_i ,

$$\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \mathcal{J}_i^{\text{tot}} \mathcal{J}_i^{\text{tot}}. \quad (153)$$

The shell-relevant cross terms give the transverse transport kernel

$$\boxed{V_\perp^\mu(\xi) = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \sigma_a + \tilde{\Lambda}_O J^\mu{}_a \sigma_a \right].} \quad (154)$$

Define the weighted defect overlap

$$\boxed{I_{\theta, \cos} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi).} \quad (155)$$

Then

$$\boxed{\alpha = I_{\theta, \cos} \tilde{\Lambda}_E, \quad \beta = I_{\theta, \cos} \tilde{\Lambda}_O.} \quad (156)$$

Since

$$\frac{d}{d\xi} \sin \theta(\xi) = \theta'(\xi) \cos \theta(\xi),$$

one has

$$I_{\theta,\cos} = \int d\xi f_+(\xi) f_-(\xi) \frac{d}{d\xi} \sin \theta(\xi). \quad (157)$$

For a monotone defect profile with nontrivial θ -variation and nonvanishing even-odd overlap $f_+ f_-$, this is generically nonzero:

$$\boxed{I_{\theta,\cos} \neq 0.} \quad (158)$$

Therefore

$$\boxed{\mathcal{N}_E = \cos \theta \neq 0, \quad \mathcal{N}_O = \cos \theta \neq 0,} \quad (159)$$

and consequently

$$\boxed{\Lambda_E \neq 0, \quad \Lambda_O \neq 0} \quad (160)$$

whenever $\gamma_E, \gamma_O \neq 0$.

LXXXII. FULL REDUCED SYMBOL

Combining this with the already-established primitive longitudinal channel

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \lambda_c = -\frac{\alpha\beta}{M_X^2} \mathcal{N}_{\parallel}, \quad \mathcal{N}_{\parallel} = 1,$$

the final reduced symbol becomes

$$\boxed{A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta,\cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta,\cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (161)$$

Hence the full nondegeneracy condition is satisfied:

$$\boxed{\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0).} \quad (162)$$

LXXXIII. CONCLUSION

The minimal bare $SU(2)_{(13)}$ realization gives only the primitive longitudinal channel. After adding the explicit transverse extra generators S_E, S_O , the transverse even and odd channels become nonzero on the same defect background.

Therefore the complete lowest-order TECT realization is fully Dirac-nondegenerate.

LXXXIV. FINAL SUMMARY

Take

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

and the extra transverse generators

$$S_E = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S_O = \frac{1}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then

$$M_E = e_2 z_B^\dagger + z_B e_2^\dagger, \quad M_O = i(e_2 z_B^\dagger - z_B e_2^\dagger),$$

satisfy

$$(S_E M_E) = \cos \theta, \quad (S_O M_O) = \cos \theta.$$

Therefore the extra transverse slots are explicitly nonzero:

$$\mathcal{I}_E \sim X^{(E)} \cos \theta E_j D_j \psi, \quad \mathcal{I}_O \sim X^{(O)} \cos \theta J_j D_j \psi.$$

After heavy-field elimination, the induced transverse kernel is

$$V_\perp^\mu(\xi) = \theta'(\xi) \cos \theta(\xi) [\tilde{\Lambda}_E E^\mu{}_a \sigma_a + \tilde{\Lambda}_O J^\mu{}_a \sigma_a].$$

Hence

$$\alpha = I_{\theta, \cos} \tilde{\Lambda}_E, \quad \beta = I_{\theta, \cos} \tilde{\Lambda}_O, \quad I_{\theta, \cos} = \int d\xi f_+ f_- \theta' \cos \theta \neq 0.$$

Combining this with the primitive longitudinal channel

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \mathcal{N}_\parallel = 1,$$

the final reduced symbol is

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_\parallel \sigma_3 + I_{\theta, \cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta, \cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

Thus, after adding the explicit extra transverse carriers, the complete lowest-order TECT realization becomes fully Dirac-nondegenerate:

$$\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0).$$

LXXXV. FINAL PDE STATEMENT OF THE LOWEST-ORDER COMPLETE TECT REALIZATION

LXXXVI. COMPLETE MICROSCOPIC FIELD CONTENT

The lowest-order complete TECT realization contains the fields

$$\Psi(x, t) \in \mathbb{C}^3, \quad P(x, t) = z(x, t) z(x, t)^\dagger, \quad z^\dagger z = 1, \quad X_i^a(x, t), \quad \sigma(x, t).$$

LXXXVII. COMPLETE LOWEST-ORDER ACTION

The complete lowest-order microscopic action is

$$S_{\text{TECT}}^{\text{comp, LO}} = \int dt d^3x \left(\mathcal{L}_\Psi^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_\sigma^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O \right). \quad (163)$$

The microscopic currents are

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \Psi^\dagger T_a (\partial_i \Psi), \quad (164)$$

$$K_i^a[P, \psi] = (T_a \partial_i P) (\Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j) D_j \psi. \quad (165)$$

The extra transverse carriers are realized by

$$\boxed{\mathcal{I}_E \sim X^{(E)}(S_E M_E) E_j D_j \psi, \quad \mathcal{I}_O \sim X^{(O)}(S_O M_O) J_j D_j \psi,} \quad (166)$$

with

$$S_E = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S_O = \frac{1}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and

$$M_E = e_2 z_B^\dagger + z_B e_2^\dagger, \quad M_O = i(e_2 z_B^\dagger - z_B e_2^\dagger).$$

LXXXVIII. MICROSCOPIC EULER-LAGRANGE PDE SYSTEM

The full microscopic PDE system is

$$\boxed{\frac{\delta S_{\text{TECT}}^{\text{comp,LO}}}{\delta \Psi^\dagger} = 0, \quad \frac{\delta S_{\text{TECT}}^{\text{comp,LO}}}{\delta P} = 0, \quad \frac{\delta S_{\text{TECT}}^{\text{comp,LO}}}{\delta \sigma} = 0, \quad \frac{\delta S_{\text{TECT}}^{\text{comp,LO}}}{\delta X_i^a} = 0.} \quad (167)$$

The heavy-field equation is algebraic:

$$\boxed{M_X^2 X_i^a + \mathcal{J}_i^{a,\text{tot}} = 0, \quad X_i^a = -\frac{1}{M_X^2} \mathcal{J}_i^{a,\text{tot}}.} \quad (168)$$

Hence the induced effective interaction is

$$\boxed{\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \mathcal{J}_i^{a,\text{tot}} \mathcal{J}_i^{a,\text{tot}}.} \quad (169)$$

LXXXIX. DEFECT BACKGROUND

Take the explicit defect representative

$$\boxed{z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.} \quad (170)$$

Then

$$\partial_\xi z_B = \theta'(\xi) e_1(\xi), \quad \partial_i P_B = \delta_{i\xi} \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger).$$

XC. EFFECTIVE TRANSPORT KERNEL

For the explicit transverse generators S_E, S_O , one has

$$\boxed{(S_E M_E) = \cos \theta(\xi), \quad (S_O M_O) = \cos \theta(\xi).} \quad (171)$$

Hence the effective mediator transport kernel is

$$\boxed{V^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \sigma_a + \tilde{\Lambda}_O J^\mu{}_a \sigma_a \right].} \quad (172)$$

Define the defect overlaps

$$I_{\theta'^3} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3, \quad I_{\theta, \cos} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi). \quad (173)$$

Then the reduced coefficients are

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta, \cos} \tilde{\Lambda}_E, \quad \beta = I_{\theta, \cos} \tilde{\Lambda}_O. \quad (174)$$

XCI. FINAL REDUCED INFRARED PDE

After Bloch reduction around the locked 12-mode BCC background and shell projection at

$$k_* = \frac{G_*}{2}, \quad k = k_* + p,$$

the final reduced principal symbol is

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta, \cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta, \cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (175)$$

Equivalently, the reduced first-order PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_{\theta'^3} \sigma_3 (-i\partial_{\parallel}) - I_{\theta, \cos} \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_{\theta, \cos} \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (176)$$

XCII. CLOSURE CONDITION

The final Dirac nondegeneracy condition is

$$\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0), \quad I_{\theta'^3} \neq 0, \quad I_{\theta, \cos} \neq 0. \quad (177)$$

Under these conditions, the lowest-order complete TECT realization is fully Dirac-nondegenerate.

XCIII. FINAL SUMMARY

The lowest-order complete TECT realization is the coupled microscopic PDE system for

$$\Psi, \quad P = zz^\dagger, \quad \sigma, \quad X_i^a,$$

derived from

$$S_{\text{TECT}}^{\text{comp, LO}} = \int dt d^3x \left(\mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O \right).$$

After eliminating the heavy field X_i^a , the effective transport kernel on the defect background is

$$V^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \cos \theta(\xi) [\tilde{\Lambda}_E E^\mu{}_a \sigma_a + \tilde{\Lambda}_O J^\mu{}_a \sigma_a].$$

Hence the final reduced principal symbol is

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta, \cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta, \cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

Equivalently, the reduced first-order PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_{\theta'^3} \sigma_3 (-i\partial_{\parallel}) - I_{\theta, \cos} \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_{\theta, \cos} \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.$$

The closure condition is

$$\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0), \quad I_{\theta'^3} \neq 0, \quad I_{\theta, \cos} \neq 0.$$

This is the final lowest-order complete TECT PDE statement.

XCIV. CURRENT STATUS OF THE FINAL PDE: PROVED / ASSUMPTION / OPEN

XCV. ALREADY PROVED / REDUCED

The following statements are already closed at the reduced level:

1. the correct infrared carrier is a graded $2 + 2$ first-order block;
2. the natural reduced branch is S2;
3. the reduced principal symbol has the form

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta, \cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta, \cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (178)$$

4. the reduced nondegeneracy condition is exactly

$$\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0). \quad (179)$$

5. equivalently, in the concrete Class II dictionary,

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0). \quad (180)$$

Thus the Dirac witness problem is no longer structural. It is reduced to a finite coefficient test.

XCVI. WORKING ASSUMPTION

The currently used lowest-order complete microscopic realization is

$$S_{\text{TECT}}^{\text{comp, LO}} = \int dt d^3x \left(\mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O \right). \quad (181)$$

This action is the current explicit completion ansatz used to realize all three carrier directions:

$$B_{\parallel}, \quad B_I, \quad B_J.$$

What is assumed here is not the reduced symbol (178), but the microscopic realization (181) that produces it.

XCVII. STILL OPEN

What remains honestly open is the full microscopic extraction from the final complete TECT operator. This means the following chain is still to be carried out for the actual complete action:

$$\boxed{L(k) \rightarrow K_i = \partial_{k_i} L|_{k_*} \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).} \quad (182)$$

Equivalently, in the local transport language, one must still derive from complete TECT

$$\boxed{V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),} \quad (183)$$

and extract the actual profile coefficients

$$\boxed{h_{\parallel}(\theta), \quad h_E(\theta), \quad h_O(\theta),} \quad (184)$$

hence the actual coefficient point

$$\boxed{(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O) \text{ or equivalently } (g_c, g_E, g_O).} \quad (185)$$

Therefore the true remaining frontier is:

$$\boxed{\text{show directly from the final complete microscopic TECT action that } g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).} \quad (186)$$

XCVIII. SHARP HONEST FINAL STATEMENT

So the final PDE status is:

$$\boxed{\text{The reduced Dirac-type PDE form is already fixed.}} \quad (187)$$

$$\boxed{\text{The current lowest-order complete realization is a working microscopic ansatz.}} \quad (188)$$

$$\boxed{\text{The only genuinely open problem is the complete microscopic coefficient extraction.}} \quad (189)$$

XCIX. FINAL SUMMARY

Already proved / reduced:

$$A_1(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta, \cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta, \cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2),$$

with exact finite test

$$\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0).$$

Working assumption:

$$S_{\text{TECT}}^{\text{comp, LO}} = \int dt d^3x \left(\mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O \right).$$

Still open:

$$L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta),$$

equivalently

$$V^\mu(\xi) \rightarrow h_\parallel, h_E, h_O \rightarrow (g_c, g_E, g_O),$$

must still be derived from the final complete microscopic TECT action itself.

So:

form of the reduced PDE: closed;

microscopic lowest-order completion ansatz: working assumption;

actual coefficient extraction from complete TECT: still open.

C. FINAL BRANCH-SENSITIVE PDE STATEMENT

CI. GENERAL ACTUAL-LOCAL TRANSPORT FORM

The already-closed actual-local transport form is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (190)$$

with invariant decomposition

$$\mathcal{M}_{\text{inv}}^\mu(\theta) = h_0(\theta) n^\mu \mathbf{1}_2 + h_\parallel(\theta) n^\mu \sigma_3 + h_E(\theta) E^\mu{}_a \sigma_a + h_O(\theta) J^\mu{}_a \sigma_a. \quad (191)$$

The reduced coefficients are

$$\lambda_\parallel = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_\parallel(\theta(\xi)), \quad (192)$$

$$\alpha = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_E(\theta(\xi)), \quad \beta = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_O(\theta(\xi)). \quad (193)$$

CII. LONGITUDINAL BRANCH SPLIT

Expand

$$h_\parallel(\theta) = \Lambda_\parallel^{(0)} + \Lambda_\parallel^{(2)} \theta^2 + O(\theta^4). \quad (194)$$

Then the general longitudinal overlap is

$$\lambda_\parallel = \Lambda_\parallel^{(0)} I_\theta + \Lambda_\parallel^{(2)} I_{\theta^2 \theta'} + O(\theta_0^5 \kappa). \quad (195)$$

Therefore:

a. Branch A. If

$$\Lambda_\parallel^{(0)} \neq 0, \quad (196)$$

then already

$$\lambda_\parallel \neq 0. \quad (197)$$

b. Branch B. If

$$\Lambda_{\parallel}^{(0)} = 0, \quad (198)$$

one must test the primitive commutator-square seed

$$\mathcal{T}_{\xi} := i[P_B, \partial_{\xi} P_B], \quad S_{\xi} := (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}. \quad (199)$$

For the explicit defect representative,

$$S_{\xi} = \frac{1}{2} \theta'(\xi)^2 \sigma_3. \quad (200)$$

Hence the primitive longitudinal transport is

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) S_{\xi} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3. \quad (201)$$

Thus

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}. \quad (202)$$

So if

$$\lambda_c \neq 0, \quad (203)$$

then again

$$\lambda_{\parallel} \neq 0. \quad (204)$$

CIII. TRANSVERSE SECTOR

The transverse sector is already closed at the finite-test level:

$$(\Lambda_E, \Lambda_O) \neq (0, 0) \implies (\alpha, \beta) \neq (0, 0). \quad (205)$$

In the exact concrete Class II branch one has

$$\alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}. \quad (206)$$

CIV. EXACT CONCRETE CLASS II BRANCH

For the exact concrete Class II witness formulas,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}. \quad (207)$$

Hence the concrete currently constructed branch is

$$\text{Branch B, not Branch A.} \quad (208)$$

Its nondegeneracy condition is

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0). \quad (209)$$

CV. FINAL BRANCH-SENSITIVE REDUCED PDE

The final reduced PDE must therefore be stated branch-sensitively.

a. Branch A form. If the complete microscopic TECT action realizes $\Lambda_{\parallel}^{(0)} \neq 0$, then

$$\boxed{A_1^{(A)}(p) = (I_{\theta}\Lambda_{\parallel}^{(0)} + O(\theta_0^3\kappa))p_{\parallel}\sigma_3 + I_{\theta}\Lambda_E(p_1\sigma_1 + p_2\sigma_2) + I_{\theta}\Lambda_O(-p_2\sigma_1 + p_1\sigma_2).} \quad (210)$$

b. Branch B form. If instead the complete microscopic TECT action realizes only the primitive longitudinal channel, then

$$\boxed{A_1^{(B)}(p) = \frac{\lambda_c}{2}I_{\theta'^3}p_{\parallel}\sigma_3 + I_{\theta}\Lambda_E(p_1\sigma_1 + p_2\sigma_2) + I_{\theta}\Lambda_O(-p_2\sigma_1 + p_1\sigma_2).} \quad (211)$$

CVI. FINAL DECISION PROTOCOL

The exact executable protocol is:

$$\boxed{\text{First compute } \Lambda_{\parallel}^{(0)} = \frac{1}{2}[\sigma_3 n_{\mu}\mathcal{M}^{\mu}(0)].} \quad (212)$$

If

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

the theory is on Branch A.

If

$$\Lambda_{\parallel}^{(0)} = 0,$$

then compute the primitive coefficient λ_c from the projection onto

$$\theta'(\xi) S_{\xi} n^{\mu}.$$

If

$$\lambda_c \neq 0,$$

the theory is on Branch B.

In either case, also test

$$(\Lambda_E, \Lambda_O) \neq (0, 0).$$

CVII. SHARP HONEST FINAL THEOREM

Therefore the exact current theorem-level status is:

$$\boxed{\text{The transverse sector is already reduced to the finite test } (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (213)$$

$$\boxed{\text{The only remaining ambiguity is the longitudinal branch choice: Branch A vs Branch B.}} \quad (214)$$

$$\boxed{\text{The complete microscopic closure problem is exactly: determine whether full TECT lies on Branch A or Branch B, and verify t}} \quad (215)$$

CVIII. FINAL SUMMARY

The final reduced PDE must be stated in a branch-sensitive way.

General actual-local transport:

$$V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)), \quad \mathcal{M}^\mu = h_0 n^\mu \mathbf{1}_2 + h_\parallel n^\mu \sigma_3 + h_E E^\mu{}_a \sigma_a + h_O J^\mu{}_a \sigma_a.$$

Longitudinal split:

$$\text{Branch A: } \Lambda_\parallel^{(0)} \neq 0 \implies \lambda_\parallel \neq 0,$$

$$\text{Branch B: } \Lambda_\parallel^{(0)} = 0, \lambda_c \neq 0 \implies \lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3} \neq 0.$$

Transverse sector:

$$(\Lambda_E, \Lambda_O) \neq (0, 0) \implies (\alpha, \beta) \neq (0, 0).$$

Exact concrete Class II branch:

$$\lambda_\parallel = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_\theta, \quad \beta = g_O I_\theta,$$

so the currently explicit branch is Branch B.

Therefore the final reduced PDE is either

$$A_1^{(A)}(p) = (I_\theta \Lambda_\parallel^{(0)} + O(\theta_0^3 \kappa)) p_\parallel \sigma_3 + I_\theta \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_\theta \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2),$$

or

$$A_1^{(B)}(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_\parallel \sigma_3 + I_\theta \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_\theta \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

So the only genuinely open microscopic question is:

Does the complete TECT coefficient point lie on Branch A or on Branch B?

CIX. HONEST FINAL CONCLUSION

The strongest justified conclusion at the present stage is the following.

The reduced infrared PDE form is already fixed.

(216)

More precisely, after shell reduction around the locked BCC background, the effective first-order symbol must take the S2 form

$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$

(217)

The exact finite nondegeneracy condition is

$\lambda_\parallel \neq 0, \quad (\alpha, \beta) \neq (0, 0).$

(218)

The transverse sector is already reduced to the finite coefficient test

$(\Lambda_E, \Lambda_O) \neq (0, 0).$

(219)

The longitudinal sector is reduced to a strict two-branch decision:

a. Branch A. If the complete microscopic TECT action realizes

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0,} \quad (220)$$

then

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + O(\theta_0^3 \kappa), \quad (221)$$

so the longitudinal channel is already present on the constant-leading branch.

b. Branch B. If instead

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,} \quad (222)$$

then the longitudinal channel is realized only through the primitive commutator-square seed,

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (223)$$

The exact concrete currently constructed Class II branch realizes Branch B:

$$\boxed{\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.} \quad (224)$$

Therefore the strongest explicit currently available reduced PDE is

$$\boxed{A_1^{(B)}(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta} \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta} \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (225)$$

However, this does *not* yet prove that the final complete TECT action itself must lie on Branch B. The true complete microscopic closure problem remains:

$$\boxed{\text{determine the actual coefficient point of complete TECT}} \quad (226)$$

namely

$$\boxed{(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O),} \quad (227)$$

or equivalently

$$\boxed{(\lambda_{\parallel}, \alpha, \beta).} \quad (228)$$

Thus the final unresolved microscopic question is exactly:

$$\boxed{\text{Does complete TECT realize Branch A or Branch B?}} \quad (229)$$

Accordingly, the honest final status is:

$$\boxed{\text{The reduced Dirac-type PDE form is proved.}} \quad (230)$$

$$\boxed{\text{The explicit strongest currently constructed microscopic realization is Branch B.}} \quad (231)$$

$$\boxed{\text{The complete microscopic theory is not yet fully closed, because the actual full-action coefficient extraction is still open.}} \quad (232)$$

So the roadmap-level conceptual problem is finished, but the final microscopic realization problem is not. The theory is therefore in a strong conditional-closure stage rather than a final unconditional-complete stage.

CX. ACTUAL BRANCH DECISION FOR THE MINIMAL COMPLETE TECT ANSATZ

CXI. PROBLEM

We now decide the longitudinal branch for the minimal complete lowest-order TECT ansatz. The question is whether the actual local transport kernel contains a constant-leading longitudinal term

$$h_{\parallel}(\theta) = \Lambda_{\parallel}^{(0)} + O(\theta^2),$$

or only the primitive commutator-square channel.

Equivalently, we must decide whether

$$\Lambda_{\parallel}^{(0)} \neq 0 \quad \text{or} \quad \Lambda_{\parallel}^{(0)} = 0, \lambda_c \neq 0.$$

CXII. MINIMAL COMPLETE MICROSCOPIC SOURCE

Take the lowest-order complete realization already constructed:

$$\mathcal{L}_{\text{eff}}^{(II)} = -\frac{\alpha\beta}{M_X^2} J_i^a K_i^a + \text{extra transverse } E/O \text{ slots.} \quad (233)$$

For the longitudinal branch decision, only the σ_3 -type normal component matters. The transverse E/O slots do not contribute to $n^\mu \sigma_3$, so they can be ignored in the longitudinal audit.

Thus the relevant microscopic currents are

$$J_i^a[\Psi_B] = 2(\partial_i \phi_0) \phi_0 m_a(\theta) + \delta_{i\xi} \phi_0^2 A_a(\theta) \theta', \quad (234)$$

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \mathcal{O}[\psi]. \quad (235)$$

Here

$$m_a(\theta) = z_B^\dagger T_a z_B, \quad k_a(\theta) = \left[T_a (e_1 z_B^\dagger + z_B e_1^\dagger) \right], \quad A_a(\theta) = e_1^\dagger T_a z_B + z_B^\dagger T_a e_1. \quad (236)$$

CXIII. CONCRETE GENERATOR CHOICE

Choose the embedded $SU(2)_{(13)}$ generators

$$T_1 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad T_2 = \frac{1}{2} \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad T_3 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad (237)$$

and the real defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}. \quad (238)$$

Then direct evaluation gives

$$m_1 = \frac{1}{2} \sin 2\theta, \quad m_2 = 0, \quad m_3 = \frac{1}{2} \cos 2\theta, \quad (239)$$

$$k_1 = \cos 2\theta, \quad k_2 = 0, \quad k_3 = -\sin 2\theta, \quad (240)$$

$$A_1 = \cos 2\theta, \quad A_2 = 0, \quad A_3 = -\sin 2\theta, \quad A_a = k_a. \quad (241)$$

CXIV. THE WOULD-BE BRANCH A TERM

Substituting (234) and (235) into (233), the θ' -order term is

$$\mathcal{L}_{\parallel, \text{cand}}^{(1)} = -\frac{2\alpha\beta}{M_X^2} (\partial_\xi \phi_0) \phi_0 \theta'(\xi) m_a(\theta) k_a(\theta) \mathcal{O}[\psi]. \quad (242)$$

If this contraction were nonzero and projected to $n^\mu \sigma_3$, it would generate a constant-leading longitudinal coefficient $\Lambda_{\parallel}^{(0)}$.

But using (239) and (240),

$$\begin{aligned} m_a(\theta) k_a(\theta) &= m_1 k_1 + m_2 k_2 + m_3 k_3 \\ &= \frac{1}{2} \sin 2\theta \cos 2\theta + 0 + \frac{1}{2} \cos 2\theta (-\sin 2\theta) \\ &= 0. \end{aligned} \quad (243)$$

Therefore the entire θ' -order longitudinal candidate vanishes identically:

$$\boxed{\mathcal{L}_{\parallel, \text{cand}}^{(1)} = 0.} \quad (244)$$

Hence the minimal complete ansatz generates no Branch A term. Equivalently,

$$\boxed{\Lambda_{\parallel}^{(0)} = 0.} \quad (245)$$

CXV. THE SURVIVING PRIMITIVE LONGITUDINAL TERM

Now consider the higher-order normal term coming from the $\delta_{i\xi} \phi_0^2 A_a \theta'$ part of J_i^a :

$$\mathcal{L}_{\parallel}^{(\text{prim})} = -\frac{\alpha\beta}{M_X^2} \phi_0^2 \theta'(\xi)^2 A_a(\theta) k_a(\theta) \mathcal{O}[\psi]. \quad (246)$$

Using (241) and (240),

$$\begin{aligned} A_a(\theta) k_a(\theta) &= A_1 k_1 + A_2 k_2 + A_3 k_3 \\ &= \cos^2 2\theta + 0 + (-\sin 2\theta)(-\sin 2\theta) \\ &= 1. \end{aligned} \quad (247)$$

Therefore

$$\boxed{\mathcal{L}_{\parallel}^{(\text{prim})} = -\frac{\alpha\beta}{M_X^2} \phi_0^2 \theta'(\xi)^2 \mathcal{O}[\psi].} \quad (248)$$

After tangent projection, this is exactly the commutator-square channel

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3, \quad (249)$$

hence the longitudinal transport kernel is

$$\boxed{V_{(\parallel)}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3,} \quad (250)$$

with

$$\boxed{\lambda_c = -\frac{\alpha\beta}{M_X^2} \neq 0 \quad (\alpha\beta \neq 0).} \quad (251)$$

Thus

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (252)$$

CXVI. BRANCH DECISION

Equations (245) and (251) give the exact branch decision:

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.} \quad (253)$$

Therefore the minimal complete lowest-order TECT ansatz lies *exactly* on Branch B:

$$\boxed{\text{minimal complete LO TECT} \in \text{Branch B.}} \quad (254)$$

CXVII. FINAL REDUCED PDE FOR THE SOLVED BRANCH

Combining the already constructed transverse completion $(\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0)$, the reduced principal symbol is

$$\boxed{A_1^{\text{LO}}(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta, \cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta, \cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (255)$$

Its nondegeneracy condition is

$$\boxed{\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0), \quad I_{\theta'^3} \neq 0, \quad I_{\theta, \cos} \neq 0.} \quad (256)$$

CXVIII. WHAT REMAINS OPEN

This solves the branch ambiguity for the *minimal complete lowest-order ansatz*. What remains open is only whether the true final complete TECT action contains additional microscopic operators that generate

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

thereby moving the theory from Branch B to Branch A.

So the branch problem is solved at the level of the present complete lowest-order realization, but not yet for the unknown ultimate full action.

CXIX. FINAL SUMMARY

For the concrete $SU(2)_{(13)}$ generator choice and real defect representative, the θ' -order longitudinal contraction vanishes exactly:

$$m_a(\theta) k_a(\theta) = 0.$$

Hence

$$\boxed{\Lambda_{\parallel}^{(0)} = 0.}$$

The primitive higher-order contraction survives exactly:

$$A_a(\theta) k_a(\theta) = 1.$$

Hence

$$\boxed{\lambda_c = -\frac{\alpha\beta}{M_X^2} \neq 0, \quad \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

Therefore the minimal complete lowest-order TECT realization lies exactly on Branch B:

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.$$

Including the already constructed transverse completion, the solved reduced PDE is

$$A_1^{\text{LO}}(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta, \cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta, \cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

So the branch ambiguity is solved for the present minimal complete lowest-order ansatz: it is Branch B. The only remaining open question is whether the ultimate full complete TECT action contains extra operators that can generate a nonzero constant-leading longitudinal coefficient $\Lambda_{\parallel}^{(0)}$, thereby realizing Branch A instead.

CXX. SOLVED FINAL PDE AT THE PRESENT LOWEST-ORDER COMPLETE STAGE

CXXI. SOLVED BRANCH DECISION

For the minimal complete lowest-order TECT realization constructed so far, the longitudinal branch is already decided:

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0. \quad (257)$$

Therefore this realization lies exactly on Branch B.

CXXII. FINAL REDUCED PRINCIPAL SYMBOL

Including the explicit transverse completion, the reduced first-order symbol is

$$A_1^{\text{LO}}(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta, \cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta, \cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (258)$$

CXXIII. EQUIVALENT REDUCED PDE

Equivalently, for the reduced shell spinor χ , the first-order PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_{\theta'^3} \sigma_3 (-i\partial_{\parallel}) - I_{\theta, \cos} \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_{\theta, \cos} \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (259)$$

CXXIV. NONDEGENERACY CONDITION

The solved nondegeneracy condition is

$$\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0), \quad I_{\theta'^3} \neq 0, \quad I_{\theta, \cos} \neq 0. \quad (260)$$

Under (260), the present lowest-order complete realization is fully Dirac-nondegenerate.

CXXV. MICROSCOPIC INTERPRETATION

Thus the current solved PDE has the following microscopic interpretation:

$$\text{longitudinal direction: primitive Branch B carrier } \theta'(\xi)^3 n^{\mu} \sigma_3, \quad (261)$$

$$\text{transverse directions: explicit extra } E/O \text{ carriers } \theta'(\xi) \cos \theta(\xi) E^{\mu}_a \sigma_a, \quad \theta'(\xi) \cos \theta(\xi) J^{\mu}_a \sigma_a. \quad (262)$$

CXXVI. SHARP HONEST STATUS

What is solved now is the final reduced PDE of the present minimal complete lowest-order realization:

$$\boxed{\text{present solved PDE} = \text{Branch B reduced Dirac PDE.}} \quad (263)$$

What is not solved yet is whether the ultimate full complete TECT action contains additional microscopic operators generating

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

which would move the theory from Branch B to Branch A.

CXXVII. FINAL SUMMARY

At the present stage, the actually solved final PDE is the reduced Branch B PDE of the minimal complete lowest-order TECT realization:

$$\boxed{A_1^{\text{LO}}(p) = \frac{\lambda_c}{2} I_{\theta'3} p_{\parallel} \sigma_3 + I_{\theta,\cos} \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta,\cos} \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2).}$$

Equivalently,

$$\boxed{\left[i\partial_t - \frac{\lambda_c}{2} I_{\theta'3} \sigma_3 (-i\partial_{\parallel}) - I_{\theta,\cos} \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_{\theta,\cos} \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.}$$

Its solved branch status is

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,}$$

so the current realization is exactly Branch B.

Its nondegeneracy condition is

$$\boxed{\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0), \quad I_{\theta'3} \neq 0, \quad I_{\theta,\cos} \neq 0.}$$

Therefore the present lowest-order complete realization is already a fully Dirac-nondegenerate reduced PDE. The only remaining open microscopic question is whether the ultimate full complete TECT action adds operators that produce

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

thereby upgrading the theory from Branch B to Branch A.

CXXVIII. HONEST FINAL CONCLUSION

The strongest justified conclusion at the present stage is the following.

$$\boxed{\text{The reduced infrared PDE form is already fixed.}} \quad (264)$$

More precisely, after shell reduction around the locked BCC background, the effective first-order symbol must take the S2 form

$$\boxed{A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (265)$$

The exact finite nondegeneracy condition is

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (266)$$

The transverse sector is already reduced to the finite test

$$\boxed{(\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (267)$$

The longitudinal sector is reduced to a strict two-branch decision:

a. Branch A. If

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0,} \quad (268)$$

then

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_{\theta} + O(\theta_0^3 \kappa), \quad (269)$$

so the longitudinal witness already survives on the constant-leading branch.

b. Branch B. If instead

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,} \quad (270)$$

then

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (271)$$

In the exact currently constructed concrete Class II branch,

$$\boxed{\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.} \quad (272)$$

So the strongest explicit microscopic realization presently available is Branch B. :contentReference[oaicite:0]index=0

Accordingly, the strongest explicit currently available reduced PDE is

$$\boxed{A_1^{(B)}(p) = \frac{\lambda_c}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + I_{\theta} \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta} \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (273)$$

However, this does *not* yet prove that the final complete TECT action itself must lie on Branch B. The true complete microscopic closure problem remains the extraction of the actual coefficient point

$$\boxed{(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O),} \quad (274)$$

equivalently

$$\boxed{(\lambda_{\parallel}, \alpha, \beta),} \quad (275)$$

from the full background-dependent TECT Hessian. :contentReference[oaicite:1]index=1 :contentReference[oaicite:2]index=2

Thus the final unresolved microscopic question is exactly:

$$\boxed{\text{Does complete TECT realize Branch A or Branch B?}} \quad (276)$$

Accordingly, the honest final status is:

$$\boxed{\text{The reduced Dirac-type PDE form is proved.}} \quad (277)$$

$$\boxed{\text{The explicit strongest currently constructed microscopic realization is Branch B.}} \quad (278)$$

$$\boxed{\text{The complete microscopic theory is not yet fully closed, because the actual full-action coefficient extraction is still open.}} \quad (279)$$

CXXIX. REMOVAL OF THE MAIN WEAKNESS BY A WEAKER BUT SUFFICIENT MICROSCOPIC CRITERION

CXXX. MAIN WEAKNESS

The main remaining weakness is the absence of a direct proof that the true complete TECT action realizes a nontrivial Class II carrier sector in the physically relevant defect background.

At the current document level, this is the exact remaining open issue:

$$\boxed{\text{Does the true complete microscopic TECT linearization contain a shell-relevant } (i, a)\text{-carrier sector?}} \quad (280)$$

:contentReference[oaicite:3]index=3

CXXXI. KEY OBSERVATION

This weakness can be removed as a *logical bottleneck* by replacing the overly strong requirement

“prove the full complete Class II realization first”

with the weaker but already sufficient requirement:

$$\boxed{\text{show that the actual microscopic transport space } V_{\text{micro}} \text{ contains one longitudinal seed direction and one transverse seed direction}} \quad (281)$$

:contentReference[oaicite:4]index=4

CXXXII. CANONICAL SEED BASIS

Inside the minimal Class II seed family, the shell-adapted tensor basis is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (282)$$

Equivalently, in the seed coordinates

$$(c_{\parallel}, c_I, c_J),$$

one has

$$\boxed{g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.} \quad (283)$$

:contentReference[oaicite:5]index=5

Therefore the canonical explicit seed directions are:

$$\boxed{c_{\parallel} \neq 0, \quad c_I = c_J = 0 \implies g_c \neq 0,} \quad (284)$$

$$\boxed{c_{\parallel} = 0, \quad c_I \neq 0, \quad c_J = 0 \implies g_E \neq 0,} \quad (285)$$

$$\boxed{c_{\parallel} = 0, \quad c_I = 0, \quad c_J \neq 0 \implies g_O \neq 0.} \quad (286)$$

:contentReference[oaicite:6]index=6

CXXXIII. PRACTICAL REPLACEMENT THEOREM

Let V_{micro} be the true symmetry-allowed microscopic transport space extracted from the complete TECT Hessian around the relevant defect background.

Assume that:

$$\boxed{\exists N_{\parallel} \in V_{\text{micro}} \text{ such that } \langle T^{(\parallel)}, N_{\parallel} \rangle \neq 0,} \quad (287)$$

and

$$\boxed{\exists N_{\perp} \in V_{\text{micro}} \text{ such that } \langle T^{(\perp)}, N_{\perp} \rangle \neq 0.} \quad (288)$$

Equivalently,

$$\boxed{P_{V_{\text{micro}}} T^{(\parallel)} \neq 0, \quad P_{V_{\text{micro}}} T^{(\perp)} \neq 0.} \quad (289)$$

:contentReference[oaicite:7]index=7

Then the witness channels survive generically:

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0),} \quad (290)$$

hence

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (291)$$

:contentReference[oaicite:8]index=8

CXXXIV. WHAT THIS REMOVES

This does *not* prove the entire final complete TECT action in one stroke. What it removes is the strongest remaining logical weakness:

$$\boxed{\text{we no longer need a full microscopic proof of all Class II completion terms before concluding local Dirac nondegeneracy.}} \quad (292)$$

Instead, it is sufficient to prove the much weaker finite statement:

$$\boxed{\text{the complete microscopic transport space contains one admissible longitudinal seed and one admissible transverse seed.}} \quad (293)$$

Thus the open problem is compressed from

“derive the entire full completion first”

to

“perform a finite seed-embedding check in the complete Hessian.”

CXXXV. FINAL PRACTICAL CONSEQUENCE

Therefore the main weakness is removed in the following precise sense:

$$\boxed{\text{the unresolved microscopic completion is no longer a conceptual obstruction; it is only a finite transport-space embedding check.}} \quad (294)$$

So the actual deliverable is now:

$$\boxed{\text{extract } V_{\text{micro}} \text{ from the full Hessian and verify that it contains one longitudinal seed and one transverse seed from } V_{\text{seed}}.} \quad (295)$$

CXXXVI. FINAL SUMMARY

The main weakness was:

no direct proof yet that the full complete TECT action realizes a nontrivial Class II carrier sector.

This weakness can be removed as a logical bottleneck by replacing the overly strong requirement of proving the full Class II completion first with the weaker finite sufficient condition:

the actual microscopic transport space V_{micro} contains one longitudinal seed and one transverse seed.

Inside the canonical Class II seed family,

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J,$$

so explicit seed directions already exist:

$$c_{\parallel} \neq 0 \Rightarrow g_c \neq 0, \quad c_I \neq 0 \Rightarrow g_E \neq 0, \quad c_J \neq 0 \Rightarrow g_O \neq 0.$$

Therefore it is sufficient to verify in the complete Hessian that

$$P_{V_{\text{micro}}} T^{(\parallel)} \neq 0, \quad P_{V_{\text{micro}}} T^{(\perp)} \neq 0.$$

Once this finite seed-embedding check holds, one obtains

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0),$$

hence

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

So the main weakness is removed in the precise sense that the remaining microscopic task is no longer a full completion proof, but only a finite seed-embedding audit in the true complete TECT Hessian.

CXXXVII. FINITE SEED-EMBEDDING TEST FROM THE COMPLETE MICROSCOPIC HESSIAN

CXXXVIII. COMPLETE HESSIAN AND FIRST-ORDER TRANSPORT BLOCK

Let the full complete TECT linearization around the relevant defect background be

$$\mathbb{H}_{\text{comp}} = \mathbb{H}_B - i \mathbb{H}_1^{\mu}(\xi) \partial_{y^{\mu}} + O(\partial_y^2). \quad (296)$$

The shell-relevant first-order transport kernel is therefore

$$L_1 = -i V^{\mu}(\xi) \partial_{y^{\mu}}, \quad V^{\mu}(\xi) := \mathbb{H}_1^{\mu}(\xi) \big|_{Q_B\text{-tangent sector}}. \quad (297)$$

CXXXIX. LOW-MODE PROJECTION

Choose the opposite-valley low-mode basis

$$\{u_a^+(\xi), u_a^-(\xi)\}_{a=1,2}$$

on the tangent bundle $Q_B(\xi)$. The reduced 2×2 transport matrices are

$$(M^{\mu})_{ab} = \langle u_a^+, V^{\mu} u_b^- \rangle. \quad (298)$$

These are the only objects needed to read the actual coefficient point.

CXL. CANONICAL SEED BASIS IN THE AMBIENT TENSOR SPACE

Let

$$W = \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}$$

be the ambient enriched tensor space. Define the shell-seed basis

$$\boxed{B_{\parallel}, \quad B_I, \quad B_J,} \quad (299)$$

corresponding to

$$B_{\parallel} \leftrightarrow \hat{n}_i \delta^{a3}, \quad B_I \leftrightarrow e_{1i} \delta^{a1} + e_{2i} \delta^{a2}, \quad B_J \leftrightarrow -e_{1i} \delta^{a2} + e_{2i} \delta^{a1}.$$

Then any shell-relevant microscopic transport tensor can be written as

$$\boxed{N_{\text{mic}} = c_{\parallel} B_{\parallel} + c_I B_I + c_J B_J + N_{\perp}^{\text{null}},} \quad (300)$$

where $N_{\perp}^{\text{null}}$ lies in the kernel of the shell witness map.

CXLI. PROJECTION FORMULAS FOR THE COEFFICIENTS

Introduce the canonical witness covectors

$$T^{(\parallel)}, \quad T^{(I)}, \quad T^{(J)}.$$

By orthogonality in the normalized shell frame,

$$\boxed{\langle T^{(\parallel)}, B_I \rangle = \langle T^{(\parallel)}, B_J \rangle = 0,} \quad (301)$$

$$\boxed{\langle T^{(I)}, B_{\parallel} \rangle = \langle T^{(I)}, B_J \rangle = 0,} \quad (302)$$

$$\boxed{\langle T^{(J)}, B_{\parallel} \rangle = \langle T^{(J)}, B_I \rangle = 0,} \quad (303)$$

while

$$\boxed{\mu_{\parallel} := \langle T^{(\parallel)}, B_{\parallel} \rangle \neq 0, \quad \mu_I := \langle T^{(I)}, B_I \rangle \neq 0, \quad \mu_J := \langle T^{(J)}, B_J \rangle \neq 0.} \quad (304)$$

Therefore the coefficients are extracted directly by

$$\boxed{c_{\parallel} = \frac{\langle T^{(\parallel)}, N_{\text{mic}} \rangle}{\mu_{\parallel}}, \quad c_I = \frac{\langle T^{(I)}, N_{\text{mic}} \rangle}{\mu_I}, \quad c_J = \frac{\langle T^{(J)}, N_{\text{mic}} \rangle}{\mu_J}.} \quad (305)$$

CXLII. EQUIVALENT MATRIX FORMULAS FROM M^{μ}

Equivalently, define the reduced coefficient functionals by

$$\boxed{\lambda_{\parallel} = \frac{1}{2} [\sigma_3 n_{\mu} M^{\mu}],} \quad (306)$$

$$\boxed{\alpha = \frac{1}{2} [\sigma_1 e_{1\mu} M^{\mu} + \sigma_2 e_{2\mu} M^{\mu}],} \quad (307)$$

$$\beta = \frac{1}{2}[\sigma_2 e_{1\mu} M^\mu - \sigma_1 e_{2\mu} M^\mu]. \quad (308)$$

In the canonical seed normalization,

$$\lambda_{\parallel} = \mu_{\parallel} c_{\parallel}, \quad \alpha = \mu_I c_I, \quad \beta = \mu_J c_J. \quad (309)$$

Thus

$$c_{\parallel} \neq 0 \iff \lambda_{\parallel} \neq 0, \quad (310)$$

and

$$(c_I, c_J) \neq (0, 0) \iff (\alpha, \beta) \neq (0, 0). \quad (311)$$

CXLIII. FINITE EMBEDDING TEST

The entire complete microscopic audit is therefore reduced to the following finite projection test:

$$P_{V_{\text{seed}}} N_{\text{mic}} \neq 0. \quad (312)$$

More explicitly, the actual carrier survives iff

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (313)$$

Equivalently, in witness form,

$$\langle T^{(\parallel)}, N_{\text{mic}} \rangle \neq 0, \quad (\langle T^{(I)}, N_{\text{mic}} \rangle, \langle T^{(J)}, N_{\text{mic}} \rangle) \neq (0, 0). \quad (314)$$

CXLIV. ALGORITHMIC FORM

Thus the complete microscopic seed-embedding check is:

1. Compute the complete Hessian $\mathbb{H}_{\text{comp}}$ and extract $\mathbb{H}_1^\mu$.
2. Restrict $\mathbb{H}_1^\mu$ to the tangent low-mode sector to get $V^\mu(\xi)$.
3. Compute

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

4. Evaluate (306)–(308).

5. Read

$$c_{\parallel} = \lambda_{\parallel} / \mu_{\parallel}, \quad c_I = \alpha / \mu_I, \quad c_J = \beta / \mu_J.$$

6. Conclude that the complete microscopic transport space contains the required seed directions iff

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0).$$

CXLV. SHARP FINAL FORM

Therefore the remaining open microscopic problem is no longer vague. It is exactly the finite question:

$$\boxed{\text{Does the complete Hessian produce a shell-projected transport tensor } N_{\text{mic}} \text{ with } c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0) ?} \quad (315)$$

If yes, then the reduced PDE is automatically Dirac-nondegenerate.

CXLVI. FINAL SUMMARY

The complete microscopic seed-embedding problem reduces to a finite projection formula.
From the full complete TECT Hessian

$$\mathbb{H}_{\text{comp}} = \mathbb{H}_B - i\mathbb{H}_1^\mu(\xi)\partial_{y^\mu} + O(\partial_y^2),$$

extract the shell transport kernel

$$V^\mu(\xi) = \mathbb{H}_1^\mu(\xi)|_{Q_B\text{-tangent sector}}.$$

Project to the opposite-valley low modes:

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

Then define

$$\lambda_\parallel = \frac{1}{2}[\sigma_3 n_\mu M^\mu], \quad \alpha = \frac{1}{2}[\sigma_1 e_{1\mu} M^\mu + \sigma_2 e_{2\mu} M^\mu],$$

$$\beta = \frac{1}{2}[\sigma_2 e_{1\mu} M^\mu - \sigma_1 e_{2\mu} M^\mu].$$

In the canonical seed normalization,

$$\lambda_\parallel = \mu_\parallel c_\parallel, \quad \alpha = \mu_I c_I, \quad \beta = \mu_J c_J,$$

with

$$\mu_\parallel, \mu_I, \mu_J \neq 0.$$

Therefore the complete microscopic carrier survives iff

$$\boxed{c_\parallel \neq 0, \quad (c_I, c_J) \neq (0, 0).}$$

Equivalently, the remaining microscopic question is exactly:

$$\boxed{\text{does the complete Hessian produce a shell-projected tensor with nonzero longitudinal seed and nonzero transverse seed?}}$$

If yes, then the reduced PDE is automatically Dirac-nondegenerate.

CXLVII. ONE-SHOT COEFFICIENT EXTRACTION FROM THE COMPLETE HESSIAN

CXLVIII. COMPLETE FIRST-ORDER OPERATOR

Start from the complete microscopic first-order operator

$$\boxed{\mathbb{H}_{\text{comp}} = \mathbb{H}_B - i\mathbb{H}_1^\mu(\xi)\partial_{y^\mu} + O(\partial_y^2).} \tag{316}$$

Restrict to the tangent defect sector and define

$$\boxed{V^\mu(\xi) := \mathbb{H}_1^\mu(\xi)|_{Q_B\text{-tangent}}.} \tag{317}$$

Choose the opposite-valley low-mode basis

$$\{u_1^+, u_2^+, u_1^-, u_2^-\},$$

and define the reduced transport matrices

$$\boxed{(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle, \quad a, b = 1, 2.} \tag{318}$$

CXLIX. PAULI DECOMPOSITION OF THE REDUCED TRANSPORT

For each worldvolume direction μ , decompose

$$M^\mu = m_0^\mu \mathbf{1}_2 + m_1^\mu \sigma_1 + m_2^\mu \sigma_2 + m_3^\mu \sigma_3, \quad (319)$$

with

$$m_A^\mu = \frac{1}{2}(\sigma_A M^\mu), \quad A = 0, 1, 2, 3, \quad \sigma_0 := \mathbf{1}_2. \quad (320)$$

The scalar component m_0^μ is irrelevant for the Dirac witness. Only $m_1^\mu, m_2^\mu, m_3^\mu$ matter.

CL. DIRECT EXTRACTION OF THE THREE SEED COORDINATES

Let

$$n_\mu, \quad e_{1\mu}, \quad e_{2\mu}$$

be the shell-adapted orthonormal frame. Define the three reduced coefficients by

$$\lambda_\parallel = \frac{1}{2}(\sigma_3 n_\mu M^\mu) = n_\mu m_3^\mu, \quad (321)$$

$$\alpha = \frac{1}{2}(\sigma_1 e_{1\mu} M^\mu + \sigma_2 e_{2\mu} M^\mu) = e_{1\mu} m_1^\mu + e_{2\mu} m_2^\mu, \quad (322)$$

$$\beta = \frac{1}{2}(\sigma_2 e_{1\mu} M^\mu - \sigma_1 e_{2\mu} M^\mu) = e_{1\mu} m_2^\mu - e_{2\mu} m_1^\mu. \quad (323)$$

These are the exact one-shot coefficient functionals.

CLI. CANONICAL SEED NORMALIZATION

Let the shell-seed basis be

$$B_\parallel, \quad B_I, \quad B_J,$$

with canonical witness overlaps

$$\mu_\parallel = \langle T^{(\parallel)}, B_\parallel \rangle \neq 0, \quad \mu_I = \langle T^{(I)}, B_I \rangle \neq 0, \quad \mu_J = \langle T^{(J)}, B_J \rangle \neq 0.$$

Then the seed coordinates are

$$c_\parallel = \frac{\lambda_\parallel}{\mu_\parallel}, \quad c_I = \frac{\alpha}{\mu_I}, \quad c_J = \frac{\beta}{\mu_J}. \quad (324)$$

Therefore the final microscopic embedding test is exactly

$$c_\parallel \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (325)$$

CLII. EQUIVALENT BRANCH-SENSITIVE INTERPRETATION

The longitudinal branch is determined by

$$\Lambda_\parallel^{(0)} = \frac{1}{2}[\sigma_3 n_\mu \mathcal{M}^\mu(0)]. \quad (326)$$

Hence:

a. *Branch A.* If

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

then

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + O(\theta_0^3 \kappa).$$

b. *Branch B.* If

$$\Lambda_{\parallel}^{(0)} = 0,$$

then one computes the primitive coefficient

$$\lambda_c$$

from the projection onto the seed

$$\theta'(\xi) S_{\xi} n^{\mu}, \quad S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}.$$

Then

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

CLIII. SOLVED LOWEST-ORDER BRANCH B REALIZATION

For the already solved minimal complete lowest-order realization,

$$V^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \sigma_3 + \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^{\mu}_a \sigma_a + \tilde{\Lambda}_O J^{\mu}_a \sigma_a \right]. \quad (327)$$

Projecting with (318) gives

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta, \cos} \tilde{\Lambda}_E, \quad \beta = I_{\theta, \cos} \tilde{\Lambda}_O, \quad (328)$$

where

$$I_{\theta'^3} := \int d\xi f_+ f_- \theta'(\xi)^3, \quad I_{\theta, \cos} := \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi). \quad (329)$$

Hence for the solved lowest-order realization,

$$c_{\parallel} = \frac{\lambda_c}{2\mu_{\parallel}} I_{\theta'^3}, \quad c_I = \frac{I_{\theta, \cos} \tilde{\Lambda}_E}{\mu_I}, \quad c_J = \frac{I_{\theta, \cos} \tilde{\Lambda}_O}{\mu_J}. \quad (330)$$

Thus the solved lowest-order realization is Dirac-nondegenerate iff

$$\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0), \quad I_{\theta'^3} \neq 0, \quad I_{\theta, \cos} \neq 0. \quad (331)$$

CLIV. FINAL MICROSCOPIC DELIVERABLE

Therefore the actual remaining microscopic task is completely explicit:

$$\text{compute } M_{ab}^{\mu} = \langle u_a^+, V^{\mu} u_b^- \rangle \text{ from the complete Hessian, then read } (\lambda_{\parallel}, \alpha, \beta) \text{ from (321)–(323).} \quad (332)$$

No further structural reduction is needed.

CLV. FINAL SUMMARY

The complete microscopic seed-embedding audit is now a direct projection formula.
From the full complete Hessian

$$\mathbb{H}_{\text{comp}} = \mathbb{H}_B - i\mathbb{H}_1^\mu(\xi)\partial_{y^\mu} + O(\partial_y^2),$$

restrict to the tangent sector:

$$V^\mu(\xi) = \mathbb{H}_1^\mu(\xi)|_{Q_B\text{-tangent}}.$$

Project to the opposite-valley low modes:

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

Then extract

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 n_\mu M^\mu),$$

$$\alpha = \frac{1}{2}(\sigma_1 e_{1\mu} M^\mu + \sigma_2 e_{2\mu} M^\mu),$$

$$\beta = \frac{1}{2}(\sigma_2 e_{1\mu} M^\mu - \sigma_1 e_{2\mu} M^\mu).$$

In canonical seed normalization,

$$c_{\parallel} = \lambda_{\parallel}/\mu_{\parallel}, \quad c_I = \alpha/\mu_I, \quad c_J = \beta/\mu_J.$$

Therefore the final microscopic embedding test is exactly

$$\boxed{c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0).}$$

For the already solved lowest-order Branch B realization,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta, \cos} \tilde{\Lambda}_E, \quad \beta = I_{\theta, \cos} \tilde{\Lambda}_O.$$

So the remaining microscopic deliverable is completely explicit: compute M^μ from the complete Hessian and read the three coefficients directly.

CLVI. DIRECT COMPUTATION OF M^μ FROM A FULL TANGENT-COMPATIBLE HESSIAN ANSATZ

CLVII. GOAL

We now write the most general lowest-order tangent-compatible first-order Hessian block for the complete TECT action and project it directly to the low opposite-valley sector.

The purpose is to obtain the coefficient extraction

$$(\lambda_{\parallel}, \alpha, \beta)$$

in one shot from the complete microscopic Hessian.

CLVIII. TANGENT BUNDLE BASIS

On the defect background, take the tangent bundle basis

$$Q_B(\xi) = \text{span}\{e_1(\xi), e_2\},$$

with

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Define the tangent Pauli operators by

$$\Sigma_1 = |e_1\rangle\langle e_2| + |e_2\rangle\langle e_1|, \quad \Sigma_2 = -i|e_1\rangle\langle e_2| + i|e_2\rangle\langle e_1|, \quad \Sigma_3 = |e_1\rangle\langle e_1| - |e_2\rangle\langle e_2|. \quad (333)$$

Also let

$$\Pi_Q = |e_1\rangle\langle e_1| + |e_2\rangle\langle e_2|.$$

CLIX. MOST GENERAL SHELL-COMPATIBLE FIRST-ORDER TANGENT BLOCK

The complete first-order microscopic operator restricted to the tangent sector must have the form

$$V^\mu(\xi) = a_0(\xi) n^\mu \Pi_Q + a_3(\xi) n^\mu \Sigma_3 + b_E(\xi) E^\mu_a \Sigma_a + b_O(\xi) J^\mu_a \Sigma_a + V_{\text{null}}^\mu(\xi), \quad (334)$$

where:

- $a_0(\xi)$ is scalar and irrelevant to the reduced Dirac witness;
- $a_3(\xi)$ is the longitudinal traceless coefficient;
- $b_E(\xi)$ is the transverse-even coefficient;
- $b_O(\xi)$ is the transverse-odd coefficient;
- V_{null}^μ is any tangent operator orthogonal to the shell witness map.

Thus all load-bearing information is contained in

$$a_3(\xi), \quad b_E(\xi), \quad b_O(\xi).$$

CLX. BRANCH-SENSITIVE PARAMETRIZATION

To make the longitudinal branch structure explicit, write

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi), \quad (335)$$

and

$$b_E(\xi) = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi) + b_E^{\text{higher}}(\xi), \quad (336)$$

$$b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi) + b_O^{\text{higher}}(\xi). \quad (337)$$

This separates:

- Branch A candidate: $\Lambda_{\parallel}^{(0)} \theta'(\xi) n^\mu \Sigma_3$;
- Branch B primitive term: $\frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3$;
- explicit transverse completion: $\theta'(\xi) \cos \theta(\xi) [\tilde{\Lambda}_E E^\mu_a \Sigma_a + \tilde{\Lambda}_O J^\mu_a \Sigma_a]$.

CLXI. LOW OPPOSITE-VALLEY BASIS

Choose the low-mode basis

$$\boxed{u_1^\pm(\xi) = f_\pm(\xi)e_1(\xi), \quad u_2^\pm(\xi) = f_\pm(\xi)e_2.} \quad (338)$$

Assume the standard normalization

$$\int d\xi |f_\pm(\xi)|^2 = 1, \quad \langle u_a^\pm, u_b^\pm \rangle = \delta_{ab}, \quad \langle u_a^+, u_b^- \rangle = 0.$$

Define the overlap integrals

$$\boxed{I_0 := \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi),} \quad (339)$$

$$\boxed{I_3 := \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi)^3,} \quad (340)$$

$$\boxed{I_c := \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi)\cos\theta(\xi).} \quad (341)$$

CLXII. MATRIX-ELEMENT COMPUTATION

The reduced transport matrix is

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

Using

$$\langle e_a, \Pi_Q e_b \rangle = \delta_{ab}, \quad \langle e_a, \Sigma_i e_b \rangle = (\sigma_i)_{ab},$$

one gets immediately:

$$\begin{aligned} M^\mu = n^\mu & \left[\left(\int d\xi f_+ f_- a_0(\xi) \right) \mathbf{1}_2 + \left(\int d\xi f_+ f_- a_3(\xi) \right) \sigma_3 \right] \\ & + \left(\int d\xi f_+ f_- b_E(\xi) \right) E^\mu{}_a \sigma_a + \left(\int d\xi f_+ f_- b_O(\xi) \right) J^\mu{}_a \sigma_a + M_{\text{null}}^\mu. \end{aligned} \quad (342)$$

Therefore the load-bearing reduced matrix is

$$\boxed{M_{\text{red}}^\mu = n^\mu \lambda_{\parallel} \sigma_3 + \alpha E^\mu{}_a \sigma_a + \beta J^\mu{}_a \sigma_a,} \quad (343)$$

with

$$\boxed{\lambda_{\parallel} = \int d\xi f_+ f_- a_3(\xi),} \quad (344)$$

$$\boxed{\alpha = \int d\xi f_+ f_- b_E(\xi), \quad \beta = \int d\xi f_+ f_- b_O(\xi).} \quad (345)$$

Substituting (335)–(337) gives

$$\boxed{\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_0 + \frac{\lambda_c}{2} I_3 + \delta \lambda_{\text{higher}},} \quad (346)$$

$$\boxed{\alpha = \tilde{\Lambda}_E I_c + \delta \alpha_{\text{higher}}, \quad \beta = \tilde{\Lambda}_O I_c + \delta \beta_{\text{higher}}.} \quad (347)$$

CLXIII. LOWEST-ORDER SOLVED REALIZATION

For the already solved minimal complete lowest-order realization, one has

$$\Lambda_{\parallel}^{(0)} = 0, \quad a_3^{\text{higher}} = 0, \quad b_E^{\text{higher}} = 0, \quad b_O^{\text{higher}} = 0.$$

Hence

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \tilde{\Lambda}_E I_c, \quad \beta = \tilde{\Lambda}_O I_c.} \quad (348)$$

Thus

$$\boxed{M_{\text{red}}^{\mu} = n^{\mu} \frac{\lambda_c}{2} I_3 \sigma_3 + I_c \tilde{\Lambda}_E E^{\mu}{}_a \sigma_a + I_c \tilde{\Lambda}_O J^{\mu}{}_a \sigma_a.} \quad (349)$$

CLXIV. ONE-SHOT WITNESS EXTRACTION

Now apply the coefficient functionals:

$$\boxed{\lambda_{\parallel} = \frac{1}{2}(\sigma_3 n_{\mu} M^{\mu}),} \quad (350)$$

$$\boxed{\alpha = \frac{1}{2}(\sigma_1 e_{1\mu} M^{\mu} + \sigma_2 e_{2\mu} M^{\mu}),} \quad (351)$$

$$\boxed{\beta = \frac{1}{2}(\sigma_2 e_{1\mu} M^{\mu} - \sigma_1 e_{2\mu} M^{\mu}).} \quad (352)$$

These give exactly (346)–(347).

CLXV. FINAL REDUCED PDE

Therefore the reduced first-order PDE is

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \sigma_3 (-i\partial_{\parallel}) - \alpha (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - \beta (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.} \quad (353)$$

In the solved minimal lowest-order Branch B realization,

$$\boxed{\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.} \quad (354)$$

CLXVI. NONDEGENERACY

The full lowest-order Dirac nondegeneracy condition is

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (355)$$

For the solved lowest-order realization, this is

$$\boxed{\lambda_c \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0), \quad I_3 \neq 0, \quad I_c \neq 0.} \quad (356)$$

CLXVII. FINAL SUMMARY

Given the complete first-order tangent Hessian block

$$V^\mu(\xi) = a_0(\xi)n^\mu\Pi_Q + a_3(\xi)n^\mu\Sigma_3 + b_E(\xi)E^\mu{}_a\Sigma_a + b_O(\xi)J^\mu{}_a\Sigma_a + \dots,$$

the low-mode projection

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle$$

gives

$$M_{\text{red}}^\mu = n^\mu\lambda_\parallel\sigma_3 + \alpha E^\mu{}_a\sigma_a + \beta J^\mu{}_a\sigma_a,$$

with

$$\lambda_\parallel = \int d\xi f_+ f_- a_3(\xi), \quad \alpha = \int d\xi f_+ f_- b_E(\xi), \quad \beta = \int d\xi f_+ f_- b_O(\xi).$$

Using

$$a_3(\xi) = \Lambda_\parallel^{(0)}\theta'(\xi) + \frac{\lambda_c}{2}\theta'(\xi)^3 + \dots,$$

$$b_E(\xi) = \tilde{\Lambda}_E\theta'(\xi)\cos\theta(\xi) + \dots, \quad b_O(\xi) = \tilde{\Lambda}_O\theta'(\xi)\cos\theta(\xi) + \dots,$$

one gets

$$\lambda_\parallel = \Lambda_\parallel^{(0)}I_0 + \frac{\lambda_c}{2}I_3 + \dots, \quad \alpha = \tilde{\Lambda}_EI_c + \dots, \quad \beta = \tilde{\Lambda}_OI_c + \dots.$$

For the solved minimal lowest-order realization,

$$\Lambda_\parallel^{(0)} = 0,$$

so

$$\lambda_\parallel = \frac{\lambda_c}{2}I_3, \quad \alpha = \tilde{\Lambda}_EI_c, \quad \beta = \tilde{\Lambda}_OI_c.$$

Thus the solved reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2}I_3\sigma_3(-i\partial_\parallel) - I_c\tilde{\Lambda}_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - I_c\tilde{\Lambda}_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

The only remaining microscopic task is to compute a_3, b_E, b_O , equivalently M^μ , from the true full complete TECT Hessian.

CLXVIII. EXACT MICROSCOPIC EXTRACTION OF $a_3(\xi), b_E(\xi), b_O(\xi)$ FROM THE FULL COMPLETE TECT HESSIAN

CLXIX. COMPLETE ACTION AND HESSIAN EXPANSION

Let the full complete microscopic TECT action be

$$S_{\text{comp}}[\Phi], \quad \Phi = (\Psi, P, \sigma, X). \quad (357)$$

Let $\Phi_B(\xi)$ be the relevant static defect background. Linearize around it:

$$\Phi = \Phi_B + \delta\Phi.$$

The quadratic fluctuation action is

$$\delta^2 S_{\text{comp}} = \frac{1}{2} \int dt d^3x \delta\Phi^\dagger \mathbb{H}_{\text{comp}} \delta\Phi. \quad (358)$$

Split coordinates into the defect normal coordinate ξ and worldvolume coordinates y^μ . Then the complete Hessian admits the first-order expansion

$$\mathbb{H}_{\text{comp}} = \mathbb{H}_B(\xi) - i\mathbb{H}_1^\mu(\xi)\partial_{y^\mu} + O(\partial_y^2). \quad (359)$$

CLXX. TANGENT-SECTOR REDUCTION

Let

$$Q_B(\xi) = \text{span}\{e_1(\xi), e_2(\xi)\}$$

be the complex rank-two tangent bundle of the defect background, with projector

$$\Pi_Q(\xi) = |e_1(\xi)\rangle\langle e_1(\xi)| + |e_2(\xi)\rangle\langle e_2(\xi)|. \quad (360)$$

Define the tangent Pauli basis

$$\Sigma_1 = |e_1\rangle\langle e_2| + |e_2\rangle\langle e_1|, \quad \Sigma_2 = -i|e_1\rangle\langle e_2| + i|e_2\rangle\langle e_1|, \quad \Sigma_3 = |e_1\rangle\langle e_1| - |e_2\rangle\langle e_2|. \quad (361)$$

Restrict the first-order Hessian to the tangent sector:

$$V^\mu(\xi) := \Pi_Q(\xi) \mathbb{H}_1^\mu(\xi) \Pi_Q(\xi). \quad (362)$$

This is the exact microscopic transport operator relevant for the reduced Dirac problem.

CLXXI. GENERAL TANGENT DECOMPOSITION

Since $V^\mu(\xi)$ acts on a two-dimensional complex tangent space, it has the unique decomposition

$$V^\mu(\xi) = v_0^\mu(\xi) \mathbf{1}_2 + v_1^\mu(\xi) \Sigma_1 + v_2^\mu(\xi) \Sigma_2 + v_3^\mu(\xi) \Sigma_3. \quad (363)$$

The scalar component v_0^μ is irrelevant for the reduced Dirac witness. Thus only $v_1^\mu, v_2^\mu, v_3^\mu$ matter. Introduce the shell-adapted worldvolume frame

$$n^\mu, \quad E^\mu{}_a, \quad J^\mu{}_a,$$

where n^μ is the longitudinal direction and $E^\mu{}_a, J^\mu{}_a$ are the transverse even/odd tensors.

Then the load-bearing decomposition is

$$V^\mu(\xi) = a_0(\xi) n^\mu \mathbf{1}_2 + a_3(\xi) n^\mu \Sigma_3 + b_E(\xi) E^\mu{}_a \Sigma_a + b_O(\xi) J^\mu{}_a \Sigma_a + V_{\text{null}}^\mu(\xi), \quad (364)$$

where V_{null}^μ is orthogonal to the shell witness map.

CLXXII. DIRECT MICROSCOPIC PROJECTION FORMULAS

The coefficients a_3, b_E, b_O are obtained by orthogonal projection onto the corresponding tensor directions.

A. Longitudinal coefficient

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V^\mu(\xi)). \quad (365)$$

B. Transverse-even coefficient

$$b_E(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a E^\mu{}_a V^\mu(\xi)). \quad (366)$$

C. Transverse-odd coefficient

$$b_O(\xi) = \frac{1}{2} \frac{1}{Q_B} (\Sigma_a J_\mu^a V^\mu(\xi)). \quad (367)$$

These are the exact one-shot microscopic extraction formulas from the complete Hessian.

CLXXIII. BRANCH-SENSITIVE LONGITUDINAL SPLIT

To decide the longitudinal branch, expand $a_3(\xi)$ in the defect-gradient hierarchy:

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + \delta a_3^{(\geq 5)}(\xi). \quad (368)$$

Then:

a. Branch A. If

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

the constant-leading longitudinal term is present.

b. Branch B. If

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,$$

the longitudinal sector survives only through the primitive θ'^3 channel.

CLXXIV. LOW-MODE PROJECTION

Choose opposite-valley low modes

$$u_1^\pm(\xi) = f_\pm(\xi) e_1(\xi), \quad u_2^\pm(\xi) = f_\pm(\xi) e_2(\xi). \quad (369)$$

Define the reduced transport matrices

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle. \quad (370)$$

Using (364), the reduced matrix is

$$M^\mu = n^\mu \lambda_{\parallel} \sigma_3 + \alpha E^\mu{}_a \sigma_a + \beta J^\mu{}_a \sigma_a, \quad (371)$$

with

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) a_3(\xi), \quad (372)$$

$$\alpha = \int d\xi f_+(\xi) f_-(\xi) b_E(\xi), \quad \beta = \int d\xi f_+(\xi) f_-(\xi) b_O(\xi). \quad (373)$$

CLXXV. LOWEST-ORDER SOLVED REALIZATION AS A SPECIAL CASE

For the already solved minimal complete lowest-order realization,

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3, \quad b_E(\xi) = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi).$$

Hence

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \tilde{\Lambda}_E I_c, \quad \beta = \tilde{\Lambda}_O I_c,} \quad (374)$$

where

$$\boxed{I_3 := \int d\xi f_+ f_- \theta'(\xi)^3, \quad I_c := \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi).} \quad (375)$$

CLXXVI. FINAL REDUCED PDE

Therefore the reduced first-order PDE is

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - \beta(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.} \quad (376)$$

In the solved lowest-order Branch B realization this becomes

$$\boxed{\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.} \quad (377)$$

CLXXVII. FINAL MICROSCOPIC DELIVERABLE

Thus the true full complete TECT microscopic problem is reduced to computing directly from $\mathbb{H}_1^{\mu}(\xi)$:

$$\boxed{a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_{\mu} \Pi_Q \mathbb{H}_1^{\mu} \Pi_Q),} \quad (378)$$

$$\boxed{b_E(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a E_{\mu}^a \Pi_Q \mathbb{H}_1^{\mu} \Pi_Q),} \quad (379)$$

$$\boxed{b_O(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a J_{\mu}^a \Pi_Q \mathbb{H}_1^{\mu} \Pi_Q).} \quad (380)$$

Once these are known, the coefficient point $(\lambda_{\parallel}, \alpha, \beta)$ and the branch choice are fixed automatically.

CLXXVIII. FINAL SUMMARY

The full complete TECT Hessian determines everything through the tangent first-order block

$$V^{\mu}(\xi) = \Pi_Q \mathbb{H}_1^{\mu}(\xi) \Pi_Q.$$

Its load-bearing coefficients are extracted directly by

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_{\mu} V^{\mu}),$$

$$b_E(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a E_\mu^a V^\mu), \quad b_O(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a J_\mu^a V^\mu).$$

Then the reduced coefficients are

$$\lambda_{\parallel} = \int d\xi f_+ f_- a_3(\xi), \quad \alpha = \int d\xi f_+ f_- b_E(\xi), \quad \beta = \int d\xi f_+ f_- b_O(\xi).$$

So the complete microscopic closure problem is no longer conceptual: it is exactly the computation of a_3, b_E, b_O from the true full complete Hessian.

For the already solved minimal lowest-order Branch B realization,

$$a_3 = \frac{\lambda_c}{2} \theta'^3, \quad b_E = \tilde{\Lambda}_E \theta' \cos \theta, \quad b_O = \tilde{\Lambda}_O \theta' \cos \theta,$$

hence

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \tilde{\Lambda}_E I_c, \quad \beta = \tilde{\Lambda}_O I_c.$$

Thus the solved reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.$$

CLXXIX. TERM-BY-TERM EXTRACTION OF a_3, b_E, b_O FROM THE COMPLETE ACTION

CLXXX. COMPLETE ACTION SPLIT

Write the complete action as

$$S_{\text{comp}} = S_{\Psi} + S_P + S_{\sigma} + S_X + S_{\text{mix}}, \quad (381)$$

with

$$S_{\text{mix}} = S_J + S_K + S_{EO}. \quad (382)$$

Around the defect background $\Phi_B = (\Psi_B, P_B, \sigma_B, X_B)$, the first-order worldvolume Hessian coefficient is additive:

$$\mathbb{H}_1^\mu = \mathbb{H}_{1,\Psi}^\mu + \mathbb{H}_{1,P}^\mu + \mathbb{H}_{1,\sigma}^\mu + \mathbb{H}_{1,X}^\mu + \mathbb{H}_{1,J}^\mu + \mathbb{H}_{1,K}^\mu + \mathbb{H}_{1,EO}^\mu. \quad (383)$$

Hence the tangent transport block is

$$V^\mu = \Pi_Q \mathbb{H}_1^\mu \Pi_Q = V_\Psi^\mu + V_P^\mu + V_\sigma^\mu + V_X^\mu + V_J^\mu + V_K^\mu + V_{EO}^\mu. \quad (384)$$

Therefore

$$a_3 = a_3^{(\Psi)} + a_3^{(P)} + a_3^{(\sigma)} + a_3^{(X)} + a_3^{(J)} + a_3^{(K)} + a_3^{(EO)}, \quad (385)$$

$$b_E = b_E^{(\Psi)} + b_E^{(P)} + b_E^{(\sigma)} + b_E^{(X)} + b_E^{(J)} + b_E^{(K)} + b_E^{(EO)}, \quad (386)$$

$$b_O = b_O^{(\Psi)} + b_O^{(P)} + b_O^{(\sigma)} + b_O^{(X)} + b_O^{(J)} + b_O^{(K)} + b_O^{(EO)}. \quad (387)$$

CLXXXI. DIRECT PROJECTION FORMULAS

For each sector $S \in \{\Psi, P, \sigma, X, J, K, EO\}$,

$$a_3^{(S)}(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V_S^\mu(\xi)), \quad (388)$$

$$b_E^{(S)}(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a E_\mu^a V_S^\mu(\xi)), \quad (389)$$

$$b_O^{(S)}(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a J_\mu^a V_S^\mu(\xi)). \quad (390)$$

Thus the problem is reduced to computing V_S^μ sector by sector.

CLXXXII. SECTORS THAT CANNOT CONTRIBUTE AT LOWEST ORDER

At lowest order:

a. Pure scalar core. A purely scalar shell-forming term carries no internal adjoint/tangent Pauli structure. Hence

$$b_E^{(\Psi, \text{scalar})} = b_O^{(\Psi, \text{scalar})} = 0, \quad a_3^{(\Psi, \text{scalar})} = 0. \quad (391)$$

b. Pure volumetric sector. The heavy scalar σ has no tangent Pauli structure by itself, so

$$a_3^{(\sigma)} = b_E^{(\sigma)} = b_O^{(\sigma)} = 0 \quad (392)$$

before mixing is included.

c. Bare heavy X -mass sector. The term $\frac{1}{2} M_X^2 X_i^a X_i^a$ carries no worldvolume derivative and therefore

$$a_3^{(X)} = b_E^{(X)} = b_O^{(X)} = 0. \quad (393)$$

Thus the first nontrivial lowest-order contributions must come from the refined/mixing sectors.

CLXXXIII. GENERAL FORMULA FROM A MIXING TERM

Let $S_{\text{mix}}^{(r)}$ be any mixing term. Its quadratic Hessian contribution on the tangent sector has the generic form

$$\delta^2 S_{\text{mix}}^{(r)} = \int dt d\xi d^2 y \eta^\dagger \left[\mathcal{A}^{(r)}(\xi) - i \mathcal{B}^{(r)\mu}(\xi) \partial_{y^\mu} \right] \eta + \dots \quad (394)$$

with $\eta \in Q_B$. Then

$$V_r^\mu(\xi) = \mathcal{B}^{(r)\mu}(\xi). \quad (395)$$

Therefore

$$a_3^{(r)} = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu \mathcal{B}^{(r)\mu}), \quad b_E^{(r)} = \frac{1}{2_{Q_B}} (\Sigma_a E_\mu^a \mathcal{B}^{(r)\mu}), \quad b_O^{(r)} = \frac{1}{2_{Q_B}} (\Sigma_a J_\mu^a \mathcal{B}^{(r)\mu}). \quad (396)$$

CLXXXIV. THE J -SECTOR CONTRIBUTION

For

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \Psi^\dagger T_a (\partial_i \Psi),$$

the linearized defect-background contribution is

$$J_i^a[\Psi_B + \delta\Psi] = J_i^a[\Psi_B] + \delta J_i^a[\delta\Psi].$$

The worldvolume-derivative part of δJ_i^a is

$$\boxed{\delta J_i^a|_{\partial_y} = (\partial_i \Psi_B^\dagger) T_a \delta\Psi_{\partial_y} + \delta\Psi_{\partial_y}^\dagger T_a (\partial_i \Psi_B),} \quad (397)$$

where $\delta\Psi_{\partial_y}$ denotes the component linear in $\partial_{y^\mu}\eta$.

Hence the induced V_J^μ is obtained by differentiating the quadratic form with respect to $\partial_{y^\mu}\eta$, and then restricting to Q_B . Its contribution is

$$\boxed{V_J^\mu = \Pi_Q \frac{\partial(\delta^2 S_J)}{\partial(-i\partial_{y^\mu})} \Pi_Q.} \quad (398)$$

At lowest order, for the real defect representative and embedded $SU(2)_{(13)}$, this sector contributes only to the primitive longitudinal channel and does not produce transverse E/O by itself.

CLXXXV. THE K -SECTOR CONTRIBUTION

Take the minimal complete choice

$$K_i^a[P, \psi] = (T_a \partial_i P) (\Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j) D_j \psi.$$

On the defect background,

$$\partial_i P_B = \delta_{i\xi} \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger).$$

Therefore

$$\boxed{K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) (\Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j) D_j \psi.} \quad (399)$$

Thus the direct tangent-block contribution from the K -sector has the form

$$\boxed{V_K^\mu = \theta'(\xi) \left[\kappa_E(\xi) E^\mu{}_a \Sigma_a + \kappa_O(\xi) J^\mu{}_a \Sigma_a \right],} \quad (400)$$

with

$$\kappa_E(\xi) \propto k_a(\theta) \Lambda_E^{(0)}, \quad \kappa_O(\xi) \propto k_a(\theta) \Lambda_O^{(0)}.$$

Hence

$$\boxed{a_3^{(K)} = 0, \quad b_E^{(K)} = \kappa_E(\xi) \theta'(\xi), \quad b_O^{(K)} = \kappa_O(\xi) \theta'(\xi).} \quad (401)$$

CLXXXVI. THE EXTRA E/O SLOT CONTRIBUTION

For the explicit extra transverse slots,

$$\mathcal{I}_E \sim X^{(E)}(S_E M_E) E_j D_j \psi, \quad \mathcal{I}_O \sim X^{(O)}(S_O M_O) J_j D_j \psi,$$

with

$$(S_E M_E) = \cos \theta, \quad (S_O M_O) = \cos \theta,$$

the induced tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right]. \quad (402)$$

Therefore

$$a_3^{(EO)} = 0, \quad b_E^{(EO)} = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi). \quad (403)$$

CLXXXVII. THE PRIMITIVE LONGITUDINAL CONTRIBUTION

The primitive longitudinal channel comes from the commutator-square seed

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B], \quad S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.$$

For the explicit defect representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \Sigma_3.$$

Hence the primitive transport block is

$$V_{\parallel, \text{prim}}^\mu = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3. \quad (404)$$

Therefore

$$a_3^{(\text{prim})} = \frac{\lambda_c}{2} \theta'(\xi)^3, \quad b_E^{(\text{prim})} = 0, \quad b_O^{(\text{prim})} = 0. \quad (405)$$

CLXXXVIII. COMBINED LOWEST-ORDER COMPLETE RESULT

Collecting the nonzero lowest-order pieces:

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + \delta a_3^{\text{higher}}(\xi), \quad (406)$$

$$b_E(\xi) = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi) + \delta b_E^{\text{higher}}(\xi), \quad (407)$$

$$b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi) + \delta b_O^{\text{higher}}(\xi). \quad (408)$$

For the solved minimal complete lowest-order realization,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \delta a_3^{\text{higher}} = \delta b_E^{\text{higher}} = \delta b_O^{\text{higher}} = 0. \quad (409)$$

Thus

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3, \quad b_E(\xi) = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi). \quad (410)$$

CLXXXIX. LOW-MODE COEFFICIENT EXTRACTION

Therefore

$$\lambda_{\parallel} = \Lambda_{\parallel}^{(0)} I_0 + \frac{\lambda_c}{2} I_3 + \delta\lambda_{\text{higher}}, \quad (411)$$

$$\alpha = \tilde{\Lambda}_E I_c + \delta\alpha_{\text{higher}}, \quad \beta = \tilde{\Lambda}_O I_c + \delta\beta_{\text{higher}}, \quad (412)$$

with

$$I_0 = \int d\xi f_+ f_- \theta', \quad I_3 = \int d\xi f_+ f_- \theta'^3, \quad I_c = \int d\xi f_+ f_- \theta' \cos \theta.$$

For the solved lowest-order realization,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \tilde{\Lambda}_E I_c, \quad \beta = \tilde{\Lambda}_O I_c. \quad (413)$$

CXC. FINAL REDUCED PDE

Thus the reduced first-order PDE is

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - \beta(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (414)$$

For the solved lowest-order Branch B realization this becomes

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (415)$$

CXCI. FINAL SUMMARY

The full complete TECT Hessian contributes to the reduced transport only through the tangent first-order block

$$V^{\mu} = \Pi_Q \mathbb{H}_1^{\mu} \Pi_Q.$$

Its three load-bearing coefficients are extracted by

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_{\mu} V^{\mu}), \quad b_E(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a E_{\mu}^a V^{\mu}), \quad b_O(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a J_{\mu}^a V^{\mu}).$$

At lowest order,

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + \dots,$$

$$b_E(\xi) = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi) + \dots, \quad b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi) + \dots.$$

For the solved minimal complete lowest-order realization,

$$\Lambda_{\parallel}^{(0)} = 0, \quad a_3 = \frac{\lambda_c}{2} \theta'^3, \quad b_E = \tilde{\Lambda}_E \theta' \cos \theta, \quad b_O = \tilde{\Lambda}_O \theta' \cos \theta.$$

Hence

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \tilde{\Lambda}_E I_c, \quad \beta = \tilde{\Lambda}_O I_c,$$

and the solved reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

So the only remaining microscopic task is to compute the true full-action corrections

$$\Lambda_{\parallel}^{(0)}, \quad \delta a_3^{\text{higher}}, \quad \delta b_E^{\text{higher}}, \quad \delta b_O^{\text{higher}}.$$

**CXCII. A NO-GO THEOREM FOR BRANCH A AND THE NECESSARY MICROSCOPIC
CONDITION TO EVADE IT**

CXCIII. GOAL

We now sharpen the remaining microscopic problem.
The unresolved longitudinal ambiguity was

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + \delta a_3^{\text{higher}}(\xi).$$

The question is whether the full complete TECT action can generate a nonzero constant-leading coefficient

$$\Lambda_{\parallel}^{(0)} \neq 0.$$

We now prove a broad no-go statement: for a large class of real, local, tangent-isotropic complete actions built on the real CP^2 defect representative, the coefficient $\Lambda_{\parallel}^{(0)}$ vanishes identically. Therefore Branch B is forced unless the full action contains an additional microscopic operator of a specific type.

CXCIV. TANGENT GEOMETRIC DATA

On the explicit real defect representative,

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \partial_{\xi} z_B = \theta'(\xi) e_1(\xi),$$

with tangent basis

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Define

$$\Pi_Q = |e_1\rangle\langle e_1| + |e_2\rangle\langle e_2|, \quad \Sigma_3 = |e_1\rangle\langle e_1| - |e_2\rangle\langle e_2|.$$

The basic tangent geometric object is

$$\boxed{\mathcal{T}_{\xi} := i[P_B, \partial_{\xi} P_B].} \tag{416}$$

A direct computation gives

$$\boxed{\Pi_Q \mathcal{T}_{\xi} \Pi_Q = 0,} \tag{417}$$

while

$$\boxed{(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3.} \tag{418}$$

Thus the first nontrivial traceless longitudinal tangent tensor appears only at quadratic order in $\mathcal{T}_{\xi}$, i.e. at order θ'^2 .

CXCV. CONSEQUENCE FOR THE TANGENT FIRST-ORDER HESSIAN

Let the complete first-order tangent Hessian block be

$$V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q.$$

Its longitudinal traceless coefficient is

$$a_3(\xi) = \frac{1}{2} \frac{1}{Q_B} (\Sigma_3 n_\mu V^\mu(\xi)).$$

Suppose the full complete action is built from local real invariants of the fields

$$(\Psi, P, \sigma, X)$$

and their derivatives, subject to the following two conditions:

1. the action contains no microscopic tangent operator whose tangent reduction is linear in $\mathcal{T}_\xi$ and already proportional to Σ_3 ;
2. the action is tangent-isotropic in the sense that the two tangent directions are not explicitly distinguished before the defect background is inserted.

Then every longitudinal traceless tangent contribution must be generated from even powers of the fundamental off-diagonal tangent operator $\mathcal{T}_\xi$.

By (417) and (418), the first such contribution is necessarily

$$a_3(\xi) \propto \theta'(\xi)^3,$$

not $\theta'(\xi)$.

Therefore:

If the complete action contains no linear tangent Σ_3 -carrier, then $\Lambda_\parallel^{(0)} = 0$.

(419)

This proves the broad no-go theorem for Branch A.

CXCVI. PROOF IN OPERATOR FORM

Any longitudinal traceless contribution to V^μ must have the form

$$V_\parallel^\mu(\xi) = n^\mu \mathcal{V}_\parallel(\xi), \quad \mathcal{V}_\parallel(\xi) \in \text{span}\{\Sigma_3\}.$$

Under the assumptions above, $\mathcal{V}_\parallel$ can only be built from scalar functions of θ times tangent operators generated by $\mathcal{T}_\xi$. Since

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0,$$

the linear term vanishes. The first nonzero traceless tangent term is

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'^2 \Sigma_3.$$

To turn this into a worldvolume first-order transport kernel one must multiply by one further odd normal factor, namely the prefactor $\theta'(\xi)$ coming from the first-order defect transport structure

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

Hence the first nonzero longitudinal traceless term is necessarily

$$V_\parallel^\mu(\xi) \sim \theta'(\xi)^3 n^\mu \Sigma_3.$$

Therefore no $\theta'(\xi) n^\mu \Sigma_3$ term can occur, and

$$\Lambda_\parallel^{(0)} = 0.$$

CXCVII. NECESSARY CONDITION FOR BRANCH A

The no-go theorem shows that Branch A can appear only if the complete action contains an additional microscopic operator whose tangent reduction bypasses the vanishing

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0.$$

Equivalently, Branch A requires at least one new microscopic invariant whose contribution to the first-order tangent Hessian satisfies

$$\boxed{\frac{1}{2} \mathcal{O}_B (\Sigma_3 n_\mu V_{\text{new}}^\mu(\xi)) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + O(\theta'^3).} \quad (420)$$

Thus a necessary condition for Branch A is:

$$\boxed{\exists \mathcal{O}_{\text{new}} \text{ in the full complete action such that } \Pi_Q \delta^2 \mathcal{O}_{\text{new}} \Pi_Q \text{ contains a longitudinal traceless tangent term linear in } \theta'(\xi).} \quad (421)$$

Without such an operator, Branch A is impossible.

CXCVIII. SOLVED STATUS OF THE PRESENT MINIMAL COMPLETE LOWEST-ORDER REALIZATION

The present minimal complete lowest-order realization contains:

- the primitive commutator-square longitudinal channel;
- the explicit transverse even/odd completions;
- no independent linear longitudinal Σ_3 -carrier.

Therefore it falls under the no-go theorem above, and one concludes

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.} \quad (422)$$

Hence the solved lowest-order complete realization is necessarily on Branch B.

CXCIX. FINAL REDUCED PDE UNDER THE NO-GO THEOREM

Under the no-go theorem assumptions, the reduced first-order PDE is forced to be

$$\boxed{\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.} \quad (423)$$

CC. SHARP CONCLUSION

So the remaining microscopic task is now even sharper than before.
One no longer merely asks:

“Does the full complete action give Branch A or Branch B?”

Instead one asks:

$$\boxed{\text{Does the full complete TECT action contain an explicit linear longitudinal tangent carrier that violates the no-go theorem assumption?}} \quad (424)$$

If the answer is no, then Branch B is forced for the full theory as well. If the answer is yes, then Branch A becomes possible and must be checked explicitly.

CCI. FINAL SUMMARY

For the real CP^2 defect representative, the basic tangent operator satisfies

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0,$$

while

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3.$$

Therefore, for any complete local real action that does not contain an additional linear longitudinal tangent carrier, the first nonzero longitudinal traceless transport term is necessarily

$$V_\parallel^\mu(\xi) \sim \theta'(\xi)^3 n^\mu \Sigma_3.$$

Hence:

$$\boxed{\Lambda_\parallel^{(0)} = 0}$$

for this entire class of actions.

So Branch A is impossible unless the full complete TECT action contains an extra microscopic operator whose tangent Hessian produces a term

$$\Pi_Q \delta^2 \mathcal{O}_{\text{new}} \Pi_Q \supset \Lambda_\parallel^{(0)} \theta'(\xi) n^\mu \Sigma_3.$$

Without such an operator, Branch B is forced.

Since the present minimal complete lowest-order realization contains no such linear longitudinal carrier, it is necessarily on Branch B:

$$\boxed{\Lambda_\parallel^{(0)} = 0, \quad \lambda_c \neq 0.}$$

Thus the solved reduced PDE remains

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_\parallel) - I_c \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.$$

So the remaining microscopic question is now:

does the ultimate full complete TECT action contain a linear longitudinal tangent carrier beyond the present lowest-order realization?

CCII. CLASSIFICATION OF POSSIBLE LINEAR LONGITUDINAL TANGENT CARRIERS

CCIII. PROBLEM

We seek all possible local microscopic operators that could generate a longitudinal tangent contribution of the form

$$\boxed{V_{\text{new}}^\mu(\xi) \supset \Lambda_\parallel^{(0)} \theta'(\xi) n^\mu \Sigma_3.} \tag{425}$$

Equivalently, we want to classify all possible local invariants whose tangent Hessian contribution contains a term linear in the defect gradient and traceless in the tangent 2×2 sector.

CCIV. ALGEBRA GENERATED BY P AND $\partial_\xi P$

Let

$$P = P_B(\xi) = z_B z_B^\dagger, \quad \mathcal{T}_\xi := i[P, \partial_\xi P].$$

For the real defect representative,

$$z_B = \begin{pmatrix} \cos \theta \\ 0 \\ \sin \theta \end{pmatrix}, \quad \partial_\xi z_B = \theta' e_1.$$

Inside the tangent bundle Q_B , the basic identities are

$$\boxed{\Pi_Q \mathcal{T}_\xi \Pi_Q = 0}, \quad (426)$$

and

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'^2 \Sigma_3}. \quad (427)$$

Hence the local algebra generated purely by P and one power of $\partial_\xi P$ has no nonzero tangent-traceless element. The first tangent-traceless element appears only at quadratic order in $\partial_\xi P$.

CCV. GENERAL LOCAL OPERATOR ANSATZ AT LINEAR DERIVATIVE ORDER

Consider any local microscopic invariant density $\mathcal{O}_{\text{lin}}$ which is:

1. local in the fields,
2. linear in the normal derivative ∂_ξ ,
3. built from the complete field set (Ψ, P, σ, X) ,
4. tangent-isotropic before inserting the defect background.

Its tangent Hessian contribution must have the schematic form

$$\boxed{V_{\text{lin}}^\mu(\xi) = \theta'(\xi) n^\mu \mathcal{A}_Q(\xi)}, \quad (428)$$

where $\mathcal{A}_Q(\xi)$ is a tangent endomorphism.

The question is whether $\mathcal{A}_Q$ can contain a nonzero traceless part proportional to Σ_3 .

CCVI. NO-GO FOR ISOTROPIC LOCAL OPERATORS

Assume $\mathcal{O}_{\text{lin}}$ is built only from:

$$P, \quad \partial_\xi P, \quad \Psi^\dagger \Psi, \quad \sigma, \quad \text{and isotropic scalar contractions.}$$

Then after restriction to the tangent bundle, $\mathcal{A}_Q$ is generated by the tangent reductions of:

$$\Pi_Q, \quad \Pi_Q \mathcal{T}_\xi \Pi_Q, \quad \Pi_Q \mathcal{T}_\xi^2 \Pi_Q, \quad \dots$$

At linear derivative order, only $\Pi_Q \mathcal{T}_\xi \Pi_Q$ can contribute. But by (426),

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0.$$

Therefore:

$$\boxed{\text{No isotropic local operator built only from } (P, \partial P, \Psi^\dagger \Psi, \sigma) \text{ can generate a linear } \theta'(\xi) \Sigma_3 \text{ term.}} \quad (429)$$

Thus Branch A is impossible within this entire class.

CCVII. NECESSARY INGREDIENT FOR BRANCH A

To obtain

$$\theta'(\xi)\Sigma_3,$$

one needs an additional local tangent-anisotropy carrier H_Q satisfying

$$\boxed{H_Q \in \text{End}(Q_B), \quad (H_Q)_{\text{tr}} \propto \Sigma_3, \quad H_Q \notin \text{Alg}(P, \partial P)_{\text{iso}}.} \quad (430)$$

Then a linear longitudinal term may arise from an operator of the form

$$\boxed{\mathcal{O}_{\text{new}} \sim \theta'(\xi) n^\mu H_Q.} \quad (431)$$

Therefore a necessary condition for Branch A is the existence of a microscopic field or composite tensor whose tangent restriction already distinguishes e_1 from e_2 at zeroth derivative order.

CCVIII. COMPLETE CLASSIFICATION OF POSSIBILITIES

At lowest local order, there are only three logically distinct ways to realize such an H_Q .

A. Type I: explicit tangent anisotropy field

Introduce a new microscopic field Y whose defect-background value induces

$$\boxed{\Pi_Q Y_B \Pi_Q = y_0 \mathbf{1}_2 + y_3 \Sigma_3, \quad y_3 \neq 0.} \quad (432)$$

Then an invariant such as

$$\boxed{\mathcal{O}_I = n_\mu \Pi_Q (Y \mathcal{T}^\mu)} \quad (433)$$

can generate

$$\theta'(\xi) n^\mu \Sigma_3.$$

B. Type II: explicit tangent-frame selector

Introduce a microscopic structure R that defines a preferred tangent line inside Q_B :

$$\boxed{\Pi_Q R \Pi_Q = r_1 |e_1\rangle\langle e_1| + r_2 |e_2\rangle\langle e_2|, \quad r_1 \neq r_2.} \quad (434)$$

Then

$$\Pi_Q R \Pi_Q = \frac{r_1 + r_2}{2} \mathbf{1}_2 + \frac{r_1 - r_2}{2} \Sigma_3,$$

so any linear derivative invariant coupling R to $\partial_\xi P$ can produce a Branch A term.

C. Type III: non-isotropic condensate refinement

A refinement of the condensate sector $\mathcal{L}_\Psi^{\text{ref}}$ may itself induce a tangent anisotropy tensor

$$H_Q^{(\Psi)}$$

after background insertion, provided it does not reduce to an isotropic scalar function of P and ∂P .

Then a linear transport contribution of the form

$$\boxed{V_{\Psi,\text{new}}^\mu(\xi) = \theta'(\xi) n^\mu H_Q^{(\Psi)}} \quad (435)$$

with

$$(H_Q^{(\Psi)})_{\text{tr}} \propto \Sigma_3$$

would generate Branch A.

CCIX. EXHAUSTION AT LOWEST LOCAL ORDER

Therefore the lowest-order microscopic classification is exhausted by:

$$\boxed{\text{Branch A is impossible unless the full action contains a tangent-anisotropy carrier of Type I, Type II, or Type III.}} \quad (436)$$

Equivalently, Branch A requires an additional microscopic structure whose tangent restriction distinguishes e_1 and e_2 before the first nontrivial power of $\partial_\xi P$ is taken.

CCX. APPLICATION TO THE PRESENT MINIMAL COMPLETE LO REALIZATION

The present minimal complete lowest-order TECT realization contains:

- the real projector defect P_B ,
- the heavy scalar σ ,
- the heavy carrier X_i^a ,
- the transverse E/O completion,
- no explicit tangent-anisotropy carrier of Type I, Type II, or Type III.

Hence it satisfies the no-go assumptions, and therefore

$$\boxed{\Lambda_{\parallel}^{(0)} = 0.} \quad (437)$$

So the present minimal complete LO realization is necessarily on Branch B.

CCXI. FINAL SHARPENED OPEN PROBLEM

The remaining microscopic problem is no longer vague. It is exactly:

$$\boxed{\text{Does the ultimate full complete TECT action contain a Type I, Type II, or Type III tangent-anisotropy carrier?}} \quad (438)$$

If no, then the full theory is forced to remain on Branch B. If yes, then Branch A becomes possible and must be checked explicitly by direct Hessian projection.

CCXII. FINAL SUMMARY

For the real CP^2 defect representative,

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0, \quad (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'^2 \Sigma_3.$$

Therefore no isotropic local operator built only from

$$(P, \partial P, \Psi^\dagger \Psi, \sigma)$$

can produce a linear longitudinal tangent term

$$\theta'(\xi) \Sigma_3.$$

Hence Branch A is impossible unless the full complete action contains an additional tangent-anisotropy carrier H_Q with

$$(H_Q)_{\text{tr}} \propto \Sigma_3.$$

At lowest local order, there are only three possibilities:

Type I: an explicit tangent-anisotropy field,

Type II: an explicit tangent-frame selector,

Type III: a non-isotropic condensate refinement.

This classification is exhaustive at lowest local order.

Since the present minimal complete lowest-order realization contains none of these,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so it is necessarily on Branch B.

Therefore the remaining microscopic question is exactly:

does the ultimate full complete TECT action contain a Type I, II, or III tangent-anisotropy carrier?

CCXIII. TYPE II COMPLETION AND BRANCH A

CCXIV. TYPE II FRAME SELECTOR

Introduce a microscopic frame-selector tensor R such that on the defect background

$$\Pi_Q R \Pi_Q = r_1 |e_1\rangle\langle e_1| + r_2 |e_2\rangle\langle e_2|, \quad r_1 \neq r_2. \quad (439)$$

Equivalently,

$$\Pi_Q R \Pi_Q = r_0 \mathbf{1}_2 + r_3 \Sigma_3, \quad r_0 = \frac{r_1 + r_2}{2}, \quad r_3 = \frac{r_1 - r_2}{2} \neq 0. \quad (440)$$

CCXV. MINIMAL TYPE II INVARIANT

The lowest-order longitudinal Type II mixing is

$$\mathcal{L}_A^{(II)} = \gamma_A X_i^a n^i (T_a R) \partial_\xi \phi_0 + \text{h.c.} \quad (441)$$

After integrating out the heavy field X_i^a ,

$$\delta \mathcal{L}_{\text{eff}}^{(A)} \sim -\frac{\gamma_A \beta_X}{M_X^2} (\partial_\xi \phi_0) (T_a R) (T_a \partial_\xi P) D_\mu \psi. \quad (442)$$

On the defect background,

$$(T_a \partial_\xi P) \propto \theta'(\xi),$$

so the induced tangent transport is linear in $\theta'(\xi)$.

CCXVI. DIRECT TANGENT REDUCTION

Projecting to the tangent sector gives

$$V_A^\mu(\xi) = \theta'(\xi) n^\mu \Pi_Q R \Pi_Q = \theta'(\xi) n^\mu (r_0 \mathbf{1}_2 + r_3 \Sigma_3). \quad (443)$$

Therefore the longitudinal traceless coefficient is

$$a_3^{(A)}(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V_A^\mu(\xi)) = r_3 \theta'(\xi). \quad (444)$$

Hence

$$\Lambda_{\parallel}^{(0)} = r_3 \neq 0. \quad (445)$$

So Type II immediately realizes Branch A.

CCXVII. REDUCED COEFFICIENT

Projecting to the low modes,

$$\lambda_{\parallel}^{(A)} = \int d\xi f_+ f_- a_3^{(A)}(\xi) = r_3 I_\theta. \quad (446)$$

Therefore the Branch A reduced symbol is

$$A_1^{(A)}(p) = r_3 I_\theta p_{\parallel} \sigma_3 + I_c \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_c \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (447)$$

Its nondegeneracy condition is

$$r_3 \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0), \quad I_\theta \neq 0, \quad I_c \neq 0. \quad (448)$$

CCXVIII. INTERNAL TYPE II FRAME SELECTOR FROM TECT VARIABLES

CCXIX. GOAL

We want to realize the Type II selector

$$\Pi_Q R \Pi_Q = r_0 \mathbf{1}_2 + r_3 \Sigma_3, \quad r_3 \neq 0$$

using only internal TECT variables, without introducing an external ad hoc field.

CCXX. TANGENT-BUNDLE DATA

On the real defect background,

$$z_B = \begin{pmatrix} \cos \theta \\ 0 \\ \sin \theta \end{pmatrix}, \quad e_1 = \begin{pmatrix} -\sin \theta \\ 0 \\ \cos \theta \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

with

$$\Pi_Q = |e_1\rangle\langle e_1| + |e_2\rangle\langle e_2|, \quad \Sigma_3 = |e_1\rangle\langle e_1| - |e_2\rangle\langle e_2|.$$

So Type II means:

$$R_Q := \Pi_Q R \Pi_Q = r_1 |e_1\rangle\langle e_1| + r_2 |e_2\rangle\langle e_2|, \quad r_1 \neq r_2.$$

CCXXI. CANDIDATE 1: PROJECTED HESSIAN ANISOTROPY FROM THE CONDENSATE SECTOR

Let

$$\mathcal{H}_{\Psi,B}^{(\xi)} := \Pi_Q \left. \frac{\delta^2 S_\Psi}{\delta \eta^\dagger \delta \eta} \right|_{\Phi_B} \Pi_Q$$

be the tangent restriction of the static ξ -normal Hessian of the condensate sector.

Define its traceless tangent part

$$R^{(\Psi)} := \mathcal{H}_{\Psi,B}^{(\xi)} - \frac{1}{2_{Q_B}} (\mathcal{H}_{\Psi,B}^{(\xi)}) \Pi_Q. \quad (449)$$

If

$$R^{(\Psi)} \neq 0, \quad (450)$$

then automatically

$$R^{(\Psi)} = r_3^{(\Psi)} \Sigma_3$$

for some $r_3^{(\Psi)} \neq 0$, because any traceless Hermitian 2×2 operator on Q_B is a linear combination of $\Sigma_{1,2,3}$, and the longitudinal Type II selector keeps only the Σ_3 component.

Thus a non-isotropic condensate Hessian already furnishes an internal Type II selector.

CCXXII. CANDIDATE 2: PROJECTED HEAVY-CARRIER ANISOTROPY

Let the heavy carrier sector contain a background-dependent quadratic kernel

$$\mathcal{K}_{X,B}^{ij,ab}.$$

After contraction with the shell-normal direction $n_i n_j$ and the relevant internal channel, define

$$R^{(X)} := \Pi_Q \mathcal{K}_{X,B}^\parallel \Pi_Q - \frac{1}{2_{Q_B}} (\Pi_Q \mathcal{K}_{X,B}^\parallel \Pi_Q) \Pi_Q. \quad (451)$$

If

$$R^{(X)} \neq 0, \quad (452)$$

then again

$$R^{(X)} = r_3^{(X)} \Sigma_3, \quad r_3^{(X)} \neq 0.$$

So an anisotropic heavy-carrier kernel also gives an internal Type II selector.

CCXXIII. MINIMAL INTERNAL REALIZATION

Therefore the minimal internal Type II selector is

$$\boxed{R = R^{(\Psi)} + R^{(X)}}. \quad (453)$$

Its tangent reduction is

$$\boxed{\Pi_Q R \Pi_Q = r_0 \Pi_Q + r_3 \Sigma_3, \quad r_3 = r_3^{(\Psi)} + r_3^{(X)}}. \quad (454)$$

Hence the necessary and sufficient lowest-order internal condition for Type II is simply

$$\boxed{r_3^{(\Psi)} + r_3^{(X)} \neq 0}. \quad (455)$$

CCXXIV. INDUCED BRANCH A TERM

Insert this R into the Type II longitudinal mixing

$$\mathcal{L}_A^{(II)} = \gamma_A X_i^a n^i (T_a R) \partial_\xi \phi_0 + \text{h.c.}$$

and integrate out X . The tangent transport block is

$$\boxed{V_A^\mu(\xi) = \theta'(\xi) n^\mu (r_0 \mathbf{1}_2 + r_3 \Sigma_3)}. \quad (456)$$

Therefore

$$\boxed{a_3^{(A)}(\xi) = r_3 \theta'(\xi), \quad \Lambda_\parallel^{(0)} = r_3}. \quad (457)$$

So Branch A occurs iff

$$\boxed{r_3 \neq 0}. \quad (458)$$

CCXXV. RESULTING BRANCH-SENSITIVE PDE

If $r_3 = 0$, the theory stays on Branch B:

$$A_1^{(B)}(p) = \frac{\lambda_c}{2} I_3 p_\parallel \sigma_3 + I_c \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_c \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

If $r_3 \neq 0$, the theory is upgraded to Branch A:

$$\boxed{A_1^{(A)}(p) = r_3 I_\theta p_\parallel \sigma_3 + I_c \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + I_c \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2)}. \quad (459)$$

CCXXVI. FINAL MICROSCOPIC TEST

Thus the true full-action branch decision is reduced to the explicit internal test

$$\boxed{r_3 = \frac{1}{2_{Q_B}} (\Sigma_3 \Pi_Q (R^{(\Psi)} + R^{(X)}) \Pi_Q)}. \quad (460)$$

If this trace vanishes, full TECT remains on Branch B. If it is nonzero, full TECT lies on Branch A.

CCXXVII. FINAL SUMMARY

Type II does not require a new external field. It can be realized internally from TECT variables as a tangent anisotropy selector

$$\Pi_Q R \Pi_Q = r_0 \mathbf{1}_2 + r_3 \Sigma_3.$$

The two minimal internal sources are:

$$R^{(\Psi)} = \mathcal{H}_{\Psi,B}^{(\xi)} - \frac{1}{2_{Q_B}} (\mathcal{H}_{\Psi,B}^{(\xi)}) \Pi_Q,$$

coming from condensate-sector tangent Hessian anisotropy, and

$$R^{(X)} = \Pi_Q \mathcal{K}_{X,B}^{\parallel} \Pi_Q - \frac{1}{2_{Q_B}} (\Pi_Q \mathcal{K}_{X,B}^{\parallel} \Pi_Q) \Pi_Q,$$

coming from heavy-carrier kernel anisotropy.

The minimal internal selector is

$$R = R^{(\Psi)} + R^{(X)}.$$

Its branch-deciding coefficient is

$$r_3 = \frac{1}{2_{Q_B}} (\Sigma_3 \Pi_Q (R^{(\Psi)} + R^{(X)}) \Pi_Q).$$

If

$$r_3 = 0,$$

the theory remains on Branch B.

If

$$r_3 \neq 0,$$

then

$$\Lambda_{\parallel}^{(0)} = r_3 \neq 0,$$

and the theory is promoted to Branch A.

So the ultimate full-action branch decision is now reduced to one explicit internal trace test for r_3 .

CCXXVIII. ACTUAL COMPUTATION OF THE TYPE II BRANCH TRACE r_3

CCXXIX. DEFINITION

The Type II branch-deciding coefficient is

$$r_3 = \frac{1}{2_{Q_B}} (\Sigma_3 \Pi_Q (R^{(\Psi)} + R^{(X)}) \Pi_Q). \tag{461}$$

We now evaluate this for the present minimal complete lowest-order realization.

CCXXX. CONDENSATE-SECTOR CONTRIBUTION

By definition,

$$R^{(\Psi)} = \mathcal{H}_{\Psi,B}^{(\xi)} - \frac{1}{2_{Q_B}} (\mathcal{H}_{\Psi,B}^{(\xi)}) \Pi_Q, \quad (462)$$

where $\mathcal{H}_{\Psi,B}^{(\xi)}$ is the tangent restriction of the static normal Hessian of the condensate sector.

For the present minimal lowest-order condensate core, the tangent quadratic operator is isotropic on Q_B :

$$\boxed{\mathcal{H}_{\Psi,B}^{(\xi)} = h_{\Psi}(\xi) \Pi_Q.} \quad (463)$$

Therefore

$$\begin{aligned} R^{(\Psi)} &= h_{\Psi}(\xi) \Pi_Q - \frac{1}{2_{Q_B}} (h_{\Psi}(\xi) \Pi_Q) \Pi_Q \\ &= h_{\Psi}(\xi) \Pi_Q - \frac{1}{2} (2h_{\Psi}(\xi)) \Pi_Q \\ &= 0. \end{aligned} \quad (464)$$

Hence

$$\boxed{r_3^{(\Psi)} = \frac{1}{2_{Q_B}} (\Sigma_3 \Pi_Q R^{(\Psi)} \Pi_Q) = 0.} \quad (465)$$

CCXXXI. HEAVY-CARRIER CONTRIBUTION

By definition,

$$R^{(X)} = \Pi_Q \mathcal{K}_{X,B}^{\parallel} \Pi_Q - \frac{1}{2_{Q_B}} (\Pi_Q \mathcal{K}_{X,B}^{\parallel} \Pi_Q) \Pi_Q. \quad (466)$$

In the present minimal lowest-order realization, the heavy-carrier sector is just the isotropic mass term

$$\frac{1}{2} M_X^2 X_i^a X_i^a,$$

so its background-parallel kernel is proportional to the identity on the tangent bundle:

$$\boxed{\Pi_Q \mathcal{K}_{X,B}^{\parallel} \Pi_Q = h_X(\xi) \Pi_Q.} \quad (467)$$

Therefore

$$\begin{aligned} R^{(X)} &= h_X(\xi) \Pi_Q - \frac{1}{2_{Q_B}} (h_X(\xi) \Pi_Q) \Pi_Q \\ &= h_X(\xi) \Pi_Q - \frac{1}{2} (2h_X(\xi)) \Pi_Q \\ &= 0. \end{aligned} \quad (468)$$

Hence

$$\boxed{r_3^{(X)} = \frac{1}{2_{Q_B}} (\Sigma_3 \Pi_Q R^{(X)} \Pi_Q) = 0.} \quad (469)$$

CCXXXII. TOTAL BRANCH TRACE

Combining the two sectors,

$$r_3 = r_3^{(\Psi)} + r_3^{(X)} = 0. \quad (470)$$

Therefore

$$\Lambda_{\parallel}^{(0)} = r_3 = 0. \quad (471)$$

So the present minimal complete lowest-order realization cannot lie on Branch A.

CCXXXIII. BRANCH DECISION

Since the primitive channel was already shown to survive,

$$\lambda_c \neq 0, \quad (472)$$

while

$$\Lambda_{\parallel}^{(0)} = 0, \quad (473)$$

the present realization lies exactly on Branch B:

$$\text{minimal complete LO TECT} \in \text{Branch B}. \quad (474)$$

CCXXXIV. INTERPRETATION

Thus the only way to obtain Branch A is to add a genuine tangent-anisotropic contribution to either

$$\mathcal{H}_{\Psi,B}^{(\xi)} \quad (475)$$

or

$$\mathcal{K}_{X,B}^{\parallel}. \quad (476)$$

Without such an anisotropic correction, the branch trace remains

$$r_3 = 0,$$

and Branch B is forced.

CCXXXV. CALCULATION: LOCAL $SO(2)$ INVARIANCE FORCES $r_3 = 0$

CCXXXVI. SETUP

Let $Q_B(\xi) = \text{span}\{e_1(\xi), e_2\}$ be the tangent two-plane on the real defect background, and let

$$\Sigma_3 = |e_1\rangle\langle e_1| - |e_2\rangle\langle e_2|.$$

The Type II branch-deciding coefficient is

$$r_3 = \frac{1}{2} R_Q(\Sigma_3 R_Q), \quad R_Q := \Pi_Q(R^{(\Psi)} + R^{(X)})\Pi_Q. \quad (477)$$

We now compute r_3 under the natural TECT assumption that the tangent low-energy action is local and $SO(2)$ -invariant.

CCXXXVII. REPRESENTATION-THEORETIC LEMMA

Let $U(\varphi) \in SO(2)$ act on Q_B by

$$e_1 \mapsto \cos \varphi e_1 + \sin \varphi e_2, \quad e_2 \mapsto -\sin \varphi e_1 + \cos \varphi e_2.$$

Suppose $A \in \text{End}(Q_B)$ is $SO(2)$ -equivariant:

$$\boxed{U(\varphi) A U(\varphi)^{-1} = A \quad \forall \varphi.} \quad (478)$$

Then, on the real two-dimensional irreducible representation of $SO(2)$, Schur's lemma implies

$$\boxed{A = a \mathbf{1}_2.} \quad (479)$$

Therefore every local tangent Hessian kernel that preserves the transverse $SO(2)$ symmetry is proportional to the identity on Q_B .

CCXXXVIII. CONDENSATE CONTRIBUTION

Let

$$\mathcal{H}_{\Psi,B}^{(\xi)} := \Pi_Q \left. \frac{\delta^2 S_\Psi}{\delta \eta^\dagger \delta \eta} \right|_{\Phi_B} \Pi_Q.$$

If the condensate-sector tangent action is local and $SO(2)$ -invariant, then by (479),

$$\boxed{\mathcal{H}_{\Psi,B}^{(\xi)} = h_\Psi(\xi) \mathbf{1}_2.} \quad (480)$$

Hence its traceless part vanishes:

$$\boxed{R^{(\Psi)} = \mathcal{H}_{\Psi,B}^{(\xi)} - \frac{1}{2_{Q_B}} (\mathcal{H}_{\Psi,B}^{(\xi)}) \mathbf{1}_2 = 0.} \quad (481)$$

So

$$\boxed{r_3^{(\Psi)} = \frac{1}{2_{Q_B}} (\Sigma_3 R^{(\Psi)}) = 0.} \quad (482)$$

CCXXXIX. HEAVY-CARRIER CONTRIBUTION

Likewise let

$$\mathcal{K}_{X,B}^\parallel$$

be the shell-parallel heavy-carrier quadratic kernel restricted to Q_B . If the heavy-carrier kinetic sector is also tangent $SO(2)$ -invariant, then

$$\boxed{\Pi_Q \mathcal{K}_{X,B}^\parallel \Pi_Q = h_X(\xi) \mathbf{1}_2.} \quad (483)$$

Therefore

$$\boxed{R^{(X)} = \Pi_Q \mathcal{K}_{X,B}^\parallel \Pi_Q - \frac{1}{2_{Q_B}} (\Pi_Q \mathcal{K}_{X,B}^\parallel \Pi_Q) \mathbf{1}_2 = 0,} \quad (484)$$

and hence

$$\boxed{r_3^{(X)} = \frac{1}{2_{Q_B}} (\Sigma_3 R^{(X)}) = 0.} \quad (485)$$

CCXL. TOTAL RESULT

Combining the two sectors,

$$\boxed{r_3 = r_3^{(\Psi)} + r_3^{(X)} = 0.} \quad (486)$$

Therefore

$$\boxed{\Lambda_{\parallel}^{(0)} = r_3 = 0.} \quad (487)$$

Hence Branch A is impossible under local tangent $SO(2)$ invariance.

CCXLI. CONSEQUENCE

Under the natural TECT polarization-bundle condition that the tangent effective action is local and $SO(2)$ -invariant, the only surviving longitudinal channel is the primitive one:

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}.} \quad (488)$$

So the theory is forced onto Branch B unless the full action contains an explicit term that breaks tangent $SO(2)$ down to a frame selector on Q_B .

CCXLII. NECESSARY EVASION CONDITION

To evade the result above, the full action must contain a tangent anisotropy operator R_Q with

$$\boxed{[U(\varphi), R_Q] \neq 0 \quad \text{for some } \varphi,} \quad (489)$$

so that

$$R_Q = r_0 \mathbf{1}_2 + r_3 \Sigma_3, \quad r_3 \neq 0.$$

Without such an explicit $SO(2)$ -breaking tangent selector, one always has $r_3 = 0$.

CCXLIII. FINAL SUMMARY

Assume the tangent low-energy sector of TECT is local and $SO(2)$ -invariant on the transverse polarization bundle Q_B . Then every tangent Hessian kernel on Q_B is $SO(2)$ -equivariant, hence by irreducibility,

$$A = a \mathbf{1}_2.$$

Therefore both the condensate tangent Hessian and the heavy-carrier tangent kernel are proportional to the identity:

$$\mathcal{H}_{\Psi,B}^{(\xi)} = h_{\Psi}(\xi) \mathbf{1}_2, \quad \Pi_Q \mathcal{K}_{X,B}^{\parallel} \Pi_Q = h_X(\xi) \mathbf{1}_2.$$

Their traceless parts vanish:

$$R^{(\Psi)} = 0, \quad R^{(X)} = 0.$$

Hence the Type II branch trace is

$$\boxed{r_3 = \frac{1}{2} \frac{1}{Q_B} (\Sigma_3 (R^{(\Psi)} + R^{(X)})) = 0.}$$

So

$$\Lambda_{\parallel}^{(0)} = 0.$$

Therefore, under local tangent $SO(2)$ invariance, Branch A is impossible and the theory is forced onto Branch B:

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.$$

To realize Branch A, the full action must contain an explicit tangent $SO(2)$ -breaking selector $R_Q = r_0 \mathbf{1}_2 + r_3 \Sigma_3$ with $r_3 \neq 0$.

CCXLIV. CONDITIONAL BRANCH FIXING FROM LOCAL $SO(2)$ INVARIANCE

CCXLV. AIM

We do not yet prove the full complete microscopic PDE unconditionally. Instead, we prove the following stronger conditional statement:

If the full tangent low-energy sector of TECT remains local and $SO(2)$ -invariant on Q_B , then Branch A is impossible and Branch B is realized. (490)

This removes the longitudinal branch ambiguity under one concrete microscopic symmetry assumption.

CCXLVI. TANGENT BUNDLE AND BRANCH TRACE

Let

$$Q_B(\xi) = \text{span}\{e_1(\xi), e_2\}, \quad \Sigma_3 = |e_1\rangle\langle e_1| - |e_2\rangle\langle e_2|.$$

The branch-deciding Type II trace is

$$r_3 = \frac{1}{2_{Q_B}} (\Sigma_3 R_Q), \quad R_Q := \Pi_Q(R^{(\Psi)} + R^{(X)})\Pi_Q. \quad (491)$$

Branch A requires

$$r_3 \neq 0, \quad \Lambda_{\parallel}^{(0)} = r_3.$$

Branch B corresponds to

$$r_3 = 0$$

with surviving primitive coefficient $\lambda_c \neq 0$.

CCXLVII. $SO(2)$ -EQUIVARIANCE LEMMA

Let $U(\varphi) \in SO(2)$ act on Q_B by

$$e_1 \mapsto \cos \varphi e_1 + \sin \varphi e_2, \quad e_2 \mapsto -\sin \varphi e_1 + \cos \varphi e_2.$$

Suppose $A \in \text{End}(Q_B)$ is $SO(2)$ -equivariant:

$$U(\varphi) A U(\varphi)^{-1} = A \quad \forall \varphi. \quad (492)$$

Since Q_B is the real irreducible 2-dimensional representation of $SO(2)$, Schur's lemma implies

$$A = a \mathbf{1}_2. \quad (493)$$

Therefore no $SO(2)$ -equivariant tangent operator can contain a traceless Σ_3 part.

CCXLVIII. APPLICATION TO THE CONDENSATE TANGENT HESSIAN

Let

$$\mathcal{H}_{\Psi,B}^{(\xi)} := \Pi_Q \frac{\delta^2 S_{\Psi}}{\delta \eta^\dagger \delta \eta} \Big|_{\Phi_B} \Pi_Q.$$

Assume the full condensate tangent sector is local and $SO(2)$ -invariant. Then by (493),

$$\boxed{\mathcal{H}_{\Psi,B}^{(\xi)} = h_{\Psi}(\xi) \mathbf{1}_2.} \quad (494)$$

Hence its traceless part vanishes:

$$\boxed{R^{(\Psi)} = \mathcal{H}_{\Psi,B}^{(\xi)} - \frac{1}{2_{Q_B}} (\mathcal{H}_{\Psi,B}^{(\xi)}) \mathbf{1}_2 = 0.} \quad (495)$$

CCXLIX. APPLICATION TO THE HEAVY-CARRIER KERNEL

Likewise let

$$\mathcal{K}_{X,B}^{\parallel}$$

be the shell-parallel heavy-carrier quadratic kernel restricted to Q_B . Assume the heavy-carrier tangent sector is also local and $SO(2)$ -invariant. Then again

$$\boxed{\Pi_Q \mathcal{K}_{X,B}^{\parallel} \Pi_Q = h_X(\xi) \mathbf{1}_2,} \quad (496)$$

so

$$\boxed{R^{(X)} = \Pi_Q \mathcal{K}_{X,B}^{\parallel} \Pi_Q - \frac{1}{2_{Q_B}} (\Pi_Q \mathcal{K}_{X,B}^{\parallel} \Pi_Q) \mathbf{1}_2 = 0.} \quad (497)$$

CCL. BRANCH CONSEQUENCE

Combining the two sectors,

$$\boxed{R_Q = 0, \quad r_3 = \frac{1}{2_{Q_B}} (\Sigma_3 R_Q) = 0.} \quad (498)$$

Therefore

$$\boxed{\Lambda_{\parallel}^{(0)} = 0.} \quad (499)$$

Hence Branch A is impossible under local tangent $SO(2)$ invariance.

CCLI. PRIMITIVE LONGITUDINAL SURVIVAL

The primitive commutator-square channel is still available:

$$S_{\xi} = (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3.$$

Therefore the longitudinal transport survives through

$$\boxed{V_{\parallel, \text{prim}}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \Sigma_3,} \quad (500)$$

so that

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}. \quad (501)$$

Thus, under the $SO(2)$ -invariance assumption,

$$\boxed{\text{the complete theory is forced onto Branch B.}} \quad (502)$$

CCLII. RESULTING REDUCED PDE

If in addition the transverse sector remains nonzero,

$$(\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0),$$

the reduced PDE is

$$\boxed{\left[i\partial_t - \frac{\lambda_c}{2} I_{\theta'^3} \sigma_3 (-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.} \quad (503)$$

CCLIII. HONEST SCOPE

This does *not* finish the full microscopic PDE unconditionally. What it proves is:

$$\boxed{\text{full branch ambiguity disappears if the complete tangent sector preserves local } SO(2).} \quad (504)$$

So the remaining microscopic question is now sharpened to:

$$\boxed{\text{Does the true full complete TECT Hessian preserve local tangent } SO(2), \text{ or does it contain an explicit } SO(2)\text{-breaking frame s}} \quad (505)$$

CCLIV. TERM-BY-TERM $SO(2)$ TEST FOR THE FULL COMPLETE TECT HESSIAN

CCLV. GOAL

We now classify the microscopic operators of the full complete TECT action according to whether they preserve or break the local tangent $SO(2)$ symmetry on the polarization bundle Q_B .

This directly decides whether the full theory is forced onto Branch B or can realize Branch A.

CCLVI. TANGENT $SO(2)$ STRUCTURE

Let

$$Q_B(\xi) = \text{span}\{e_1(\xi), e_2\}$$

be the tangent two-plane on the defect background.

Any local tangent operator on Q_B can be written as

$$A = a_0 \mathbf{1}_2 + a_1 \Sigma_1 + a_2 \Sigma_2 + a_3 \Sigma_3.$$

Under local tangent $SO(2)$, the pair (e_1, e_2) rotates, so:

- $\mathbf{1}_2$ is an $SO(2)$ -scalar;
- Σ_2 is the $SO(2)$ -invariant antisymmetric generator;
- (Σ_1, Σ_3) transform as a spin-2 traceless symmetric doublet.

Therefore a Branch A carrier requires a nonzero Σ_3 -component, i.e. a tangent symmetric-traceless selector.

CCLVII. MICROSCOPIC CRITERION

Let the complete action be

$$S_{\text{comp}} = S_{\Psi} + S_P + S_{\sigma} + S_X + S_{\text{mix}}.$$

For each sector S , let

$$V_S^{\mu} = \Pi_Q \mathbb{H}_{1,S}^{\mu} \Pi_Q$$

be its tangent first-order Hessian block.

Then:

$$\boxed{\text{Sector } S \text{ preserves local tangent } SO(2) \iff V_S^{\mu} = u_S^{\mu} \mathbf{1}_2 + v_S^{\mu} \Sigma_2} \quad (506)$$

and

$$\boxed{\text{Sector } S \text{ breaks local tangent } SO(2) \iff V_S^{\mu} \text{ contains a nonzero } \Sigma_1 \text{ or } \Sigma_3 \text{ component.}} \quad (507)$$

Since Branch A needs a longitudinal traceless term,

$$V^{\mu} \supset \theta'(\xi) n^{\mu} \Sigma_3,$$

the only dangerous piece is the Σ_3 -projection:

$$\boxed{a_3^{(S)}(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_{\mu} V_S^{\mu}(\xi)).} \quad (508)$$

CCLVIII. SECTOR-BY-SECTOR CLASSIFICATION

A. Scalar condensate core

Any condensate-core term depending only on scalar contractions such as

$$\Psi^{\dagger} \Psi, \quad \partial_i \Psi^{\dagger} \partial_i \Psi, \quad (\nabla^2 \Psi)^{\dagger} (\nabla^2 \Psi), \quad (\Psi^{\dagger} \Psi)^2$$

is tangent-isotropic.

Hence

$$\boxed{V_{\Psi}^{\mu} = u_{\Psi}^{\mu} \mathbf{1}_2 + v_{\Psi}^{\mu} \Sigma_2, \quad a_3^{(\Psi)} = 0.} \quad (509)$$

So the ordinary locked Brazovskii condensate core cannot generate Branch A.

B. Projector geometry sector

Any projector term built only from isotropic scalar contractions of

$$P, \quad \partial_{\mu} P, \quad \partial_{\mu} \partial_{\nu} P, \dots$$

and without an explicit tangent-frame selector is also tangent-isotropic.

Therefore

$$\boxed{V_P^{\mu} = u_P^{\mu} \mathbf{1}_2 + v_P^{\mu} \Sigma_2, \quad a_3^{(P)} = 0.} \quad (510)$$

So the isotropic projector geometry sector cannot generate Branch A.

C. Volumetric scalar sector

The heavy scalar σ carries no tangent polarization label. Therefore any local isotropic σ -sector term satisfies

$$\boxed{V_\sigma^\mu = u_\sigma^\mu \mathbf{1}_2, \quad a_3^{(\sigma)} = 0.} \quad (511)$$

So the pure volumetric sector cannot generate Branch A.

D. Isotropic heavy-carrier kernel

If the heavy carrier has only isotropic mass/kinetic terms,

$$\frac{1}{2} M_X^2 X_i^a X_i^a \quad \text{and isotropic derivative refinements,}$$

then its tangent restriction is again scalar:

$$\boxed{V_X^\mu = u_X^\mu \mathbf{1}_2 + v_X^\mu \Sigma_2, \quad a_3^{(X)} = 0.} \quad (512)$$

So isotropic X -sector refinements also cannot generate Branch A.

E. Transverse E/O completion

The explicit E/O slots generate

$$V_{EO}^\mu = b_E(\xi) E^\mu{}_a \Sigma_a + b_O(\xi) J^\mu{}_a \Sigma_a.$$

This is exactly what produces the transverse Dirac channels, but it is not a longitudinal frame selector. Hence

$$\boxed{a_3^{(EO)} = 0.} \quad (513)$$

So E/O completion does not create Branch A.

F. Primitive longitudinal channel

The primitive commutator-square channel gives

$$V_{\parallel, \text{prim}}^\mu = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3.$$

Therefore it contributes to the longitudinal witness, but only at primitive order:

$$\boxed{a_3^{(\text{prim})}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3.} \quad (514)$$

This is Branch B, not Branch A.

CCLIX. WHAT CAN GENERATE BRANCH A?

By the classification above, Branch A requires a sector S_A whose tangent Hessian block contains

$$V_A^\mu \supset \theta'(\xi) n^\mu \Sigma_3.$$

Equivalently, the microscopic action must contain a local term whose tangent reduction carries a symmetric-traceless selector on Q_B .

At lowest order, this means:

$$\boxed{\text{Branch A requires an explicit local tangent } SO(2)\text{-breaking term.}} \quad (515)$$

Examples would be terms containing a background-dependent rank-two selector R_Q with

$$R_Q = r_0 \mathbf{1}_2 + r_3 \Sigma_3, \quad r_3 \neq 0.$$

CCLX. TOTAL CONCLUSION

Therefore, if every sector in the full complete TECT action is built from local tangent-isotropic scalar contractions, then

$$\boxed{a_3^{(\Psi)} = a_3^{(P)} = a_3^{(\sigma)} = a_3^{(X)} = a_3^{(EO)} = 0,} \quad (516)$$

and the only surviving longitudinal term is

$$\boxed{a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3.} \quad (517)$$

Hence

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,} \quad (518)$$

so the full theory is forced onto Branch B.

CCLXI. SHARP REMAINING MICROSCOPIC TEST

The only remaining full-action test is therefore:

$$\boxed{\text{Does the true complete TECT action contain any explicit local tangent } SO(2)\text{-breaking operator?}} \quad (519)$$

If no, then Branch B is exact.

If yes, then compute

$$a_3^{(A)}(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_{\mu} V_A^{\mu}(\xi))$$

and decide whether $\Lambda_{\parallel}^{(0)} \neq 0$.

CCLXII. FINAL SUMMARY

The term-by-term $SO(2)$ test is:

$$\text{sector } S \text{ preserves tangent } SO(2) \iff V_S^{\mu} = u_S^{\mu} \mathbf{1}_2 + v_S^{\mu} \Sigma_2,$$

and Branch A is possible only if some sector contains a nonzero longitudinal Σ_3 component.

For the following sectors:

S_{Ψ} (scalar condensate core), S_P (isotropic projector geometry), S_{σ} (volumetric scalar), S_X (isotropic heavy-carrier kernel),

one has

$$a_3^{(S)} = 0.$$

The primitive channel contributes only

$$a_3^{(\text{prim})} = \frac{\lambda_c}{2} \theta'^3,$$

which is Branch B.

Therefore, if the full complete TECT action contains only local tangent-isotropic sectors, then

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,}$$

so the full theory is forced onto Branch B.

Thus the only remaining microscopic question is:

$$\boxed{\text{Does the true complete TECT action contain any explicit local tangent } SO(2)\text{-breaking operator?}}$$

If no, Branch B is exact. If yes, Branch A becomes possible.

CCLXIII. EXHAUSTION OF LOWEST-ORDER LOCAL $SO(2)$ -BREAKING CANDIDATES FROM EXISTING TECT VARIABLES

CCLXIV. PROBLEM

We now test whether an explicit local tangent $SO(2)$ -breaking longitudinal selector can be built at lowest local order from the existing TECT variables

$$(\Psi, P, \sigma, X_i^a)$$

without introducing any new independent microscopic field.

The target is a tangent operator R_Q such that

$$\boxed{R_Q = r_0 \mathbf{1}_2 + r_3 \Sigma_3, \quad r_3 \neq 0.} \tag{520}$$

If no such operator exists, then Branch A is excluded at this order.

CCLXV. AVAILABLE LOCAL BUILDING BLOCKS

On the real defect background, the available local objects are:

$$P_B = z_B z_B^\dagger, \quad \partial_\xi P_B, \quad \Psi_B = \phi_0 z_B, \quad \sigma_B, \quad X_{i,B}^a.$$

At lowest local order, every candidate tangent operator must be built from algebraic combinations of:

$$\boxed{\Pi_Q, \quad P_B, \quad \partial_\xi P_B, \quad \Psi_B \Psi_B^\dagger, \quad \sigma_B, \quad X_{i,B}^a T_a.} \tag{521}$$

We test these one by one after restriction to Q_B .

CCLXVI. PROJECTOR SECTOR

Since

$$P_B|_{Q_B} = 0,$$

one has

$$\boxed{\Pi_Q P_B \Pi_Q = 0.} \quad (522)$$

Hence P_B itself provides no tangent selector.

Next,

$$\partial_\xi P_B = \theta'(\xi)(e_1 z_B^\dagger + z_B e_1^\dagger).$$

Because $z_B \perp Q_B$, one gets

$$\boxed{\Pi_Q \partial_\xi P_B \Pi_Q = 0.} \quad (523)$$

Thus neither P_B nor $\partial_\xi P_B$ yields a linear tangent selector.

At quadratic order,

$$(\partial_\xi P_B)^2 = \theta'(\xi)^2(|e_1\rangle\langle e_1| + |z_B\rangle\langle z_B|),$$

so after tangent projection

$$\boxed{\Pi_Q (\partial_\xi P_B)^2 \Pi_Q = \theta'(\xi)^2 |e_1\rangle\langle e_1|.} \quad (524)$$

Its traceless part is

$$\boxed{\left(\Pi_Q (\partial_\xi P_B)^2 \Pi_Q \right)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3,} \quad (525)$$

which is exactly the primitive Branch B carrier, not a linear Branch A carrier.

CCLXVII. CONDENSATE SECTOR

The background condensate bilinear is

$$\Psi_B \Psi_B^\dagger = |\phi_0|^2 P_B.$$

Therefore

$$\boxed{\Pi_Q (\Psi_B \Psi_B^\dagger) \Pi_Q = 0.} \quad (526)$$

Similarly, any scalar condensate invariant such as

$$\Psi_B^\dagger \Psi_B = |\phi_0|^2$$

acts only as a multiple of the identity in fluctuation space and cannot distinguish e_1 from e_2 .

Hence no lowest-order condensate scalar can generate $r_3 \neq 0$.

CCLXVIII. VOLUMETRIC SCALAR SECTOR

The background scalar σ_B is a scalar under tangent $SO(2)$. Thus any local operator built only from σ_B and scalar contractions contributes only

$$\propto \mathbf{1}_2$$

on Q_B . Therefore

$$\boxed{\sigma_B\text{-sector cannot generate a tangent } \Sigma_3 \text{ selector at lowest local order.}} \quad (527)$$

CCLXIX. HEAVY CARRIER SECTOR

Now test the existing heavy carrier X_i^a .

At lowest order in the present construction, the background $X_{i,B}^a$ is either zero or isotropically integrated out. Thus the only available local tensor from the X -sector before introducing any extra anisotropic kernel is

$$X_i^a X_i^a,$$

or isotropic quadratic kernels.

After tangent projection, such terms again produce only

$$\propto \mathbf{1}_2$$

or the already-known transverse E/O structures, but no longitudinal Σ_3 selector linear in θ' .

Hence

the existing lowest-order X -sector does not generate $r_3 \neq 0$.

(528)

CCLXX. EXHAUSTION THEOREM

Collecting the results:

1. P_B gives zero on Q_B ;
2. $\partial_\xi P_B$ gives zero on Q_B ;
3. $(\partial_\xi P_B)^2$ gives $\theta'^2 \Sigma_3$, i.e. only the primitive Branch B carrier;
4. $\Psi_B \Psi_B^\dagger$ gives zero on Q_B ;
5. scalar condensate and scalar volumetric terms give only $\mathbf{1}_2$;
6. the existing lowest-order heavy-carrier sector gives no linear longitudinal selector.

Therefore:

No explicit local tangent $SO(2)$ -breaking longitudinal selector can be built at lowest local order from the existing TECT variables.

(529)

Equivalently,

$r_3 = 0 \quad \implies \quad \Lambda_{\parallel}^{(0)} = 0.$

(530)

Thus Branch A is excluded at this level.

CCLXXI. FORCED BRANCH

Since the primitive carrier survives through

$$\left(\Pi_Q (\partial_\xi P_B)^2 \Pi_Q \right)_{\text{tr}} = \frac{1}{2} \theta'^2 \Sigma_3,$$

the surviving longitudinal coefficient is

$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \lambda_c \neq 0.$

(531)

Therefore the theory is forced onto Branch B:

$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.$

(532)

CCLXXII. FINAL REDUCED PDE AT THIS ORDER

Including the already-constructed transverse completion,

$$\left[i\partial_t - \frac{\lambda_c}{2} I_{\theta'^3} \sigma_3 (-i\partial_{\parallel}) - I_{\theta, \cos} \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_{\theta, \cos} \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (533)$$

CCLXXIII. SHARP REMAINING POSSIBILITY

The only way to reopen Branch A is to add a genuinely new local operator whose tangent reduction is not generated by the algebra of

$$(P_B, \partial_{\xi} P_B, \Psi_B, \sigma_B, X_B)$$

at lowest order and that produces a linear

$$\theta'(\xi) \Sigma_3$$

selector. Without such a new operator, Branch B is exact at this order.

CCLXXIV. FINAL SUMMARY

At lowest local order, using only the existing TECT variables

$$(\Psi, P, \sigma, X_i^a),$$

one finds:

$$\Pi_Q P_B \Pi_Q = 0, \quad \Pi_Q (\partial_{\xi} P_B) \Pi_Q = 0, \quad \Pi_Q (\Psi_B \Psi_B^{\dagger}) \Pi_Q = 0.$$

Scalar condensate and volumetric terms contribute only multiples of $\mathbf{1}_2$ on Q_B . The existing lowest-order heavy-carrier sector also gives no linear longitudinal selector.

The first nontrivial traceless tangent term is

$$\left(\Pi_Q (\partial_{\xi} P_B)^2 \Pi_Q \right)_{\text{tr}} = \frac{1}{2} \theta'^2 \Sigma_3,$$

which is exactly the primitive Branch B carrier.

Therefore no explicit local tangent $SO(2)$ -breaking longitudinal selector can be built at lowest local order from the existing TECT variables alone:

$$r_3 = 0, \quad \Lambda_{\parallel}^{(0)} = 0.$$

Since the primitive carrier survives,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \lambda_c \neq 0,$$

the theory is forced onto Branch B at this order.

Hence the reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_{\theta'^3} \sigma_3 (-i\partial_{\parallel}) - I_{\theta, \cos} \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_{\theta, \cos} \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.$$

So the only way to realize Branch A is to add a genuinely new local operator beyond the existing lowest-order TECT variable algebra.

CCLXXV. HIGHER-ORDER LOCAL OPERATOR TEST FOR BRANCH A

CCLXXVI. GOAL

We now test whether higher-order local operators built from the existing TECT variables can generate a Branch A longitudinal selector

$$\theta'(\xi)\Sigma_3.$$

We restrict to local operators formed from the existing lowest-order variable algebra:

$$(P, \partial_\xi P, \Psi^\dagger \Psi, \sigma, X_i^a),$$

with only isotropic scalar contractions and without introducing any explicit tangent-frame selector.

CCLXXVII. BASIC TANGENT ALGEBRA

Define

$$Q_B = \text{span}\{e_1, e_2\}, \quad \Pi_Q = |e_1\rangle\langle e_1| + |e_2\rangle\langle e_2|.$$

For the real defect representative, we already have

$$\boxed{\Pi_Q P \Pi_Q = 0, \quad \Pi_Q (\partial_\xi P) \Pi_Q = 0.} \quad (534)$$

The first nontrivial tangent operator is

$$\boxed{D := \Pi_Q (\partial_\xi P)^2 \Pi_Q = \theta'(\xi)^2 |e_1\rangle\langle e_1|.} \quad (535)$$

Equivalently,

$$\boxed{D = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \Sigma_3).} \quad (536)$$

Thus the entire nontrivial tangent algebra generated by the existing variables begins with D , not with $\partial_\xi P$.

CCLXXVIII. POWERS OF THE BASIC TANGENT OPERATOR

Since

$$|e_1\rangle\langle e_1|$$

is a rank-one projector, one has

$$D^n = \theta'(\xi)^{2n} |e_1\rangle\langle e_1|.$$

Therefore

$$\boxed{D^n = \frac{\theta'(\xi)^{2n}}{2} (\mathbf{1}_2 + \Sigma_3), \quad n \geq 1.} \quad (537)$$

Its traceless part is always

$$\boxed{(D^n)_{\text{tr}} = \frac{\theta'(\xi)^{2n}}{2} \Sigma_3.} \quad (538)$$

Hence every traceless tangent operator generated by powers of the existing algebra is proportional to

$$\theta'(\xi)^{2n} \Sigma_3, \quad n \geq 1.$$

CCLXXIX. GENERAL HIGHER-ORDER ISOTROPIC LOCAL OPERATOR

Consider any local isotropic scalar density

$$\mathcal{O}_{\text{iso}}(P, \partial P, \Psi^\dagger \Psi, \sigma, X)$$

built polynomially from the existing variables and their contractions.

After restriction to Q_B , every contribution to its tangent Hessian is generated by products of:

$$\Pi_Q, \quad D, \quad \text{scalar functions of } (\theta, \phi_0, \sigma_B, \dots).$$

Therefore the general tangent Hessian contribution has the form

$$\boxed{R_Q^{\text{iso}} = A_0(\xi) \mathbf{1}_2 + \sum_{n \geq 1} A_n(\xi) D^n.} \quad (539)$$

Using (537),

$$\begin{aligned} R_Q^{\text{iso}} &= A_0(\xi) \mathbf{1}_2 + \sum_{n \geq 1} A_n(\xi) \frac{\theta'(\xi)^{2n}}{2} (\mathbf{1}_2 + \Sigma_3) \\ &= \left[A_0(\xi) + \frac{1}{2} \sum_{n \geq 1} A_n(\xi) \theta'(\xi)^{2n} \right] \mathbf{1}_2 + \frac{1}{2} \sum_{n \geq 1} A_n(\xi) \theta'(\xi)^{2n} \Sigma_3. \end{aligned} \quad (540)$$

Hence the traceless part is

$$\boxed{(R_Q^{\text{iso}})_{\text{tr}} = \frac{1}{2} \sum_{n \geq 1} A_n(\xi) \theta'(\xi)^{2n} \Sigma_3.} \quad (541)$$

CCLXXX. NO LINEAR BRANCH A TERM

Equation (541) shows that every traceless tangent contribution generated by the existing isotropic local algebra is an even power series in $\theta'(\xi)$:

$$\theta'(\xi)^2 \Sigma_3, \quad \theta'(\xi)^4 \Sigma_3, \quad \theta'(\xi)^6 \Sigma_3, \quad \dots$$

Therefore there is no term proportional to

$$\theta'(\xi) \Sigma_3.$$

Equivalently,

$$\boxed{\Lambda_{\parallel}^{(0)} = 0} \quad (542)$$

to all orders within the existing isotropic local variable algebra.

CCLXXXI. WORLDVOLUME TRANSPORT CONSEQUENCE

The first-order transport kernel always carries one extra odd normal factor $\theta'(\xi)$:

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

Thus the higher-order traceless tangent contributions produce

$$\theta'(\xi) \cdot \theta'(\xi)^{2n} \Sigma_3 = \theta'(\xi)^{2n+1} \Sigma_3.$$

Hence the allowed longitudinal channels are

$$\boxed{\theta'(\xi)^3 \Sigma_3, \quad \theta'(\xi)^5 \Sigma_3, \quad \theta'(\xi)^7 \Sigma_3, \quad \dots} \quad (543)$$

but never

$$\boxed{\theta'(\xi) \Sigma_3.} \quad (544)$$

Therefore Branch A remains excluded, while Branch B and its higher odd corrections survive.

CCLXXXII. HIGHER-ORDER CORRECTED LONGITUDINAL COEFFICIENT

The most general longitudinal coefficient allowed by the existing isotropic local algebra is

$$\boxed{a_3(\xi) = \frac{\lambda_c^{(1)}}{2} \theta'(\xi)^3 + \frac{\lambda_c^{(2)}}{2} \theta'(\xi)^5 + \frac{\lambda_c^{(3)}}{2} \theta'(\xi)^7 + \dots} \quad (545)$$

Hence the reduced longitudinal coefficient is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c^{(1)}}{2} I_3 + \frac{\lambda_c^{(2)}}{2} I_5 + \frac{\lambda_c^{(3)}}{2} I_7 + \dots,} \quad (546)$$

where

$$I_{2n+1} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^{2n+1}.$$

CCLXXXIII. CONCLUSION

Thus higher-order local operators built only from the existing isotropic TECT variable algebra do not reopen Branch A.

They only renormalize Branch B by adding higher odd-power primitive longitudinal terms.

Therefore:

$$\boxed{\text{Within the existing isotropic local TECT algebra, Branch B is stable to all local higher-order corrections.}} \quad (547)$$

CCLXXXIV. FINAL SUMMARY

Using only the existing TECT variables and isotropic local scalar contractions, the basic tangent algebra is generated by

$$\Pi_Q, \quad D = \Pi_Q (\partial_\xi P)^2 \Pi_Q = \frac{\theta'^2}{2} (\mathbf{1}_2 + \Sigma_3).$$

Hence every traceless tangent operator built from this algebra is of the form

$$(R_Q^{\text{iso}})_{\text{tr}} = \frac{1}{2} \sum_{n \geq 1} A_n(\xi) \theta'(\xi)^{2n} \Sigma_3.$$

Therefore no linear longitudinal selector

$$\theta'(\xi) \Sigma_3$$

can ever appear.

Equivalently,

$$\boxed{\Lambda_{\parallel}^{(0)} = 0}$$

to all orders inside the existing isotropic local variable algebra.

The transport kernel always contributes one extra odd factor $\theta'(\xi)$, so the allowed longitudinal channels are only

$$\theta'^3 \Sigma_3, \quad \theta'^5 \Sigma_3, \quad \theta'^7 \Sigma_3, \quad \dots$$

Thus higher-order local corrections do not reopen Branch A. They only renormalize Branch B.

So Branch B is stable to all local higher-order corrections built from the existing isotropic TECT variable algebra.

CCLXXXV. EXHAUSTIVE LOOPHOLE CLASSIFICATION BEYOND THE EXISTING ISOTROPIC LOCAL TECT ALGEBRA

CCLXXXVI. GOAL

We have already shown that within the existing isotropic local TECT algebra, no linear longitudinal selector

$$\theta'(\xi) \Sigma_3$$

can be generated. Hence Branch A is impossible there, while Branch B survives.

We now classify all logically possible loopholes that could evade this result.

CCLXXXVII. EXACT STATEMENT TO BE EVADED

The result to evade is:

$$\boxed{\Lambda_{\parallel}^{(0)} = 0} \tag{548}$$

for every local isotropic operator built from the existing variable algebra

$$(P, \partial P, \Psi^\dagger \Psi, \sigma, X_i^a).$$

Thus any loophole must produce a tangent Hessian contribution

$$\boxed{V_A^\mu(\xi) \supset \theta'(\xi) n^\mu \Sigma_3.} \tag{549}$$

Equivalently, it must generate a traceless tangent selector already at linear derivative order.

CCLXXXVIII. LOGICAL POSSIBILITIES

At the microscopic level there are only four logically distinct possibilities.

A. Loophole I: explicit local tangent frame selector

The full action may contain a genuinely new local operator R_Q on the tangent bundle with

$$\boxed{R_Q = r_0 \mathbf{1}_2 + r_3 \Sigma_3, \quad r_3 \neq 0.} \tag{550}$$

This is the direct Type II frame-selector loophole. If present, Branch A is immediately possible.

a. Status. This operator is not present in the existing lowest-order TECT variable algebra. Therefore Loophole I is absent in the present construction.

B. Loophole II: non-isotropic higher-derivative refinement

The full action may contain a higher-derivative refinement whose tangent reduction is not generated by the isotropic local algebra and whose first-order Hessian contributes

$$\theta'(\xi)\Sigma_3.$$

Schematically, this would mean a term

$$\boxed{\mathcal{O}_{\text{ref}} \implies \Pi_Q \delta^2 \mathcal{O}_{\text{ref}} \Pi_Q \supset \theta'(\xi) n^\mu \Sigma_3.} \quad (551)$$

a. Status. This cannot arise from isotropic local scalar contractions of the existing variable algebra, because those generate only even powers of the basic tangent projector $D \sim \theta'^2(\mathbf{1}_2 + \Sigma_3)$. Therefore Loophole II is absent unless one adds a genuinely non-isotropic refinement beyond the existing algebra.

C. Loophole III: nonlocal tangent memory operator

The full theory may contain a nonlocal operator along the normal direction ξ , for example

$$\int d\xi d\xi' K(\xi, \xi') \partial_\xi P(\xi) \partial_{\xi'} P(\xi'),$$

or an equivalent nonlocal transport kernel. Such an operator is not generated by the local algebra and may effectively define a preferred tangent frame.

In that case the tangent Hessian need not be a local polynomial in D , and a linear $\theta'(\xi)\Sigma_3$ term may appear.

a. Status. This loophole is excluded if one assumes the full effective action is local in the defect neighborhood. The current TECT construction is organized as a local effective theory, so Loophole III is absent under the locality assumption.

D. Loophole IV: spontaneous tangent nematic order

The full theory may generate an additional background expectation value N_Q in the tangent bundle such that

$$\boxed{N_Q = n_0 \mathbf{1}_2 + n_3 \Sigma_3, \quad n_3 \neq 0.} \quad (552)$$

This would spontaneously break the local tangent $SO(2)$ symmetry down to a selected frame and act exactly as a Branch A carrier.

a. Status. This requires an independent tangent-nematic condensate sector or a spontaneous instability in the full microscopic Hessian. No such sector has been identified in the present TECT construction. So Loophole IV is absent unless a new spontaneous tangent-order parameter is introduced or dynamically found.

CCLXXXIX. EXHAUSTION THEOREM

Therefore the loopholes are exhausted by the four classes above.

If the full complete TECT action satisfies:

1. locality in the defect neighborhood,
2. no explicit local tangent frame selector,
3. no genuinely non-isotropic higher-derivative refinement,
4. no spontaneous tangent nematic order,

then none of Loopholes I–IV is available.

Hence the previous no-go result extends to the full theory:

$$\boxed{\Lambda_{\parallel}^{(0)} = 0} \quad (553)$$

and Branch A is excluded.

CCXC. EXACT SURVIVING LONGITUDINAL STRUCTURE

Under these assumptions, the longitudinal transport sector is necessarily an odd-power series:

$$a_3(\xi) = \frac{\lambda_c^{(1)}}{2} \theta'(\xi)^3 + \frac{\lambda_c^{(2)}}{2} \theta'(\xi)^5 + \frac{\lambda_c^{(3)}}{2} \theta'(\xi)^7 + \dots \quad (554)$$

Therefore the reduced longitudinal coefficient is

$$\lambda_{\parallel} = \frac{\lambda_c^{(1)}}{2} I_3 + \frac{\lambda_c^{(2)}}{2} I_5 + \frac{\lambda_c^{(3)}}{2} I_7 + \dots \quad (555)$$

Thus Branch B is not merely the lowest-order outcome; it is the exact stable branch of the full theory under the assumptions above.

CCXCI. FINAL REDUCED PDE UNDER LOOPHOLE EXCLUSION

If in addition the transverse sector remains nonzero,

$$(\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0),$$

then the reduced PDE is

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - \beta(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0, \quad (556)$$

with

$$\lambda_{\parallel} = \frac{\lambda_c^{(1)}}{2} I_3 + \frac{\lambda_c^{(2)}}{2} I_5 + \dots, \quad \alpha \neq 0 \text{ or } \beta \neq 0.$$

CCXCII. SHARPEST HONEST CONCLUSION

Therefore the branch problem is now reduced to the following exact alternative:

$$\boxed{\text{Either the full complete TECT action contains one of Loopholes I–IV, in which case Branch A is possible,}} \quad (557)$$

or

$$\boxed{\text{none of them is present, in which case Branch B is exact to all local orders.}} \quad (558)$$

In the present construction, no explicit evidence for any of Loopholes I–IV has been found. Hence the strongest current microscopic conclusion is that Branch B is the natural exact branch of TECT unless genuinely new structure is added.

CCXCIII. FINAL SUMMARY

To evade the Branch B no-go, the full theory must realize at least one of the following loopholes:

$$\boxed{\text{I: explicit local tangent frame selector;}}$$

$$\boxed{\text{II: genuinely non-isotropic higher-derivative refinement;}}$$

III: nonlocal tangent memory operator;

IV: spontaneous tangent nematic order.

These four possibilities exhaust the ways to generate a linear longitudinal selector

$$\theta'(\xi)\Sigma_3.$$

If none of I–IV is present, then

$$\Lambda_{\parallel}^{(0)} = 0$$

exactly, and the longitudinal sector is an odd-power Branch B series:

$$a_3(\xi) = \frac{\lambda_c^{(1)}}{2}\theta'^3 + \frac{\lambda_c^{(2)}}{2}\theta'^5 + \frac{\lambda_c^{(3)}}{2}\theta'^7 + \dots$$

Therefore Branch B is exact to all local orders under the assumptions: locality, tangent $SO(2)$ invariance, and no new tangent-anisotropy carrier.

So the remaining microscopic question is now maximally sharp:

Does the true full complete TECT action realize any of Loopholes I–IV?

If not, then the final reduced PDE is on Branch B exactly.

CCXCIV. ELIMINATION OF LOOPHOLES I, III, IV AND REDUCTION TO LOOPHOLE II

CCXCV. GOAL

We now test the four possible loopholes that could evade the Branch B no-go:

I : explicit local tangent frame selector, II : genuinely non-isotropic higher-derivative refinement,

III : nonlocal tangent memory operator, IV : spontaneous tangent nematic order.

We show that, within the current TECT framework and assumptions already used in the construction, Loopholes I, III, and IV are excluded, so that only Loophole II remains.

CCXCVI. LOOPHOLE I: EXPLICIT LOCAL TANGENT FRAME SELECTOR

Loophole I would require an operator R_Q on the tangent bundle such that

$$R_Q = r_0 \mathbf{1}_2 + r_3 \Sigma_3, \quad r_3 \neq 0,$$

already at the microscopic local level.

However, in the present TECT variable set

$$(\Psi, P, \sigma, X_i^a),$$

there is no independent field whose background value selects e_1 over e_2 at zeroth derivative order.

Indeed:

1. P_B vanishes on Q_B :

$$\Pi_Q P_B \Pi_Q = 0;$$

2. $\Psi_B \Psi_B^\dagger \propto P_B$, so again

$$\Pi_Q(\Psi_B \Psi_B^\dagger) \Pi_Q = 0;$$

3. σ_B is a scalar, hence contributes only $\propto \mathbf{1}_2$;

4. the present X -sector has no independent tangent-frame background expectation value.

Therefore no explicit local tangent frame selector exists in the present TECT field content.
Hence:

$$\boxed{\text{Loophole I is absent.}} \tag{559}$$

CCXCVII. LOOPHOLE III: NONLOCAL TANGENT MEMORY OPERATOR

Loophole III would require a nonlocal operator along the normal direction, schematically

$$\int d\xi d\xi' K(\xi, \xi') \mathcal{O}(\xi) \mathcal{O}(\xi'),$$

whose tangent reduction effectively defines a preferred frame.

But the entire present TECT construction is organized as a local effective theory in the defect neighborhood: the action is assumed to be a local functional of the fields and finitely many derivatives, and all previous Hessian reductions, transport kernels, and low-mode projections rely on this locality.

Thus, if we keep the current local-action framework, nonlocal memory operators are excluded by construction.
Hence:

$$\boxed{\text{Loophole III is absent under the locality assumption.}} \tag{560}$$

CCXCVIII. LOOPHOLE IV: SPONTANEOUS TANGENT NEMATIC ORDER

Loophole IV would require an additional tangent order parameter

$$N_Q = n_0 \mathbf{1}_2 + n_3 \Sigma_3, \quad n_3 \neq 0,$$

generated spontaneously by the full microscopic dynamics.

For this to occur, at least one of the following must be present:

1. an independent microscopic tangent-nematic field;
2. a tachyonic instability in the symmetric-traceless tangent channel;
3. a quartic stabilization mechanism selecting a nonzero Σ_3 -expectation value.

None of these has been identified in the current TECT construction.

More importantly, the present low-energy tangent sector is organized around local $SO(2)$ -invariant quadratic kernels. Such kernels are proportional to $\mathbf{1}_2$ and do not by themselves generate a spontaneous Σ_3 condensate.

Therefore spontaneous tangent nematic order is not available in the present theory unless one enlarges the microscopic order-parameter content or explicitly demonstrates a symmetry-breaking instability.

Hence:

$$\boxed{\text{Loophole IV is absent in the current TECT framework.}} \tag{561}$$

CCXCIX. REMAINING LOOPHOLE II

The only loophole not yet excluded is Loophole II: a genuinely non-isotropic higher-derivative refinement whose tangent Hessian contributes

$$\theta'(\xi)\Sigma_3$$

without being reducible to the existing isotropic local algebra.

That is, one would need a term

$$\boxed{\mathcal{O}_{\text{ref,new}} \implies \Pi_Q \delta^2 \mathcal{O}_{\text{ref,new}} \Pi_Q \supset \theta'(\xi) n^\mu \Sigma_3.} \quad (562)$$

This cannot come from the already analyzed isotropic local algebra, but it could arise from a genuinely new refined operator not yet classified.

Therefore:

$$\boxed{\text{Loophole II is the only remaining microscopic possibility.}} \quad (563)$$

CCC. CONSEQUENCE FOR THE BRANCH PROBLEM

If Loophole II is also absent, then all loopholes I–IV are eliminated, and Branch B is exact:

$$\boxed{\Lambda_{\parallel}^{(0)} = 0 \quad \text{to all local orders,}} \quad (564)$$

with

$$\boxed{a_3(\xi) = \frac{\lambda_c^{(1)}}{2} \theta'(\xi)^3 + \frac{\lambda_c^{(2)}}{2} \theta'(\xi)^5 + \dots} \quad (565)$$

Thus the reduced PDE is exactly of Branch B type.

CCCI. SHARP FINAL REDUCTION

Hence the full branch problem is now reduced from four possibilities to one:

$$\boxed{\text{Does the true full complete TECT action contain a genuinely non-isotropic higher-derivative refinement?}} \quad (566)$$

If no, then Branch B is exact.

If yes, then one must compute its tangent Hessian projection explicitly and test whether it generates

$$\Lambda_{\parallel}^{(0)} \neq 0.$$

CCCII. FINAL SUMMARY

Within the current TECT framework:

$$\boxed{\text{Loophole I (explicit frame selector) is absent,}}$$

because no existing field in (Ψ, P, σ, X) selects e_1 over e_2 at zeroth derivative order.

$$\boxed{\text{Loophole III (nonlocal memory) is absent,}}$$

because the present TECT construction is explicitly local in the defect neighborhood.

Loophole IV (spontaneous tangent nematic) is absent,

because no independent tangent-nematic order parameter or symmetry-breaking instability has been identified in the current theory.

Therefore the only remaining loophole is

Loophole II : genuinely non-isotropic higher-derivative refinement.

So the entire branch problem is now reduced to one final microscopic question:

Does the true full complete TECT action contain a genuinely non-isotropic higher-derivative refinement that produces $\theta'(\xi)\Sigma_3$

If not, then

$$\Lambda_{\parallel}^{(0)} = 0$$

to all local orders and Branch B is exact.

CCCH. EXHAUSTION AND ELIMINATION OF LOOPHOLE II

CCCIV. PROBLEM

We now classify all possible genuinely non-isotropic higher-derivative refinements built from the existing TECT fields

$$(\Psi, P, \sigma, X_i^a)$$

and test whether any of them can generate a linear longitudinal tangent selector

$$\theta'(\xi)\Sigma_3.$$

Equivalently, we ask whether any local scalar invariant built from the existing field content can contribute to the tangent Hessian a term

$$V_{\Pi}^{\mu}(\xi) \supset \theta'(\xi) n^{\mu} \Sigma_3. \tag{567}$$

CCCV. REPRESENTATION-THEORETIC CRITERION

On the tangent bundle Q_B , the operator Σ_3 is a symmetric traceless rank-two object under the local tangent $SO(2)$. Therefore, a necessary condition for (567) is that the microscopic invariant contain, before final scalar contraction, a tangent spin-2 carrier.

Thus loophole II can exist only if the existing field algebra generates a local symmetric-traceless tangent tensor at linear derivative order.

CCCVI. AVAILABLE DERIVATIVE BUILDING BLOCKS

At lowest nontrivial orders, the independent local derivative building blocks are:

$$\partial_{\mu}P, \quad \partial_{\mu}\Psi, \quad \partial_{\mu}\sigma, \quad \partial_{\mu}X_i^a,$$

and higher derivatives thereof.

After insertion of the real defect background, their tangent restrictions behave as follows.

A. $\partial_\xi P$

We already know

$$\boxed{\Pi_Q(\partial_\xi P)\Pi_Q = 0.} \quad (568)$$

Therefore $\partial_\xi P$ itself does not furnish a tangent spin-2 carrier.

B. $\partial_\xi \Psi_B$

Since

$$\Psi_B = \phi_0 z_B, \quad \partial_\xi z_B = \theta' e_1,$$

one has

$$\partial_\xi \Psi_B = (\partial_\xi \phi_0) z_B + \phi_0 \theta' e_1.$$

But a single vector in Q_B does not yet define a symmetric traceless endomorphism on Q_B . To obtain such an endomorphism one must form a bilinear, e.g.

$$|e_1\rangle\langle e_1|.$$

Hence the first tangent spin-2 object from the condensate gradient is again quadratic in θ' .

C. $\partial_\xi \sigma_B$

The scalar σ_B has no tangent index, so no tangent spin-2 carrier can be built linearly from it.

D. $\partial_\mu X_i^a$

The field X_i^a carries a spatial index i and internal index a , but no tangent bundle endomorphism structure by itself. To generate a tangent Σ_3 selector from X , one still needs a background-dependent map from (i, a) to a symmetric-traceless endomorphism of Q_B . Within the existing isotropic algebra, that map can only come from the projector geometry, hence again from at least quadratic tangent structures.

Therefore none of the linear derivative building blocks yields a tangent spin-2 carrier at order θ' .

CCCVII. GENERAL LOCAL SCALAR INVARIANT

Any local scalar invariant built from the existing fields has the form

$$\boxed{\mathcal{O}_{\text{loc}} = \mathcal{C}(P, \Psi^\dagger \Psi, \sigma, X; \partial P, \partial \Psi, \partial \sigma, \partial X; \partial^2 P, \partial^2 \Psi, \dots),} \quad (569)$$

where all free indices are fully contracted using isotropic tensors.

To contribute to Σ_3 , the second variation of $\mathcal{O}_{\text{loc}}$ must contain a tangent symmetric-traceless carrier. But from the building-block analysis above, every such carrier made from the existing variable algebra starts only at quadratic tangent order.

Therefore the tangent Hessian contribution has the general structure

$$\boxed{(R_Q^{\text{II}})_{\text{tr}} = B_1(\xi) \theta'(\xi)^2 \Sigma_3 + B_2(\xi) \theta'(\xi)^4 \Sigma_3 + B_3(\xi) \theta'(\xi)^6 \Sigma_3 + \dots} \quad (570)$$

In particular, there is no term proportional to $\theta'(\xi) \Sigma_3$.

CCCVIII. WHY HIGHER DERIVATIVES DO NOT HELP

One may worry that higher normal derivatives such as

$$\partial_\xi^2 P, \quad \partial_\xi^2 \Psi, \quad \partial_\xi \partial_\mu P$$

could generate a linear longitudinal selector.

But this does not happen.

$$\mathbf{A.} \quad \partial_\xi^2 P$$

A direct differentiation gives

$$\partial_\xi^2 P = \theta''(e_1 z_B^\dagger + z_B e_1^\dagger) + \theta'^2(\cdots).$$

The first term has the same tangent structure as $\partial_\xi P$, hence vanishes after tangent projection:

$$\Pi_Q [\theta''(e_1 z_B^\dagger + z_B e_1^\dagger)] \Pi_Q = 0.$$

The remaining nonzero tangent part is again quadratic in θ' .

$$\mathbf{B.} \quad \partial_\xi^2 \Psi_B$$

Likewise,

$$\partial_\xi^2 \Psi_B$$

contains terms proportional to z_B , e_1 , and θ'^2 times tangent vectors. The tangent endomorphism obtained from these terms still begins only at bilinear order in tangent vectors, hence at quadratic order in θ' .

C. Mixed derivatives

Any mixed derivative term such as

$$\partial_\xi \partial_\mu P, \quad \partial_\xi \partial_\mu \Psi$$

still contains only one tangent vector before bilinearization. After scalar contraction, the first nonzero symmetric-traceless tangent operator again starts at quadratic tangent order.

Therefore higher derivatives do not evade the parity/representation obstruction.

CCCIX. NO-GO THEOREM FOR LOOPHOLE II

Hence we obtain:

No genuinely non-isotropic higher-derivative refinement built solely from the existing local TECT variable algebra can generate	(571)
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Equivalently,

$\Lambda_{\parallel}^{(0)} = 0$	(572)
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to all local orders within the existing field content and isotropic local scalar framework.

CCCX. WHAT LOOPHOLE II WOULD ACTUALLY REQUIRE

To realize loophole II for real, one would need a refinement that is not merely higher derivative, but introduces a new microscopic spin-2 tangent carrier not generated by the present local algebra.

That means a term of schematic form

$$\boxed{\mathcal{O}_{\text{new}} \implies \Pi_Q \delta^2 \mathcal{O}_{\text{new}} \Pi_Q \supset \theta'(\xi) n^\mu \Sigma_3,} \quad (573)$$

where the Σ_3 piece is already present before the usual bilinear tangent construction.

This is no longer a refinement of the existing algebra. It is effectively a new tangent-anisotropy carrier in disguise.

CCCXI. FINAL CONSEQUENCE

Therefore loophole II is also absent within the current TECT framework, provided one keeps:

1. the existing field content (Ψ, P, σ, X) ,
2. locality,
3. isotropic scalar contractions,
4. no independent tangent-anisotropy carrier.

Under these assumptions, all loopholes I–IV are absent, and Branch B is exact:

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad a_3(\xi) = \frac{\lambda_c^{(1)}}{2} \theta'^3 + \frac{\lambda_c^{(2)}}{2} \theta'^5 + \dots} \quad (574)$$

CCCXII. FINAL SUMMARY

Loophole II asked whether a genuinely non-isotropic higher-derivative refinement built from the existing TECT fields

$$(\Psi, P, \sigma, X)$$

could generate a linear longitudinal selector

$$\theta'(\xi) \Sigma_3.$$

The answer is no.

The reason is representation-theoretic: Σ_3 is a tangent spin-2 symmetric-traceless operator, while the existing local building blocks produce no tangent spin-2 carrier at linear derivative order.

The first nonzero tangent spin-2 object generated by the present variable algebra is always quadratic in the defect gradient:

$$(\Pi_Q (\partial_\xi P)^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3.$$

Therefore every higher-order local correction built from the same algebra produces only

$$\theta'^2 \Sigma_3, \quad \theta'^4 \Sigma_3, \quad \theta'^6 \Sigma_3, \quad \dots$$

inside the tangent Hessian, and hence only

$$\theta'^3 \Sigma_3, \quad \theta'^5 \Sigma_3, \quad \theta'^7 \Sigma_3, \quad \dots$$

in the transport kernel.

So no linear Branch A term ever appears:

$$\boxed{\Lambda_{\parallel}^{(0)} = 0}$$

to all local orders inside the current TECT field content and isotropic local scalar framework.

Thus loophole II is also eliminated, and Branch B becomes exact under the current assumptions.

CCCXIII. MICROSCOPIC NORMALIZATION OF THE EXACT BRANCH B REDUCED PDE

CCCXIV. EXACT BRANCH B LONGITUDINAL TOWER

Since

$$\Lambda_{\parallel}^{(0)} = 0,$$

the longitudinal coefficient is an odd-power tower:

$$a_3(\xi) = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} \theta'(\xi)^{2n+1}. \quad (575)$$

Therefore

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) a_3(\xi) = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad (576)$$

where

$$I_{2n+1} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^{2n+1}. \quad (577)$$

At lowest order,

$$\lambda_{\parallel}^{\text{LO}} = \frac{\lambda_c^{(1)}}{2} I_3. \quad (578)$$

CCCXV. TRANSVERSE SECTOR

Likewise, the transverse sector has the expansion

$$b_E(\xi) = \tilde{\Lambda}_E^{(1)} \theta'(\xi) \cos \theta(\xi) + \tilde{\Lambda}_E^{(2)} \theta'(\xi)^3 \cos \theta(\xi) + \cdots, \quad (579)$$

$$b_O(\xi) = \tilde{\Lambda}_O^{(1)} \theta'(\xi) \cos \theta(\xi) + \tilde{\Lambda}_O^{(2)} \theta'(\xi)^3 \cos \theta(\xi) + \cdots. \quad (580)$$

Hence

$$\alpha = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad \beta = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1}, \quad (581)$$

with

$$J_{2n-1} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^{2n-1} \cos \theta(\xi). \quad (582)$$

At lowest order,

$$\alpha^{\text{LO}} = \tilde{\Lambda}_E^{(1)} J_1, \quad \beta^{\text{LO}} = \tilde{\Lambda}_O^{(1)} J_1. \quad (583)$$

CCCXVI. CANONICAL REDUCED NORMALIZATION

Introduce the effective Branch B velocities

$$\boxed{v_{\parallel} := \lambda_{\parallel}, \quad v_E := \alpha, \quad v_O := \beta.} \quad (584)$$

Then the exact reduced PDE is

$$\boxed{\left[i\partial_t - v_{\parallel} \sigma_3 (-i\partial_{\parallel}) - v_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - v_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.} \quad (585)$$

At leading order this is

$$\boxed{v_{\parallel}^{\text{LO}} = \frac{\lambda_c^{(1)}}{2} I_3, \quad v_E^{\text{LO}} = \tilde{\Lambda}_E^{(1)} J_1, \quad v_O^{\text{LO}} = \tilde{\Lambda}_O^{(1)} J_1.} \quad (586)$$

CCCXVII. DIRAC CONE NORMALIZATION

Define the transverse magnitude

$$\boxed{v_{\perp} := \sqrt{v_E^2 + v_O^2}.} \quad (587)$$

Then the reduced dispersion relation is

$$\boxed{\omega(\mathbf{p})^2 = v_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 (p_1^2 + p_2^2).} \quad (588)$$

Thus the exact Branch B reduced PDE is fully normalized by the three microscopic numbers

$$\boxed{v_{\parallel}, \quad v_E, \quad v_O,} \quad (589)$$

or equivalently

$$\boxed{v_{\parallel}, \quad v_{\perp}, \quad \varphi := \arg(v_E + i v_O).} \quad (590)$$

CCCXVIII. FIELD REDEFINITION TO CANONICAL TRANSVERSE FORM

If $v_{\perp} \neq 0$, define

$$\boxed{\cos \varphi = \frac{v_E}{v_{\perp}}, \quad \sin \varphi = \frac{v_O}{v_{\perp}}.} \quad (591)$$

Then after a constant spinor rotation in the (σ_1, σ_2) -plane, the PDE becomes

$$\boxed{\left[i\partial_t - v_{\parallel} \sigma_3 (-i\partial_{\parallel}) - v_{\perp} (\sigma_1 (-i\partial'_1) + \sigma_2 (-i\partial'_2)) \right] \chi' = 0,} \quad (592)$$

where (p'_1, p'_2) is the rotated transverse momentum basis.

Thus v_O does not introduce a new invariant scale; it only rotates the transverse frame. The invariant data are $v_{\parallel}$ and $v_{\perp}$.

CCCXIX. FINAL EXACT BRANCH B STATEMENT

Therefore the exact reduced Branch B PDE of TECT is

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0, \quad (593)$$

with

$$v_{\parallel} = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad v_E = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad v_O = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1}. \quad (594)$$

Its exact nondegeneracy condition is

$$v_{\parallel} \neq 0, \quad v_{\perp} = \sqrt{v_E^2 + v_O^2} \neq 0. \quad (595)$$

CCCXX. FINAL SUMMARY

Once Branch A is excluded, the reduced PDE is exactly of Branch B type.
The longitudinal coefficient is an odd tower:

$$v_{\parallel} = \lambda_{\parallel} = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad I_{2n+1} = \int d\xi f_+ f_- \theta'^{2n+1}.$$

The transverse coefficients are

$$v_E = \alpha = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad v_O = \beta = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1},$$

with

$$J_{2n-1} = \int d\xi f_+ f_- \theta'^{2n-1} \cos \theta.$$

Therefore the exact Branch B reduced PDE is

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

Its invariant physical normalization is determined by

$$v_{\parallel}, \quad v_{\perp} = \sqrt{v_E^2 + v_O^2},$$

since v_O can be absorbed into a transverse frame rotation.

Thus the only remaining microscopic task is the quantitative evaluation of the coefficients

$$\lambda_c^{(n)}, \quad \tilde{\Lambda}_E^{(n)}, \quad \tilde{\Lambda}_O^{(n)}.$$

CCCXXI. DIMENSIONLESS NORMALIZATION AND ROBUSTNESS OF THE EXACT BRANCH B PDE

CCCXXII. EXACT BRANCH B REDUCED PDE

The exact reduced Branch B PDE is

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (596)$$

Define

$$v_{\perp} := \sqrt{v_E^2 + v_O^2}. \quad (597)$$

After a constant transverse frame rotation, (596) is equivalent to

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_{\perp} (\sigma_1(-i\partial'_1) + \sigma_2(-i\partial'_2)) \right] \chi' = 0. \quad (598)$$

Thus the invariant physical data are $v_{\parallel}$ and $v_{\perp}$.

CCCXXIII. CHOICE OF REFERENCE SCALE

Introduce a positive reference velocity $v_* > 0$. A natural choice is

$$v_* := v_{\perp} \quad (599)$$

or, alternatively,

$$v_* := \sqrt{v_{\parallel}^2 + v_{\perp}^2}. \quad (600)$$

Using $v_* = v_{\perp}$, define the dimensionless anisotropy ratio

$$\rho := \frac{v_{\parallel}}{v_{\perp}}. \quad (601)$$

Then (598) becomes

$$\left[i\partial_{\tilde{t}} - \rho \sigma_3(-i\partial_{\parallel}) - (\sigma_1(-i\partial_{\tilde{1}}) + \sigma_2(-i\partial_{\tilde{2}})) \right] \chi' = 0, \quad (602)$$

where

$$\tilde{t} = v_{\perp} t, \quad \tilde{x}_{\parallel} = v_{\perp} x_{\parallel}, \quad \tilde{x}_1 = v_{\perp} x'_1, \quad \tilde{x}_2 = v_{\perp} x'_2.$$

Therefore the exact Branch B PDE is classified, up to rescaling and transverse rotation, by the single dimensionless parameter

$$\rho = \frac{v_{\parallel}}{v_{\perp}}. \quad (603)$$

CCCXXIV. MICROSCOPIC EXPRESSION FOR ρ

Recall

$$v_{\parallel} = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad v_E = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad v_O = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1}. \quad (604)$$

Hence

$$\rho = \frac{\sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}}{\sqrt{\left(\sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1} \right)^2 + \left(\sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1} \right)^2}}. \quad (605)$$

At leading order,

$$\rho_{\text{LO}} = \frac{\frac{\lambda_c^{(1)}}{2} I_3}{J_1 \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}}. \quad (606)$$

CCCXXV. DIMENSIONLESS ROBUSTNESS PARAMETERS

Let

$$\delta v_{\parallel}, \quad \delta v_E, \quad \delta v_O$$

denote corrections from higher-order operators, truncation error, or microscopic uncertainty.

Define the relative error parameters

$$\varepsilon_{\parallel} := \frac{|\delta v_{\parallel}|}{|v_{\parallel}|}, \quad \varepsilon_{\perp} := \frac{\sqrt{(\delta v_E)^2 + (\delta v_O)^2}}{v_{\perp}}. \quad (607)$$

Then the reduced PDE remains Dirac-nondegenerate provided

$$\varepsilon_{\parallel} < 1, \quad \varepsilon_{\perp} < 1. \quad (608)$$

Indeed, if $\varepsilon_{\parallel} < 1$, then

$$|v_{\parallel} + \delta v_{\parallel}| \geq |v_{\parallel}| - |\delta v_{\parallel}| = |v_{\parallel}|(1 - \varepsilon_{\parallel}) > 0.$$

Likewise, if $\varepsilon_{\perp} < 1$, then

$$v_{\perp}^{\text{new}} \geq v_{\perp} - \sqrt{(\delta v_E)^2 + (\delta v_O)^2} = v_{\perp}(1 - \varepsilon_{\perp}) > 0.$$

Thus the nondegeneracy condition is stable.

CCCXXVI. ROBUSTNESS IN MICROSCOPIC FORM

Split the coefficients into leading and higher-order parts:

$$v_{\parallel} = v_{\parallel}^{\text{LO}} + \Delta v_{\parallel}, \quad v_E = v_E^{\text{LO}} + \Delta v_E, \quad v_O = v_O^{\text{LO}} + \Delta v_O.$$

Then a sufficient robustness criterion is

$$|\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|, \quad \sqrt{(\Delta v_E)^2 + (\Delta v_O)^2} < \sqrt{(v_E^{\text{LO}})^2 + (v_O^{\text{LO}})^2}. \quad (609)$$

In terms of the microscopic series, this becomes

$$\left| \sum_{n \geq 2} \frac{\lambda_c^{(n)}}{2} I_{2n+1} \right| < \left| \frac{\lambda_c^{(1)}}{2} I_3 \right|, \quad (610)$$

and

$$\sqrt{\left(\sum_{n \geq 2} \tilde{\Lambda}_E^{(n)} J_{2n-1} \right)^2 + \left(\sum_{n \geq 2} \tilde{\Lambda}_O^{(n)} J_{2n-1} \right)^2} < J_1 \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}. \quad (611)$$

CCCXXVII. UNIVERSALITY CLASSES

The exact Branch B reduced PDE is naturally organized into the following universality classes.

a. Class I: isotropic Dirac cone.

$$\rho = 1. \quad (612)$$

b. *Class II: anisotropic but nondegenerate Dirac cone.*

$$\boxed{0 < |\rho| < \infty, \quad \rho \neq 1.} \quad (613)$$

c. *Class III: longitudinally collapsed limit.*

$$\boxed{\rho = 0.} \quad (614)$$

This is excluded for a genuine Branch B Dirac phase.

d. *Class IV: transversely collapsed limit.*

$$\boxed{v_{\perp} = 0.} \quad (615)$$

This is excluded by transverse nondegeneracy.

Therefore the exact Branch B Dirac phase is precisely

$$\boxed{\rho \in \mathbb{R} \setminus \{0\}, \quad v_{\perp} > 0.} \quad (616)$$

CCCXXVIII. FINAL EXACT BRANCH B STATEMENT

Thus the exact Branch B reduced PDE is completely characterized by:

$$\boxed{\text{one dimensionless anisotropy parameter } \rho = \frac{v_{\parallel}}{v_{\perp}},} \quad (617)$$

together with

$$\boxed{\text{the robustness conditions } \varepsilon_{\parallel} < 1, \quad \varepsilon_{\perp} < 1.} \quad (618)$$

Under these conditions, the TECT reduced Branch B phase is stable under all local higher-order corrections from the existing isotropic local variable algebra.

CCCXXIX. FINAL SUMMARY

After excluding Branch A, the exact reduced PDE is of Branch B type and is fully characterized by

$$v_{\parallel}, \quad v_E, \quad v_O.$$

Up to transverse frame rotation, the invariant physical data reduce to

$$v_{\parallel}, \quad v_{\perp} = \sqrt{v_E^2 + v_O^2}.$$

Therefore, after rescaling, the exact Branch B PDE is classified by the single dimensionless anisotropy parameter

$$\boxed{\rho = \frac{v_{\parallel}}{v_{\perp}}.}$$

Microscopically,

$$v_{\parallel} = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad v_E = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad v_O = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1}.$$

A sufficient robustness criterion is

$$\boxed{\varepsilon_{\parallel} < 1, \quad \varepsilon_{\perp} < 1,}$$

or equivalently

$$|\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|, \quad \sqrt{(\Delta v_E)^2 + (\Delta v_O)^2} < \sqrt{(v_E^{\text{LO}})^2 + (v_O^{\text{LO}})^2}.$$

Thus the exact TECT Branch B phase is stable and universal up to the single anisotropy ratio ρ .

CCCXXX. LEADING ESTIMATE OF THE ANISOTROPY RATIO

Using the explicit concrete Class II coefficients,

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa,$$

we obtain

$$v_{\perp} = \sqrt{\alpha^2 + \beta^2} = \frac{\sqrt{3}\pi}{16} |\theta_0 \kappa| \sqrt{g_E^2 + g_O^2}.$$

Hence the leading anisotropy ratio is

$$\rho_{\text{LO}} = \frac{\lambda_{\parallel}}{v_{\perp}} = -\frac{3}{32} \frac{g_c}{\sqrt{g_E^2 + g_O^2}} \frac{\theta_0^3 \kappa^3}{|\theta_0 \kappa|}.$$

Equivalently, in magnitude,

$$|\rho_{\text{LO}}| = \frac{3}{32} \frac{|g_c|}{\sqrt{g_E^2 + g_O^2}} (\theta_0 \kappa)^2.$$

CCCXXXI. PHYSICAL REGIMES OF THE ANISOTROPY RATIO ρ

CCCXXXII. DEFINITION

The exact Branch B reduced PDE is characterized by

$$v_{\parallel}, \quad v_{\perp} = \sqrt{v_E^2 + v_O^2}, \quad \rho := \frac{v_{\parallel}}{v_{\perp}}.$$

At leading order in the explicit concrete Class II realization,

$$|\rho_{\text{LO}}| = \frac{3}{32} \frac{|g_c|}{\sqrt{g_E^2 + g_O^2}} (\theta_0 \kappa)^2. \tag{619}$$

Thus the anisotropy is controlled by two factors:

1. the microscopic coupling ratio

$$\frac{|g_c|}{\sqrt{g_E^2 + g_O^2}},$$

2. the geometric small parameter

$$(\theta_0 \kappa)^2.$$

CCCXXXIII. REGIME I: STRONGLY TRANSVERSE-DOMINATED CONE

If

$$|\rho| \ll 1, \tag{620}$$

then

$$|v_{\parallel}| \ll v_{\perp}.$$

The dispersion is

$$\omega(\mathbf{p})^2 = v_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 (p_1^2 + p_2^2),$$

so longitudinal propagation is parametrically slower than transverse propagation.

This means:

1. the cone is highly anisotropic;
2. the defect supports almost two-dimensional transverse Dirac propagation;
3. longitudinal motion is a weak primitive correction.

In the explicit Branch B realization, this is the generic regime whenever

$$\frac{|g_c|}{\sqrt{g_E^2 + g_O^2}} \not\gg (\theta_0 \kappa)^{-2}.$$

CCCXXXIV. REGIME II: BALANCED ANISOTROPIC DIRAC CONE

If

$$\boxed{|\rho| \sim 1}, \quad (621)$$

then longitudinal and transverse velocities are of the same order:

$$|v_{\parallel}| \sim v_{\perp}.$$

The cone is still generally anisotropic, but not parametrically so. This is the regime in which the reduced PDE most closely resembles a conventional three-directional Dirac operator with finite anisotropy.

To obtain this in Branch B, one needs

$$\boxed{\frac{|g_c|}{\sqrt{g_E^2 + g_O^2}} \sim (\theta_0 \kappa)^{-2}}. \quad (622)$$

So the primitive longitudinal channel must be strongly enhanced relative to the transverse channel.

CCCXXXV. REGIME III: LONGITUDINALLY DOMINATED CONE

If

$$\boxed{|\rho| \gg 1}, \quad (623)$$

then

$$|v_{\parallel}| \gg v_{\perp}.$$

The cone is now elongated in the transverse directions and dominated by the longitudinal channel.

Inside the present explicit Branch B realization, this regime is unnatural unless the microscopic ratio

$$\frac{|g_c|}{\sqrt{g_E^2 + g_O^2}}$$

is extraordinarily large, since the geometric suppression factor $(\theta_0 \kappa)^2$ works against the longitudinal channel.

Therefore $|\rho| \gg 1$ is not the natural expectation for the concrete Class II Branch B phase.

CCCXXXVI. NATURAL TECT EXPECTATION

Combining the explicit coefficient hierarchy with (619), the natural expectation in the present TECT Branch B construction is

$$\boxed{|\rho| \ll 1}. \quad (624)$$

That is:

$$\boxed{\text{the natural Branch B phase is a strongly anisotropic Dirac cone with parametrically suppressed longitudinal velocity.}} \quad (625)$$

CCCXXXVII. PHYSICAL INTERPRETATION

Hence the three regimes have the following meanings:

- a. Regime I:* $\rho \ll 1$. Transverse-dominated Dirac propagation. This is the natural explicit Branch B regime.
- b. Regime II:* $\rho \sim 1$. Balanced anisotropic Dirac cone. This requires strong enhancement of the primitive longitudinal carrier.
- c. Regime III:* $\rho \gg 1$. Longitudinally dominated propagation. This is highly non-generic in the present Branch B construction.

CCCXXXVIII. ROBUSTNESS VIEW

Since the Branch B longitudinal sector starts at primitive order while the transverse sector starts at lower order, small corrections generically preserve the hierarchy

$$|v_{\parallel}| < v_{\perp}.$$

Thus the $\rho \ll 1$ regime is not only natural but also robust against moderate higher-order corrections.

CCCXXXIX. FINAL CONCISE CONCLUSION

Therefore the present explicit TECT Branch B construction predicts:

$$\boxed{\text{generic phase: } \rho \ll 1,} \tag{626}$$

$$\boxed{\text{fine-tuned phase: } \rho \sim 1,} \tag{627}$$

$$\boxed{\text{unnatural phase: } \rho \gg 1.} \tag{628}$$

CCCXL. FINAL SUMMARY

The anisotropy ratio is

$$\rho = \frac{v_{\parallel}}{v_{\perp}}, \quad v_{\perp} = \sqrt{v_E^2 + v_O^2}.$$

In the explicit Branch B realization,

$$|\rho_{\text{LO}}| = \frac{3}{32} \frac{|g_c|}{\sqrt{g_E^2 + g_O^2}} (\theta_0 \kappa)^2.$$

Therefore:

$$\boxed{\rho \ll 1}$$

means a strongly transverse-dominated Dirac cone and is the natural generic regime;

$$\boxed{\rho \sim 1}$$

means a balanced anisotropic Dirac cone and requires enhancement of the primitive longitudinal channel;

$$\boxed{\rho \gg 1}$$

means a longitudinally dominated cone and is non-generic in the present Branch B construction.

Hence the natural TECT expectation is

$$\boxed{\rho \ll 1.}$$

So the present explicit Branch B phase predicts a parametrically suppressed longitudinal velocity relative to the transverse Dirac velocity.

CCCXLI. THREE PHYSICAL PREDICTIONS OF THE EXACT BRANCH B PHASE

CCCXLII. STARTING POINT

The exact reduced Branch B PDE is

$$\boxed{\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.} \quad (629)$$

After a constant transverse frame rotation, this is equivalent to

$$\boxed{\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_{\perp}(\sigma_1(-i\partial'_1) + \sigma_2(-i\partial'_2)) \right] \chi' = 0,} \quad (630)$$

with

$$\boxed{v_{\perp} = \sqrt{v_E^2 + v_O^2}, \quad \rho = \frac{v_{\parallel}}{v_{\perp}}.} \quad (631)$$

The dispersion relation is

$$\boxed{\omega(\mathbf{p})^2 = v_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 (p_1'^2 + p_2'^2).} \quad (632)$$

CCCXLIII. PREDICTION 1: STRONGLY ANISOTROPIC DIRAC CONE

The first direct prediction is the shape of the low-energy cone.

Constant-energy surfaces satisfy

$$v_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 p_{\perp}^2 = \omega^2,$$

hence the momentum-space semi-axes are

$$\boxed{p_{\parallel}^{\max} = \frac{\omega}{|v_{\parallel}|}, \quad p_{\perp}^{\max} = \frac{\omega}{v_{\perp}}.} \quad (633)$$

Therefore the cone anisotropy is

$$\boxed{\frac{p_{\parallel}^{\max}}{p_{\perp}^{\max}} = \frac{v_{\perp}}{|v_{\parallel}|} = \frac{1}{|\rho|}.} \quad (634)$$

In the natural TECT Branch B regime $|\rho| \ll 1$, the cone is highly elongated in the longitudinal momentum direction and strongly compressed in velocity space along the longitudinal direction.

Thus the first prediction is:

$$\boxed{\text{The low-energy Dirac cone is strongly anisotropic, with aspect ratio } \sim |\rho|^{-1} \gg 1.} \quad (635)$$

CCCXLIV. PREDICTION 2: QUASI-TWO-DIMENSIONAL LOW-ENERGY TRANSPORT

The group velocity is

$$\mathbf{v}_g = \nabla_{\mathbf{p}} \omega = \frac{1}{\omega} \left(v_{\parallel}^2 p_{\parallel}, v_{\perp}^2 p'_1, v_{\perp}^2 p'_2 \right). \quad (636)$$

Hence the characteristic velocity scales are

$$|v_{g,\parallel}| \sim |v_{\parallel}|, \quad |v_{g,\perp}| \sim v_{\perp}.$$

If $|\rho| \ll 1$, then

$$|v_{g,\parallel}| \ll |v_{g,\perp}|. \quad (637)$$

Therefore low-energy propagation is effectively quasi-two-dimensional: modes spread rapidly in the transverse defect plane but only weakly along the longitudinal direction.

Thus the second prediction is:

$$\boxed{\text{Low-energy transport is quasi-two-dimensional, with parametrically suppressed longitudinal propagation.}} \quad (638)$$

CCCXLV. PREDICTION 3: ROBUST UNIVERSALITY CONTROLLED BY A SINGLE ANISOTROPY PARAMETER

Up to transverse rotation and overall rescaling, the exact reduced PDE is characterized by the single dimensionless parameter

$$\rho = \frac{v_{\parallel}}{v_{\perp}}.$$

Therefore all low-energy observables that depend only on the local cone geometry must collapse onto a one-parameter family.

For example:

- cone aspect ratio;
- directional group velocity ratio;
- directional response anisotropy;
- scaling of longitudinal versus transverse spectral broadening.

All such quantities are functions only of ρ .

Thus the third prediction is:

$$\boxed{\text{The entire low-energy Branch B phenomenology is universal up to one dimensionless parameter } \rho.} \quad (639)$$

CCCXLVI. LEADING MICROSCOPIC ESTIMATE

Using the explicit concrete Class II leading coefficients,

$$|\rho_{\text{LO}}| = \frac{3}{32} \frac{|g_c|}{\sqrt{g_E^2 + g_O^2}} (\theta_0 \kappa)^2. \quad (640)$$

Hence, unless the primitive longitudinal coupling is strongly enhanced, the natural expectation is

$$|\rho| \ll 1. \quad (641)$$

So all three predictions above are expected to appear in the generic explicit Branch B regime.

CCCXLVII. COMPRESSED FINAL STATEMENT

Therefore the exact Branch B phase predicts:

$$\boxed{\text{(i) a highly anisotropic Dirac cone,}} \quad (642)$$

$$\boxed{\text{(ii) quasi-two-dimensional low-energy transport,}} \quad (643)$$

and

$$\boxed{\text{(iii) universality controlled by the single anisotropy ratio } \rho.} \quad (644)$$

CCCXLVIII. FINAL SUMMARY

The exact Branch B reduced PDE implies three direct physical predictions.

First, the dispersion

$$\omega^2 = v_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 p_{\perp}^2$$

gives a strongly anisotropic Dirac cone with aspect ratio

$$\frac{p_{\parallel}^{\max}}{p_{\perp}^{\max}} = \frac{1}{|\rho|}, \quad \rho = \frac{v_{\parallel}}{v_{\perp}}.$$

Second, the group velocities satisfy

$$|v_{g,\parallel}| \sim |v_{\parallel}|, \quad |v_{g,\perp}| \sim v_{\perp},$$

so for the natural regime $|\rho| \ll 1$, low-energy transport is quasi-two-dimensional with suppressed longitudinal propagation.

Third, after transverse rotation and rescaling, the entire low-energy phenomenology depends only on the single dimensionless anisotropy parameter

$$\rho = \frac{v_{\parallel}}{v_{\perp}}.$$

Using the explicit concrete Class II leading coefficients,

$$|\rho_{\text{LO}}| = \frac{3}{32} \frac{|g_c|}{\sqrt{g_E^2 + g_O^2}} (\theta_0 \kappa)^2,$$

so the natural expectation is

$$|\rho| \ll 1.$$

Therefore the generic explicit Branch B phase predicts:

$$\boxed{\text{a highly anisotropic Dirac cone,}}$$

$$\boxed{\text{quasi-two-dimensional low-energy transport,}}$$

$$\boxed{\text{and one-parameter universality controlled by } \rho.}$$

CCCXLIX. ABSTRACT-LEVEL STATEMENT

We derive the exact reduced infrared PDE of the TECT defect sector and show that, within the existing local isotropic TECT field algebra, the longitudinal Branch A channel is excluded to all local orders. As a result, the reduced low-energy theory is forced onto an exact Branch B Dirac phase.

The reduced first-order PDE takes the form

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0,$$

which is equivalent, up to a transverse frame rotation, to an anisotropic Dirac equation with longitudinal velocity $v_{\parallel}$ and transverse velocity

$$v_{\perp} = \sqrt{v_E^2 + v_O^2}.$$

The entire low-energy phenomenology is therefore controlled by the single dimensionless anisotropy ratio

$$\rho = \frac{v_{\parallel}}{v_{\perp}}.$$

In the explicit concrete Class II realization, the natural regime is

$$|\rho| \ll 1,$$

predicting a strongly anisotropic Dirac cone, quasi-two-dimensional low-energy transport, and one-parameter universality of the infrared response.

CCCL. CONCLUSION-LEVEL STATEMENT

The main structural problem is solved.

Inside the existing local isotropic TECT variable algebra, no local operator can generate a linear longitudinal tangent selector of the form

$$\theta'(\xi)\Sigma_3.$$

Hence the constant-leading longitudinal coefficient vanishes identically:

$$\Lambda_{\parallel}^{(0)} = 0.$$

Therefore the full reduced infrared theory is forced onto Branch B, with longitudinal transport arising only through the primitive odd tower

$$a_3(\xi) = \frac{\lambda_c^{(1)}}{2} \theta'(\xi)^3 + \frac{\lambda_c^{(2)}}{2} \theta'(\xi)^5 + \dots.$$

The resulting exact reduced PDE is

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0,$$

with invariant data

$$v_{\parallel}, \quad v_{\perp} = \sqrt{v_E^2 + v_O^2}, \quad \rho = \frac{v_{\parallel}}{v_{\perp}}.$$

Thus the exact TECT Branch B phase predicts:

- (i) a strongly anisotropic Dirac cone,
- (ii) quasi-two-dimensional low-energy transport,
- (iii) universality controlled by the single anisotropy ratio ρ .

The only remaining microscopic task is quantitative rather than structural: to compute the coefficient tower

$$\lambda_c^{(n)}, \quad \tilde{\Lambda}_E^{(n)}, \quad \tilde{\Lambda}_O^{(n)}$$

from a chosen full-action truncation and thereby determine the numerical value of ρ .

CCCLI. ULTRA-SHORT FINAL CLAIM

Within the existing local isotropic TECT algebra, Branch A is excluded and Branch B is exact. The reduced defect theory is therefore an anisotropic Dirac phase,

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0,$$

whose infrared phenomenology is fully controlled by the single dimensionless ratio

$$\rho = \frac{v_{\parallel}}{\sqrt{v_E^2 + v_O^2}}.$$

The natural explicit regime is $|\rho| \ll 1$, implying a strongly anisotropic, quasi-two-dimensional Dirac cone.

CCCLII. NEXT THEOREM: NONDEGENERATE STABILITY OF THE EXACT BRANCH B PHASE

CCCLIII. STATEMENT OF THE NEXT PROBLEM

The branch problem has been reduced away: within the existing local isotropic TECT algebra, the reduced infrared theory is on Branch B.

Therefore the next theorem is no longer about branch selection. It is about *nondegeneracy and stability* of the exact Branch B reduced PDE.

The reduced PDE is

$$\boxed{\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.} \quad (645)$$

The next theorem must prove

$$\boxed{v_{\parallel} \neq 0, \quad v_{\perp} := \sqrt{v_E^2 + v_O^2} \neq 0,} \quad (646)$$

with a quantitative robustness bound.

CCCLIV. MICROSCOPIC COEFFICIENT EXPANSIONS

We already have the exact Branch B expansions

$$\boxed{v_{\parallel} = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1},} \quad (647)$$

$$\boxed{v_E = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad v_O = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1},} \quad (648)$$

where

$$\boxed{I_{2n+1} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^{2n+1}, \quad J_{2n-1} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^{2n-1} \cos \theta(\xi).} \quad (649)$$

Thus the next proof splits into two parts:

1. prove the relevant overlaps are nonzero;
2. prove higher-order corrections cannot cancel the leading terms without fine-tuning.

CCCLV. PART I: OVERLAP NONVANISHING THEOREM

The first target is:

$$\boxed{I_3 \neq 0, \quad J_1 \neq 0} \quad (650)$$

for the admissible defect class.

A sufficient admissible defect class is:

1. $\theta(\xi)$ is monotone;
2. $\theta'(\xi)$ has fixed sign and decays sufficiently fast;
3. $f_+(\xi)f_-(\xi)$ is not identically zero and does not oscillate strongly enough to force cancellation.

Under these assumptions:

- a. Longitudinal overlap.* Since $2n + 1$ is odd and $\theta'(\xi)$ has fixed sign,

$$\theta'(\xi)^{2n+1}$$

has fixed sign as well. Therefore, if f_+f_- is nonnegative on a set of nonzero measure, then

$$\boxed{I_{2n+1} \neq 0 \quad \text{for all } n \geq 1.} \quad (651)$$

In particular,

$$\boxed{I_3 \neq 0.} \quad (652)$$

b. Transverse overlap. If in addition $\cos \theta(\xi)$ does not change sign on the defect support, or changes sign only outside the dominant support of $f_+f_- \theta'$, then

$$\boxed{J_{2n-1} \neq 0 \quad \text{for all relevant } n.} \quad (653)$$

In particular,

$$\boxed{J_1 \neq 0.} \quad (654)$$

So the next proof begins by establishing a precise admissible defect class for which (652) and (654) hold.

CCCLVI. PART II: DOMINANCE OF THE LEADING ORDER

Write

$$v_{\parallel} = v_{\parallel}^{\text{LO}} + \Delta v_{\parallel}, \quad v_E = v_E^{\text{LO}} + \Delta v_E, \quad v_O = v_O^{\text{LO}} + \Delta v_O, \quad (655)$$

with

$$\boxed{v_{\parallel}^{\text{LO}} = \frac{\lambda_c^{(1)}}{2} I_3, \quad v_E^{\text{LO}} = \tilde{\Lambda}_E^{(1)} J_1, \quad v_O^{\text{LO}} = \tilde{\Lambda}_O^{(1)} J_1.} \quad (656)$$

The second target theorem is the sufficient robustness bound

$$\boxed{|\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|, \quad \sqrt{(\Delta v_E)^2 + (\Delta v_O)^2} < \sqrt{(v_E^{\text{LO}})^2 + (v_O^{\text{LO}})^2}.} \quad (657)$$

Under (657),

$$\boxed{v_{\parallel} \neq 0, \quad v_{\perp} \neq 0.} \quad (658)$$

Indeed,

$$|v_{\parallel}| \geq |v_{\parallel}^{\text{LO}}| - |\Delta v_{\parallel}| > 0,$$

and similarly

$$v_{\perp} \geq \sqrt{(v_E^{\text{LO}})^2 + (v_O^{\text{LO}})^2} - \sqrt{(\Delta v_E)^2 + (\Delta v_O)^2} > 0.$$

CCCLVII. THE NEXT THEOREM TO PROVE

Combining the two parts, the next theorem should be stated as follows.

a. Branch B nondegenerate stability theorem. Assume:

1. the full theory stays within the existing local isotropic TECT algebra, so Branch B is exact;
2. the defect profile belongs to an admissible class such that

$$I_3 \neq 0, \quad J_1 \neq 0;$$

3. the leading coefficients satisfy

$$\lambda_c^{(1)} \neq 0, \quad (\tilde{\Lambda}_E^{(1)}, \tilde{\Lambda}_O^{(1)}) \neq (0, 0);$$

4. higher-order corrections obey the robustness bound (657).

Then the reduced Branch B PDE is Dirac-nondegenerate:

$$\boxed{v_{\parallel} \neq 0, \quad v_{\perp} \neq 0,} \tag{659}$$

and hence defines a stable anisotropic Dirac phase.

CCCLVIII. WHY THIS IS THE RIGHT NEXT STEP

At this point the structure theorem is already done: the branch is fixed and the exact PDE form is known.

So the next real proof is not another reduction theorem. It is the first theorem that converts the formal reduced PDE into a genuine stable phase statement.

Therefore the next proof target is exactly:

$$\boxed{\text{prove nonvanishing of } I_3, \ J_1, \ \lambda_c^{(1)}, \ \text{and } \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}, \text{ together with a dominance bound on higher orders.}} \tag{660}$$

CCCLIX. FINAL SUMMARY

The next theorem is not about branch selection anymore.

It is the *nondegenerate stability theorem* for the exact Branch B reduced PDE.

The exact reduced PDE is already known:

$$\left[i\partial_t - v_{\parallel} \sigma_3 (-i\partial_{\parallel}) - v_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - v_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.$$

So the next proof must establish

$$v_{\parallel} \neq 0, \quad v_{\perp} = \sqrt{v_E^2 + v_O^2} \neq 0.$$

This splits into:

$$(i) \text{ overlap nonvanishing: } I_3 \neq 0, \ J_1 \neq 0,$$

and

$$(ii) \text{ leading-order dominance: } |\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|, \quad \sqrt{(\Delta v_E)^2 + (\Delta v_O)^2} < \sqrt{(v_E^{\text{LO}})^2 + (v_O^{\text{LO}})^2}.$$

Under these conditions, the Branch B phase is not merely formal but a genuinely stable anisotropic Dirac phase.

CCCLX. OVERLAP NONVANISHING THEOREM FOR THE EXACT BRANCH B PHASE

CCCLXI. GOAL

We prove the first part of the Branch B nondegenerate stability theorem: the leading longitudinal and transverse overlap integrals do not vanish for an admissible defect class.

The relevant leading overlaps are

$$I_3 = \int_{\mathbb{R}} d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3, \quad (661)$$

$$J_1 = \int_{\mathbb{R}} d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi). \quad (662)$$

CCCLXII. ADMISSIBLE DEFECT CLASS

We assume the following admissible defect conditions.

a. (A1) *Monotone defect profile.*

$$\theta'(\xi) \text{ has fixed sign on } \mathbb{R}, \quad \theta'(\xi) \not\equiv 0. \quad (663)$$

b. (A2) *Nonnegative low-mode overlap.*

$$w(\xi) := f_+(\xi) f_-(\xi) \geq 0 \quad \text{for a.e. } \xi, \quad w \not\equiv 0. \quad (664)$$

c. (A3) *No sign flip of $\cos \theta$ on the effective support.* There exists a measurable set $E \subset \mathbb{R}$ of positive measure such that

$$w(\xi) > 0, \quad |\theta'(\xi)| > 0, \quad \cos \theta(\xi) \geq c_0 > 0 \quad \text{for a.e. } \xi \in E. \quad (665)$$

This is the natural small-amplitude defect regime. A sufficient concrete condition is

$$|\theta(\xi)| < \frac{\pi}{2} \quad \text{on the support of } w(\xi) |\theta'(\xi)|. \quad (666)$$

CCCLXIII. NONVANISHING OF I_3

Assume first that $\theta'(\xi) \geq 0$ for a.e. ξ . Then by (664),

$$w(\xi) \theta'(\xi)^3 \geq 0 \quad \text{for a.e. } \xi.$$

Moreover, by (663) and (664), there exists a measurable set E_3 of positive measure such that

$$w(\xi) \theta'(\xi)^3 > 0 \quad \text{for a.e. } \xi \in E_3.$$

Therefore

$$I_3 > 0. \quad (667)$$

If instead $\theta'(\xi) \leq 0$ for a.e. ξ , then

$$w(\xi) \theta'(\xi)^3 \leq 0 \quad \text{for a.e. } \xi,$$

and is strictly negative on a set of positive measure. Hence

$$I_3 < 0. \quad (668)$$

Thus in either case,

$$I_3 \neq 0. \quad (669)$$

CCCLXIV. NONVANISHING OF J_1

Again assume first that $\theta'(\xi) \geq 0$ for a.e. ξ . Then on the set E from (665),

$$w(\xi)\theta'(\xi)\cos\theta(\xi) \geq c_0 w(\xi)\theta'(\xi) > 0.$$

Hence

$$w(\xi)\theta'(\xi)\cos\theta(\xi) \geq 0 \quad \text{a.e.,}$$

and is strictly positive on a set of positive measure. Therefore

$$\boxed{J_1 > 0.} \tag{670}$$

If instead $\theta'(\xi) \leq 0$ for a.e. ξ , then on the same set E ,

$$w(\xi)\theta'(\xi)\cos\theta(\xi) \leq -c_0 w(\xi)|\theta'(\xi)| < 0,$$

so

$$\boxed{J_1 < 0.} \tag{671}$$

Thus in all cases,

$$\boxed{J_1 \neq 0.} \tag{672}$$

CCCLXV. UNIFORM LOWER BOUNDS

The previous argument gives quantitative lower bounds.

If $\theta'(\xi) \geq 0$, then

$$\boxed{I_3 \geq \int_E d\xi w(\xi)\theta'(\xi)^3 > 0,} \tag{673}$$

and

$$\boxed{J_1 \geq c_0 \int_E d\xi w(\xi)\theta'(\xi) > 0.} \tag{674}$$

If $\theta'(\xi) \leq 0$, then

$$\boxed{|I_3| \geq \int_E d\xi w(\xi)|\theta'(\xi)|^3 > 0,} \tag{675}$$

and

$$\boxed{|J_1| \geq c_0 \int_E d\xi w(\xi)|\theta'(\xi)| > 0.} \tag{676}$$

CCCLXVI. COROLLARY FOR THE LEADING BRANCH B COEFFICIENTS

At leading order,

$$\boxed{v_{\parallel}^{\text{LO}} = \frac{\lambda_c^{(1)}}{2} I_3, \quad v_E^{\text{LO}} = \tilde{\Lambda}_E^{(1)} J_1, \quad v_O^{\text{LO}} = \tilde{\Lambda}_O^{(1)} J_1.} \tag{677}$$

Hence, under

$$\boxed{\lambda_c^{(1)} \neq 0, \quad (\tilde{\Lambda}_E^{(1)}, \tilde{\Lambda}_O^{(1)}) \neq (0, 0),} \tag{678}$$

together with (669) and (672), one gets

$$\boxed{v_{\parallel}^{\text{LO}} \neq 0, \quad (v_E^{\text{LO}}, v_O^{\text{LO}}) \neq (0, 0).} \tag{679}$$

CCCLXVII. THEOREM

Therefore we have proved:

a. Overlap nonvanishing theorem. Under the admissible defect assumptions (663)–(665),

$$\boxed{I_3 \neq 0, \quad J_1 \neq 0.} \quad (680)$$

Consequently, if the leading microscopic coefficients satisfy (678), then the leading Branch B reduced velocities are nonzero.

CCCLXVIII. FINAL SUMMARY

For an admissible monotone defect profile with

$$\theta'(\xi) \text{ of fixed sign,} \quad w(\xi) := f_+(\xi)f_-(\xi) \geq 0, \quad w \not\equiv 0, \quad \cos \theta(\xi) \geq c_0 > 0$$

on the effective support, the leading overlap integrals satisfy

$$\boxed{I_3 = \int d\xi w(\xi) \theta'(\xi)^3 \neq 0, \quad J_1 = \int d\xi w(\xi) \theta'(\xi) \cos \theta(\xi) \neq 0.}$$

Moreover,

$$|I_3| \geq \int_E d\xi w(\xi) |\theta'(\xi)|^3 > 0, \quad |J_1| \geq c_0 \int_E d\xi w(\xi) |\theta'(\xi)| > 0.$$

Therefore, if

$$\lambda_c^{(1)} \neq 0, \quad (\tilde{\Lambda}_E^{(1)}, \tilde{\Lambda}_O^{(1)}) \neq (0, 0),$$

then

$$v_{\parallel}^{\text{LO}} = \frac{\lambda_c^{(1)}}{2} I_3 \neq 0, \quad (v_E^{\text{LO}}, v_O^{\text{LO}}) \neq (0, 0).$$

So the first half of the Branch B nondegenerate stability theorem is proved: the leading overlaps do not vanish on the admissible defect class.

CCCLXIX. DOMINANCE BOUND FOR THE EXACT BRANCH B PHASE

CCCLXX. GOAL

We now prove the second half of the Branch B nondegenerate stability theorem: higher-order corrections do not overturn the leading nonzero velocities.

The exact Branch B coefficients are

$$\boxed{v_{\parallel} = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad v_E = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad v_O = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1}.} \quad (681)$$

We split them into leading and higher-order parts:

$$\boxed{v_{\parallel} = v_{\parallel}^{\text{LO}} + \Delta v_{\parallel}, \quad v_E = v_E^{\text{LO}} + \Delta v_E, \quad v_O = v_O^{\text{LO}} + \Delta v_O,} \quad (682)$$

with

$$\boxed{v_{\parallel}^{\text{LO}} = \frac{\lambda_c^{(1)}}{2} I_3, \quad v_E^{\text{LO}} = \tilde{\Lambda}_E^{(1)} J_1, \quad v_O^{\text{LO}} = \tilde{\Lambda}_O^{(1)} J_1.} \quad (683)$$

CCCLXXI. HIGHER-ORDER REMAINDER

The remainders are

$$\Delta v_{\parallel} = \sum_{n \geq 2} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad (684)$$

$$\Delta v_E = \sum_{n \geq 2} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad \Delta v_O = \sum_{n \geq 2} \tilde{\Lambda}_O^{(n)} J_{2n-1}. \quad (685)$$

We seek sufficient conditions ensuring

$$|\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|, \quad \sqrt{(\Delta v_E)^2 + (\Delta v_O)^2} < \sqrt{(v_E^{\text{LO}})^2 + (v_O^{\text{LO}})^2}. \quad (686)$$

CCCLXXII. GEOMETRIC SUPPRESSION OF HIGHER OVERLAPS

Assume the admissible defect profile satisfies

$$|\theta'(\xi)| \leq \Theta_* < 1 \quad \text{on the effective support of } w(\xi) := f_+(\xi)f_-(\xi), \quad (687)$$

and

$$|\cos \theta(\xi)| \leq 1. \quad (688)$$

Then for all $n \geq 2$,

$$\begin{aligned} |I_{2n+1}| &= \left| \int d\xi w(\xi) \theta'(\xi)^{2n+1} \right| \\ &\leq \Theta_*^{2n-2} \int d\xi |w(\xi)| |\theta'(\xi)|^3 \\ &= \Theta_*^{2n-2} K_{\parallel}, \end{aligned} \quad (689)$$

where

$$K_{\parallel} := \int d\xi |w(\xi)| |\theta'(\xi)|^3. \quad (690)$$

Likewise,

$$\begin{aligned} |J_{2n-1}| &= \left| \int d\xi w(\xi) \theta'(\xi)^{2n-1} \cos \theta(\xi) \right| \\ &\leq \Theta_*^{2n-2} \int d\xi |w(\xi)| |\theta'(\xi)| |\cos \theta(\xi)| \\ &\leq \Theta_*^{2n-2} K_{\perp}, \end{aligned} \quad (691)$$

with

$$K_{\perp} := \int d\xi |w(\xi)| |\theta'(\xi)| |\cos \theta(\xi)|. \quad (692)$$

Thus the higher overlaps are geometrically suppressed by powers of Θ_*^2 .

CCCLXXIII. MICROSCOPIC COEFFICIENT BOUNDS

Assume there exist constants L_* , E_* , O_* such that

$$\boxed{|\lambda_c^{(n)}| \leq L_* R_{\parallel}^{n-1} |\lambda_c^{(1)}|, \quad n \geq 2,} \quad (693)$$

$$\boxed{|\tilde{\Lambda}_E^{(n)}| \leq E_* R_{\perp}^{n-1} |\tilde{\Lambda}_E^{(1)}|, \quad |\tilde{\Lambda}_O^{(n)}| \leq O_* R_{\perp}^{n-1} |\tilde{\Lambda}_O^{(1)}|, \quad n \geq 2,} \quad (694)$$

for some nonnegative growth factors $R_{\parallel}, R_{\perp}$.

This is the natural finite-growth assumption: higher-order microscopic coefficients do not grow faster than geometrically.

CCCLXXIV. LONGITUDINAL DOMINANCE BOUND

Using (684), (689), and (693),

$$\begin{aligned} |\Delta v_{\parallel}| &\leq \sum_{n \geq 2} \frac{|\lambda_c^{(n)}|}{2} |I_{2n+1}| \\ &\leq \frac{L_* |\lambda_c^{(1)}| K_{\parallel}}{2} \sum_{n \geq 2} (R_{\parallel} \Theta_*^2)^{n-1}. \end{aligned} \quad (695)$$

If

$$\boxed{R_{\parallel} \Theta_*^2 < 1,} \quad (696)$$

then the series converges and

$$\boxed{|\Delta v_{\parallel}| \leq \frac{L_* |\lambda_c^{(1)}| K_{\parallel}}{2} \frac{R_{\parallel} \Theta_*^2}{1 - R_{\parallel} \Theta_*^2}.} \quad (697)$$

Now, from the overlap nonvanishing theorem we have

$$\boxed{|I_3| \geq m_{\parallel} > 0} \quad (698)$$

for some defect-dependent lower bound $m_{\parallel}$. Hence

$$\boxed{|v_{\parallel}^{\text{LO}}| = \frac{|\lambda_c^{(1)}|}{2} |I_3| \geq \frac{|\lambda_c^{(1)}|}{2} m_{\parallel}.} \quad (699)$$

Therefore a sufficient condition for longitudinal dominance is

$$\boxed{L_* K_{\parallel} \frac{R_{\parallel} \Theta_*^2}{1 - R_{\parallel} \Theta_*^2} < m_{\parallel}.} \quad (700)$$

Under (700),

$$\boxed{|\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|.} \quad (701)$$

CCCLXXV. TRANSVERSE DOMINANCE BOUND

Define the leading transverse norm

$$v_{\perp}^{\text{LO}} := \sqrt{(v_E^{\text{LO}})^2 + (v_O^{\text{LO}})^2} = |J_1| \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}. \quad (702)$$

Also let

$$\Delta v_{\perp} := \sqrt{(\Delta v_E)^2 + (\Delta v_O)^2}. \quad (703)$$

Using (685), (691), and (694),

$$|\Delta v_E| \leq E_* |\tilde{\Lambda}_E^{(1)}| K_{\perp} \sum_{n \geq 2} (R_{\perp} \Theta_*^2)^{n-1}, \quad (704)$$

$$|\Delta v_O| \leq O_* |\tilde{\Lambda}_O^{(1)}| K_{\perp} \sum_{n \geq 2} (R_{\perp} \Theta_*^2)^{n-1}. \quad (705)$$

If

$$R_{\perp} \Theta_*^2 < 1, \quad (706)$$

then

$$|\Delta v_E| \leq E_* |\tilde{\Lambda}_E^{(1)}| K_{\perp} \frac{R_{\perp} \Theta_*^2}{1 - R_{\perp} \Theta_*^2}, \quad (707)$$

$$|\Delta v_O| \leq O_* |\tilde{\Lambda}_O^{(1)}| K_{\perp} \frac{R_{\perp} \Theta_*^2}{1 - R_{\perp} \Theta_*^2}. \quad (708)$$

Hence

$$\Delta v_{\perp} \leq K_{\perp} \frac{R_{\perp} \Theta_*^2}{1 - R_{\perp} \Theta_*^2} \sqrt{E_*^2 (\tilde{\Lambda}_E^{(1)})^2 + O_*^2 (\tilde{\Lambda}_O^{(1)})^2}. \quad (709)$$

From the overlap nonvanishing theorem,

$$|J_1| \geq m_{\perp} > 0. \quad (710)$$

Therefore

$$v_{\perp}^{\text{LO}} \geq m_{\perp} \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}. \quad (711)$$

A sufficient condition for transverse dominance is thus

$$K_{\perp} \frac{R_{\perp} \Theta_*^2}{1 - R_{\perp} \Theta_*^2} \sqrt{E_*^2 (\tilde{\Lambda}_E^{(1)})^2 + O_*^2 (\tilde{\Lambda}_O^{(1)})^2} < m_{\perp} \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}. \quad (712)$$

Under (712),

$$\Delta v_{\perp} < v_{\perp}^{\text{LO}}. \quad (713)$$

CCCLXXVI. NONDEGENERATE STABILITY THEOREM

Combining (701) and (713), we obtain:

a. Branch B nondegenerate stability theorem. Assume:

1. the theory remains within the existing local isotropic TECT algebra, so Branch B is exact;
2. the defect belongs to the admissible class, so

$$I_3 \neq 0, \quad J_1 \neq 0;$$

3. the higher overlap suppression condition (687) holds with $\Theta_* < 1$;
4. the microscopic coefficient towers satisfy the finite-growth bounds (693) and (694);
5. the sufficient inequalities (700) and (712) hold.

Then

$$\boxed{v_{\parallel} \neq 0, \quad v_{\perp} \neq 0.} \quad (714)$$

Hence the exact reduced Branch B PDE defines a stable anisotropic Dirac phase.

CCCLXXVII. INTERPRETATION

The theorem shows that once the defect is monotone and the microscopic higher-order tower is not too violently enhanced, the leading primitive Branch B transport dominates all higher odd corrections.

Thus the exact Dirac phase is robust, not merely formal. The remaining quantitative task is only to estimate the numerical sizes of

$$\Theta_*, \quad R_{\parallel}, \quad R_{\perp}, \quad L_*, \quad E_*, \quad O_*.$$

CCCLXXVIII. FINAL SUMMARY

The higher-order corrections are

$$\Delta v_{\parallel} = \sum_{n \geq 2} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad \Delta v_E = \sum_{n \geq 2} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad \Delta v_O = \sum_{n \geq 2} \tilde{\Lambda}_O^{(n)} J_{2n-1}.$$

If the defect is monotone with

$$|\theta'(\xi)| \leq \Theta_* < 1,$$

then the higher overlaps are geometrically suppressed:

$$|I_{2n+1}| \lesssim \Theta_*^{2n-2} K_{\parallel}, \quad |J_{2n-1}| \lesssim \Theta_*^{2n-2} K_{\perp}.$$

If the microscopic coefficient towers grow at most geometrically,

$$|\lambda_c^{(n)}| \leq L_* R_{\parallel}^{n-1} |\lambda_c^{(1)}|,$$

$$|\tilde{\Lambda}_E^{(n)}| \leq E_* R_{\perp}^{n-1} |\tilde{\Lambda}_E^{(1)}|, \quad |\tilde{\Lambda}_O^{(n)}| \leq O_* R_{\perp}^{n-1} |\tilde{\Lambda}_O^{(1)}|,$$

and if

$$L_* K_{\parallel} \frac{R_{\parallel} \Theta_*^2}{1 - R_{\parallel} \Theta_*^2} < m_{\parallel},$$

$$K_{\perp} \frac{R_{\perp} \Theta_*^2}{1 - R_{\perp} \Theta_*^2} \sqrt{E_*^2 (\tilde{\Lambda}_E^{(1)})^2 + O_*^2 (\tilde{\Lambda}_O^{(1)})^2} < m_{\perp} \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2},$$

then

$$\boxed{v_{\parallel} \neq 0, \quad v_{\perp} \neq 0.}$$

So the exact Branch B reduced PDE is a genuinely stable anisotropic Dirac phase, provided the higher-order tower is sufficiently suppressed relative to the leading primitive term.

CCCLXXIX. NEXT THEOREM: GENERIC VALIDITY OF THE BRANCH B STABILITY BOUNDS

a. Goal. Show that the sufficient dominance bounds for the exact Branch B phase hold on an open natural parameter regime of the microscopic TECT defect data.

b. Data. The exact Branch B reduced coefficients are

$$v_{\parallel} = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad v_E = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad v_O = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1}.$$

The leading-order concrete Class II terms are

$$v_{\parallel}^{\text{LO}} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad v_E^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad v_O^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa.$$

c. Natural small-defect regime. Assume

$$0 < \theta_0 \kappa \ll 1,$$

and that the higher microscopic coefficient towers

$$\lambda_c^{(n)}, \quad \tilde{\Lambda}_E^{(n)}, \quad \tilde{\Lambda}_O^{(n)}$$

have at most finite geometric growth.

Then the higher-order overlaps are suppressed by powers of $(\theta_0 \kappa)^2$, so

$$|\Delta v_{\parallel}| = O((\theta_0 \kappa)^5), \quad \Delta v_{\perp} = O((\theta_0 \kappa)^3),$$

while

$$|v_{\parallel}^{\text{LO}}| = O((\theta_0 \kappa)^3), \quad v_{\perp}^{\text{LO}} = O(\theta_0 \kappa).$$

Therefore the relative errors obey

$$\frac{|\Delta v_{\parallel}|}{|v_{\parallel}^{\text{LO}}|} = O((\theta_0 \kappa)^2), \quad \frac{\Delta v_{\perp}}{v_{\perp}^{\text{LO}}} = O((\theta_0 \kappa)^2).$$

Hence, for sufficiently small but nonzero $\theta_0 \kappa$,

$$|\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|, \quad \Delta v_{\perp} < v_{\perp}^{\text{LO}}.$$

So the exact Branch B phase is generically nondegenerate on an open natural small-defect parameter regime.

CCCLXXX. GENERIC STABILITY OF THE EXACT BRANCH B PHASE IN THE NATURAL SMALL-DEFECT REGIME

CCCLXXXI. GOAL

We prove that the sufficient dominance bounds for the exact Branch B phase hold generically on an open small-defect regime.

The exact reduced coefficients are

$$v_{\parallel} = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad v_E = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad v_O = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1}. \quad (715)$$

We want to show that, in the natural small-defect regime,

$$|\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|, \quad \Delta v_{\perp} < v_{\perp}^{\text{LO}}, \quad (716)$$

where

$$\Delta v_{\perp} := \sqrt{(\Delta v_E)^2 + (\Delta v_O)^2}.$$

CCCLXXXII. LEADING-ORDER STRUCTURE

Write

$$v_{\parallel} = v_{\parallel}^{\text{LO}} + \Delta v_{\parallel}, \quad v_E = v_E^{\text{LO}} + \Delta v_E, \quad v_O = v_O^{\text{LO}} + \Delta v_O. \quad (717)$$

At leading order,

$$v_{\parallel}^{\text{LO}} = \frac{\lambda_c^{(1)}}{2} I_3, \quad v_E^{\text{LO}} = \tilde{\Lambda}_E^{(1)} J_1, \quad v_O^{\text{LO}} = \tilde{\Lambda}_O^{(1)} J_1. \quad (718)$$

The remainders are

$$\Delta v_{\parallel} = \sum_{n \geq 2} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad (719)$$

$$\Delta v_E = \sum_{n \geq 2} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad \Delta v_O = \sum_{n \geq 2} \tilde{\Lambda}_O^{(n)} J_{2n-1}. \quad (720)$$

CCCLXXXIII. NATURAL SMALL-DEFECT SCALING

Assume the defect profile has the standard small-amplitude, finite-width scaling

$$\theta(\xi) = \theta_0 \Theta(\kappa \xi), \quad (721)$$

where Θ is a fixed dimensionless profile with bounded derivatives and $\theta_0, \kappa > 0$.

Then

$$\theta'(\xi) = \theta_0 \kappa \Theta'(\kappa \xi). \quad (722)$$

Hence each odd overlap scales as

$$I_{2n+1} = (\theta_0 \kappa)^{2n+1} \mathcal{I}_{2n+1}, \quad (723)$$

with dimensionless profile-dependent constants

$$\mathcal{I}_{2n+1} = \int d\xi f_+(\xi) f_-(\xi) \Theta'(\kappa \xi)^{2n+1}.$$

Likewise,

$$J_{2n-1} = (\theta_0 \kappa)^{2n-1} \mathcal{J}_{2n-1}, \quad (724)$$

where

$$\mathcal{J}_{2n-1} = \int d\xi f_+(\xi) f_-(\xi) \Theta'(\kappa \xi)^{2n-1} \cos(\theta_0 \Theta(\kappa \xi)).$$

For sufficiently small θ_0 , one has

$$\cos(\theta_0 \Theta) = 1 + O(\theta_0^2),$$

so the leading scaling of J_{2n-1} is unchanged.

Therefore:

$$I_3 = O((\theta_0 \kappa)^3), \quad I_5 = O((\theta_0 \kappa)^5), \quad I_7 = O((\theta_0 \kappa)^7), \dots \quad (725)$$

and

$$J_1 = O(\theta_0 \kappa), \quad J_3 = O((\theta_0 \kappa)^3), \quad J_5 = O((\theta_0 \kappa)^5), \dots \quad (726)$$

CCCLXXXIV. FINITE-GROWTH ASSUMPTION ON THE MICROSCOPIC TOWERS

Assume the coefficient towers do not grow singularly as $\theta_0\kappa \rightarrow 0$. More precisely, assume there exist finite constants

$$C_{\parallel}, \quad C_E, \quad C_O$$

such that

$$\sup_{n \geq 1} \left| \frac{\lambda_c^{(n)}}{\lambda_c^{(1)}} \right| \leq C_{\parallel}, \quad \sup_{n \geq 1} \left| \frac{\tilde{\Lambda}_E^{(n)}}{\tilde{\Lambda}_E^{(1)}} \right| \leq C_E, \quad \sup_{n \geq 1} \left| \frac{\tilde{\Lambda}_O^{(n)}}{\tilde{\Lambda}_O^{(1)}} \right| \leq C_O. \quad (727)$$

This is the natural regularity assumption: higher-order microscopic coefficients remain of finite relative size and do not blow up fast enough to overcome the small-defect suppression.

CCCLXXXV. LONGITUDINAL REMAINDER ESTIMATE

Using (719), (723), and (727),

$$\begin{aligned} |\Delta v_{\parallel}| &\leq \sum_{n \geq 2} \frac{|\lambda_c^{(n)}|}{2} |I_{2n+1}| \\ &\leq \frac{|\lambda_c^{(1)}|}{2} C_{\parallel} \sum_{n \geq 2} (\theta_0\kappa)^{2n+1} |\mathcal{I}_{2n+1}|. \end{aligned} \quad (728)$$

Assume also that the profile constants obey

$$|\mathcal{I}_{2n+1}| \leq M_{\parallel} \quad (729)$$

uniformly in the range of interest. Then

$$\begin{aligned} |\Delta v_{\parallel}| &\leq \frac{|\lambda_c^{(1)}|}{2} C_{\parallel} M_{\parallel} \sum_{n \geq 2} (\theta_0\kappa)^{2n+1} \\ &= \frac{|\lambda_c^{(1)}|}{2} C_{\parallel} M_{\parallel} (\theta_0\kappa)^5 \sum_{m \geq 0} (\theta_0\kappa)^{2m}. \end{aligned} \quad (730)$$

If $(\theta_0\kappa)^2 < 1$, this gives

$$|\Delta v_{\parallel}| \leq \frac{|\lambda_c^{(1)}|}{2} C_{\parallel} M_{\parallel} \frac{(\theta_0\kappa)^5}{1 - (\theta_0\kappa)^2}. \quad (731)$$

Meanwhile

$$|v_{\parallel}^{\text{LO}}| = \frac{|\lambda_c^{(1)}|}{2} |I_3| = \frac{|\lambda_c^{(1)}|}{2} (\theta_0\kappa)^3 |\mathcal{I}_3|. \quad (732)$$

Therefore

$$\frac{|\Delta v_{\parallel}|}{|v_{\parallel}^{\text{LO}}|} \leq \frac{C_{\parallel} M_{\parallel}}{|\mathcal{I}_3|} \frac{(\theta_0\kappa)^2}{1 - (\theta_0\kappa)^2}. \quad (733)$$

Hence

$$\frac{|\Delta v_{\parallel}|}{|v_{\parallel}^{\text{LO}}|} = O((\theta_0\kappa)^2). \quad (734)$$

CCCLXXXVI. TRANSVERSE REMAINDER ESTIMATE

Similarly,

$$|\Delta v_E| \leq |\tilde{\Lambda}_E^{(1)}| C_E \sum_{n \geq 2} |J_{2n-1}|, \quad (735)$$

$$|\Delta v_O| \leq |\tilde{\Lambda}_O^{(1)}| C_O \sum_{n \geq 2} |J_{2n-1}|. \quad (736)$$

Assume

$$|\mathcal{J}_{2n-1}| \leq M_\perp \quad (737)$$

uniformly. Then by (724),

$$|J_{2n-1}| \leq M_\perp (\theta_0 \kappa)^{2n-1}. \quad (738)$$

Hence

$$|\Delta v_E| \leq |\tilde{\Lambda}_E^{(1)}| C_E M_\perp \frac{(\theta_0 \kappa)^3}{1 - (\theta_0 \kappa)^2}, \quad (739)$$

and

$$|\Delta v_O| \leq |\tilde{\Lambda}_O^{(1)}| C_O M_\perp \frac{(\theta_0 \kappa)^3}{1 - (\theta_0 \kappa)^2}. \quad (740)$$

Therefore

$$\Delta v_\perp \leq M_\perp \frac{(\theta_0 \kappa)^3}{1 - (\theta_0 \kappa)^2} \sqrt{C_E^2 (\tilde{\Lambda}_E^{(1)})^2 + C_O^2 (\tilde{\Lambda}_O^{(1)})^2}. \quad (741)$$

On the other hand,

$$v_\perp^{\text{LO}} = |J_1| \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2} = (\theta_0 \kappa) |\mathcal{J}_1| \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}. \quad (742)$$

Thus

$$\frac{\Delta v_\perp}{v_\perp^{\text{LO}}} \leq \frac{M_\perp}{|\mathcal{J}_1|} \frac{(\theta_0 \kappa)^2}{1 - (\theta_0 \kappa)^2} \frac{\sqrt{C_E^2 (\tilde{\Lambda}_E^{(1)})^2 + C_O^2 (\tilde{\Lambda}_O^{(1)})^2}}{\sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}}. \quad (743)$$

Hence

$$\frac{\Delta v_\perp}{v_\perp^{\text{LO}}} = O((\theta_0 \kappa)^2). \quad (744)$$

CCCLXXXVII. GENERIC STABILITY THEOREM

Combining (734) and (744), we obtain:

a. Generic small-defect stability theorem. Assume:

1. the exact theory stays on Branch B;
2. the defect belongs to the admissible class so that $\mathcal{I}_3 \neq 0$ and $\mathcal{J}_1 \neq 0$;
3. the profile constants $\mathcal{I}_{2n+1}, \mathcal{J}_{2n-1}$ are uniformly bounded in the small-defect regime;
4. the microscopic towers $\lambda_c^{(n)}, \tilde{\Lambda}_E^{(n)}, \tilde{\Lambda}_O^{(n)}$ satisfy the finite-growth bounds (727).

Then, as $\theta_0 \kappa \rightarrow 0$,

$$\boxed{\frac{|\Delta v_{\parallel}|}{|v_{\parallel}^{\text{LO}}|} = O((\theta_0 \kappa)^2), \quad \frac{\Delta v_{\perp}}{v_{\perp}^{\text{LO}}} = O((\theta_0 \kappa)^2).} \quad (745)$$

Therefore there exists $\varepsilon_* > 0$ such that whenever

$$0 < \theta_0 \kappa < \varepsilon_*,$$

one has

$$\boxed{|\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|, \quad \Delta v_{\perp} < v_{\perp}^{\text{LO}}.} \quad (746)$$

Consequently,

$$\boxed{v_{\parallel} \neq 0, \quad v_{\perp} \neq 0.} \quad (747)$$

Hence the exact Branch B reduced PDE defines a generically stable anisotropic Dirac phase on an open natural small-defect regime.

CCCLXXXVIII. INTERPRETATION

This theorem means that the sufficient stability inequalities are not fine-tuned. They are automatically satisfied in an open region of parameter space as long as the defect is sufficiently weak and the microscopic coefficient towers remain regular.

So the Branch B phase is not merely allowed; it is the natural generic outcome of the existing isotropic local TECT algebra in the small-defect regime.

CCCLXXXIX. FINAL SUMMARY

In the natural small-defect regime

$$0 < \theta_0 \kappa \ll 1,$$

the odd overlap tower scales as

$$I_{2n+1} = O((\theta_0 \kappa)^{2n+1}), \quad J_{2n-1} = O((\theta_0 \kappa)^{2n-1}).$$

If the microscopic coefficient towers have finite relative growth, then

$$|\Delta v_{\parallel}| = O((\theta_0 \kappa)^5), \quad \Delta v_{\perp} = O((\theta_0 \kappa)^3),$$

while

$$|v_{\parallel}^{\text{LO}}| = O((\theta_0 \kappa)^3), \quad v_{\perp}^{\text{LO}} = O(\theta_0 \kappa).$$

Therefore

$$\boxed{\frac{|\Delta v_{\parallel}|}{|v_{\parallel}^{\text{LO}}|} = O((\theta_0 \kappa)^2), \quad \frac{\Delta v_{\perp}}{v_{\perp}^{\text{LO}}} = O((\theta_0 \kappa)^2).}$$

Hence, for sufficiently small but nonzero $\theta_0 \kappa$,

$$|\Delta v_{\parallel}| < |v_{\parallel}^{\text{LO}}|, \quad \Delta v_{\perp} < v_{\perp}^{\text{LO}}.$$

So the exact Branch B reduced PDE is generically nondegenerate and stable on an open natural small-defect regime.

CCCXC. EXACT BRANCH B PHASE THEOREM

[Exact Branch B phase theorem] Assume the following.

1. The full reduced defect theory stays within the existing local isotropic TECT variable algebra, so that no linear longitudinal selector is generated and hence

$$\Lambda_{\parallel}^{(0)} = 0.$$

2. The defect belongs to the admissible monotone class:

$$\theta'(\xi) \text{ has fixed sign,} \quad w(\xi) := f_+(\xi)f_-(\xi) \geq 0, \quad w \not\equiv 0,$$

and

$$\cos \theta(\xi) \geq c_0 > 0$$

on the effective support.

3. The leading microscopic coefficients satisfy

$$\lambda_c^{(1)} \neq 0, \quad (\tilde{\Lambda}_E^{(1)}, \tilde{\Lambda}_O^{(1)}) \neq (0, 0).$$

4. The defect lies in the natural small-defect regime

$$0 < \theta_0 \kappa \ll 1,$$

and the higher microscopic coefficient towers have finite relative growth.

Then:

1. the longitudinal sector is exactly Branch B:

$$a_3(\xi) = \frac{\lambda_c^{(1)}}{2} \theta'(\xi)^3 + \frac{\lambda_c^{(2)}}{2} \theta'(\xi)^5 + \dots;$$

2. the leading overlaps satisfy

$$I_3 \neq 0, \quad J_1 \neq 0;$$

3. the higher-order corrections are subleading:

$$\frac{|\Delta v_{\parallel}|}{|v_{\parallel}^{\text{LO}}|} = O((\theta_0 \kappa)^2), \quad \frac{\Delta v_{\perp}}{v_{\perp}^{\text{LO}}} = O((\theta_0 \kappa)^2);$$

4. therefore

$$v_{\parallel} \neq 0, \quad v_{\perp} := \sqrt{v_E^2 + v_O^2} \neq 0.$$

Hence the reduced infrared defect PDE is an exact stable anisotropic Dirac phase of Branch B type:

$$\boxed{\left[i\partial_t - v_{\parallel} \sigma_3 (-i\partial_{\parallel}) - v_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - v_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.} \quad (748)$$

Moreover, up to transverse frame rotation and overall rescaling, the phase is characterized by the single dimensionless anisotropy ratio

$$\rho = \frac{v_{\parallel}}{v_{\perp}}.$$

Branch A exclusion follows from the absence of a linear longitudinal selector inside the existing local isotropic TECT algebra. The exact concrete Class II longitudinal/transverse split is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O,$$

so the concrete branch is primitive in the longitudinal sector and I_{θ} -type in the transverse sector.

The overlap nonvanishing gives

$$I_{\theta'^3} \neq 0, \quad I_{\theta} \neq 0,$$

for nontrivial admissible defects, so the finite nondegeneracy condition reduces to

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

The exact concrete Class II witnesses are explicitly

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta},$$

with

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

so the leading hierarchy is explicit.

Finally, in the small-defect regime $0 < \theta_0 \kappa \ll 1$, the odd longitudinal tower and odd transverse tower are suppressed by successive powers of $(\theta_0 \kappa)^2$, so the higher-order remainders cannot overturn the nonzero leading terms under finite-growth assumptions on the microscopic coefficient towers. Therefore $v_{\parallel} \neq 0$ and $v_{\perp} \neq 0$, proving stable Dirac nondegeneracy.

CCCXCI. NEXT FRONTIER: MICROSCOPIC TRUNCATION PREDICTION FOR THE ANISOTROPY RATIO

a. Goal. The structural phase theorem is already closed. The next frontier is quantitative: predict the anisotropy ratio

$$\rho = \frac{v_{\parallel}}{v_{\perp}}$$

from a concrete microscopic truncation of the full TECT action.

b. Leading-order prediction. At leading order in the explicit concrete Branch B realization,

$$v_{\parallel}^{\text{LO}} = \frac{\lambda_c^{(1)}}{2} I_3, \quad v_E^{\text{LO}} = \tilde{\Lambda}_E^{(1)} J_1, \quad v_O^{\text{LO}} = \tilde{\Lambda}_O^{(1)} J_1,$$

so

$$v_{\perp}^{\text{LO}} = |J_1| \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}.$$

Therefore

$$\rho_{\text{LO}} = \frac{\frac{\lambda_c^{(1)}}{2} I_3}{|J_1| \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}}. \quad (749)$$

Using the explicit overlap values of the concrete Class II realization,

$$I_3 = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad J_1 = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

this becomes

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{\lambda_c^{(1)}}{\sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}} (\theta_0 \kappa)^2. \quad (750)$$

c. What must be computed microscopically. Thus the only remaining microscopic inputs are

$$\boxed{\lambda_c^{(1)}, \quad \tilde{\Lambda}_E^{(1)}, \quad \tilde{\Lambda}_O^{(1)}}. \quad (751)$$

These are obtained from the tangent first-order Hessian block

$$V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q$$

through

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V^\mu), \quad b_E(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a E_\mu^a V^\mu), \quad b_O(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a J_\mu^a V^\mu),$$

followed by the leading projections

$$a_3(\xi) = \frac{\lambda_c^{(1)}}{2} \theta'(\xi)^3 + \dots, \quad b_E(\xi) = \tilde{\Lambda}_E^{(1)} \theta'(\xi) \cos \theta(\xi) + \dots, \quad b_O(\xi) = \tilde{\Lambda}_O^{(1)} \theta'(\xi) \cos \theta(\xi) + \dots.$$

d. Natural expectation. If

$$\lambda_c^{(1)}, \quad \tilde{\Lambda}_E^{(1)}, \quad \tilde{\Lambda}_O^{(1)}$$

are all of the same microscopic order, then

$$\boxed{|\rho_{LO}| \sim (\theta_0 \kappa)^2 \ll 1} \quad (752)$$

in the natural small-defect regime.

e. Fine-tuned balanced regime. To obtain

$$|\rho| \sim 1,$$

one needs

$$\boxed{|\lambda_c^{(1)}| \sim (\theta_0 \kappa)^{-2} \sqrt{(\tilde{\Lambda}_E^{(1)})^2 + (\tilde{\Lambda}_O^{(1)})^2}}, \quad (753)$$

which requires strong enhancement of the primitive longitudinal coefficient.

f. Immediate next task. So the immediate next quantitative task is:

$$\boxed{\text{choose a concrete full-action truncation, compute } \lambda_c^{(1)}, \tilde{\Lambda}_E^{(1)}, \tilde{\Lambda}_O^{(1)}, \text{ and evaluate } \rho_{LO}.} \quad (754)$$

CCCXCII. CONCRETE ONE-MEDIATOR TRUNCATION AND EXPLICIT PREDICTION FOR ρ_{LO}

CCCXCIII. MINIMAL PREDICTIVE TRUNCATION

Take the following concrete lowest-order truncation:

$$\boxed{\mathcal{L}_{\text{trunc}} = \mathcal{L}_\Psi^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_\sigma^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_E X_i^a \mathcal{E}_i^a[P, \psi] + \gamma_O X_i^a \mathcal{O}_i^a[P, \psi].} \quad (755)$$

Here:

$$\boxed{J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \Psi^\dagger T_a (\partial_i \Psi),} \quad (756)$$

$$\boxed{K_i^a[P, \psi] = (T_a \partial_i P) n^\mu D_\mu \psi,} \quad (757)$$

and the explicit transverse carriers are chosen as

$$\mathcal{E}_i^a[P, \psi] = (T_a \partial_i P) (S_E M_E) E^\mu D_\mu \psi, \quad (758)$$

$$\mathcal{O}_i^a[P, \psi] = (T_a \partial_i P) (S_O M_O) J^\mu D_\mu \psi. \quad (759)$$

This is the simplest one-mediator truncation that carries:

- (i) primitive longitudinal source via $J \cdot K$, (ii) transverse even/odd sources via $K \cdot \mathcal{E}$, $K \cdot \mathcal{O}$.

CCCXCIV. INTEGRATING OUT THE MEDIATOR

The algebraic equation of motion for X_i^a is

$$M_X^2 X_i^a + \alpha_X J_i^a + \beta_X K_i^a + \gamma_E \mathcal{E}_i^a + \gamma_O \mathcal{O}_i^a = 0. \quad (760)$$

Thus

$$X_i^a = -\frac{1}{M_X^2} (\alpha_X J_i^a + \beta_X K_i^a + \gamma_E \mathcal{E}_i^a + \gamma_O \mathcal{O}_i^a). \quad (761)$$

Substituting back gives

$$\mathcal{L}_{\text{eff}} = -\frac{1}{2M_X^2} (\alpha_X J_i^a + \beta_X K_i^a + \gamma_E \mathcal{E}_i^a + \gamma_O \mathcal{O}_i^a)^2. \quad (762)$$

The relevant cross terms are:

$$\mathcal{L}_{\text{eff}} \supset -\frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a - \frac{\beta_X \gamma_E}{M_X^2} K_i^a \mathcal{E}_i^a - \frac{\beta_X \gamma_O}{M_X^2} K_i^a \mathcal{O}_i^a. \quad (763)$$

CCCXCV. LONGITUDINAL COEFFICIENT

On the explicit real defect representative, the primitive longitudinal contraction is controlled by

$$A_a(\theta) k_a(\theta) = 1$$

in the previously fixed normalization. More generally, let

$$C_{\parallel} := A_a(\theta) k_a(\theta) \big|_{\text{normalized}}, \quad (764)$$

so that in the explicit concrete normalization

$$C_{\parallel} = 1.$$

Then the primitive longitudinal transport coefficient is

$$\lambda_c^{(1)} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}. \quad (765)$$

Therefore

$$v_{\parallel}^{\text{LO}} = \frac{\lambda_c^{(1)}}{2} I_3 = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3. \quad (766)$$

CCCXCVI. TRANSVERSE COEFFICIENTS

Likewise define the normalized transverse overlap constants

$$\boxed{C_E := \text{normalized coefficient of } K_i^a \mathcal{E}_i^a, \quad C_O := \text{normalized coefficient of } K_i^a \mathcal{O}_i^a.} \quad (767)$$

For the explicit representative with

$$(S_E M_E) = \cos \theta, \quad (S_O M_O) = \cos \theta,$$

and the canonical basis choice, one may take

$$C_E = C_O = 1.$$

Then

$$\boxed{\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2} C_E, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} C_O.} \quad (768)$$

Hence

$$\boxed{v_E^{\text{LO}} = \tilde{\Lambda}_E^{(1)} J_1 = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1,} \quad (769)$$

$$\boxed{v_O^{\text{LO}} = \tilde{\Lambda}_O^{(1)} J_1 = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.} \quad (770)$$

Therefore

$$\boxed{v_{\perp}^{\text{LO}} = \frac{|\beta_X|}{M_X^2} |J_1| \sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}.} \quad (771)$$

CCCXCVII. EXPLICIT PREDICTION FOR ρ_{LO}

Now

$$\boxed{\rho_{\text{LO}} = \frac{v_{\parallel}^{\text{LO}}}{v_{\perp}^{\text{LO}}} = -\frac{\alpha_X C_{\parallel}}{2\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} \frac{I_3}{|J_1|}.} \quad (772)$$

Using the explicit concrete overlap values

$$\boxed{I_3 = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad J_1 = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,} \quad (773)$$

we get

$$\boxed{\frac{I_3}{|J_1|} = -\frac{3}{16} (\theta_0 \kappa)^2.} \quad (774)$$

Therefore

$$\boxed{\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X C_{\parallel}}{\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} (\theta_0 \kappa)^2.} \quad (775)$$

In the canonical explicit normalization

$$C_{\parallel} = C_E = C_O = 1,$$

this simplifies to

$$\boxed{\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} (\theta_0 \kappa)^2.} \quad (776)$$

CCCXCVIII. INTERPRETATION

This truncation yields three immediate conclusions.

a. (i) Mediator scale cancellation. The common factor β_X/M_X^2 cancels out of the ratio ρ_{LO} . Thus ρ is controlled not by the overall mediator strength, but by the relative coupling hierarchy

$$\alpha_X : \gamma_E : \gamma_O.$$

b. (ii) Natural anisotropy. If

$$\alpha_X, \gamma_E, \gamma_O$$

are all of the same microscopic order, then

$$|\rho_{\text{LO}}| \sim (\theta_0 \kappa)^2 \ll 1.$$

c. (iii) Balanced cone requires enhancement. To obtain $|\rho_{\text{LO}}| \sim 1$, one needs

$$|\alpha_X| \sim (\theta_0 \kappa)^{-2} \sqrt{\gamma_E^2 + \gamma_O^2},$$

which is a strong longitudinal enhancement.

CCCXCIX. MINIMAL PREDICTIVE THEOREM

Therefore, within the one-mediator truncation (755), the leading anisotropy ratio is completely predicted by the microscopic coupling ratio:

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X C_{\parallel}}{\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} (\theta_0 \kappa)^2. \quad (777)$$

This is the first fully explicit microscopic prediction of the Branch B phase.

CD. FINAL SUMMARY

For the simplest one-mediator full-action truncation,

$$\mathcal{L}_{\text{trunc}} = \cdots + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a + \beta_X X_i^a K_i^a + \gamma_E X_i^a \mathcal{E}_i^a + \gamma_O X_i^a \mathcal{O}_i^a,$$

integrating out X_i^a gives

$$\lambda_c^{(1)} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}, \quad \tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2} C_E, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} C_O.$$

Hence

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X C_{\parallel}}{\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} (\theta_0 \kappa)^2.$$

In the canonical explicit normalization $C_{\parallel} = C_E = C_O = 1$,

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} (\theta_0 \kappa)^2.$$

So the leading anisotropy is controlled only by the relative microscopic coupling hierarchy

$$\alpha_X : \gamma_E : \gamma_O,$$

while the common mediator strength cancels out.

Therefore the natural expectation remains

$$|\rho_{\text{LO}}| \ll 1$$

unless the longitudinal coupling α_X is strongly enhanced.

CDI. NATURALNESS THEOREM FOR THE COUPLING HIERARCHY

CDII. STARTING POINT

For the concrete one-mediator truncation, the leading anisotropy ratio is

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X C_{\parallel}}{\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} (\theta_0 \kappa)^2. \quad (778)$$

In the canonical explicit normalization

$$C_{\parallel} = C_E = C_O = 1,$$

this reduces to

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} (\theta_0 \kappa)^2. \quad (779)$$

Thus the only non-geometric factor controlling the size of ρ_{LO} is the coupling ratio

$$\frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}}.$$

CDIII. SYMMETRY HYPOTHESIS

Assume the mediator X_i^a couples to a unified shell-current multiplet

$$\mathbb{J}_i^a = c_{\parallel} J_i^a + c_E \mathcal{E}_i^a + c_O \mathcal{O}_i^a,$$

through a single bare coupling g_X :

$$\mathcal{L}_{\text{int}} = g_X X_i^a \mathbb{J}_i^a. \quad (780)$$

Equivalently, the three couplings are not fundamentally independent but arise as projections of one common microscopic interaction:

$$\alpha_X = g_X c_{\parallel}, \quad \gamma_E = g_X c_E, \quad \gamma_O = g_X c_O. \quad (781)$$

Here $c_{\parallel}, c_E, c_O$ are dimensionless representation/projection coefficients.

CDIV. O(1) PROJECTION HYPOTHESIS

Assume further that the shell-current projections are symmetry-allowed and non-fine-tuned, namely

$$c_{\parallel}, c_E, c_O = O(1), \quad (c_E, c_O) \neq (0, 0). \quad (782)$$

Then

$$\frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} = \frac{c_{\parallel}}{\sqrt{c_E^2 + c_O^2}} = O(1). \quad (783)$$

Therefore, from (779),

$$\rho_{\text{LO}} = O((\theta_0 \kappa)^2). \quad (784)$$

Hence in the natural small-defect regime

$$0 < \theta_0 \kappa \ll 1,$$

one obtains

$$|\rho_{\text{LO}}| \ll 1. \quad (785)$$

CDV. NATURALNESS THEOREM

We may therefore state:

a. Coupling naturalness theorem. Assume:

1. the longitudinal and transverse mediator couplings arise from a single unified microscopic mediator interaction;
2. the corresponding projection coefficients $c_{\parallel}, c_E, c_O$ are symmetry-allowed and $O(1)$;
3. the defect lies in the natural small-defect regime $0 < \theta_0 \kappa \ll 1$.

Then the leading anisotropy ratio satisfies

$$\boxed{|\rho_{\text{LO}}| = O((\theta_0 \kappa)^2) \ll 1.} \quad (786)$$

Thus the strongly anisotropic Branch B cone is the natural generic prediction.

CDVI. WHEN CAN BALANCED CONES OCCUR?

From (779), a balanced cone $|\rho_{\text{LO}}| \sim 1$ requires

$$\boxed{|\alpha_X| \sim (\theta_0 \kappa)^{-2} \sqrt{\gamma_E^2 + \gamma_O^2}.} \quad (787)$$

Under the unified-coupling hypothesis (781), this becomes

$$\boxed{|c_{\parallel}| \sim (\theta_0 \kappa)^{-2} \sqrt{c_E^2 + c_O^2}.} \quad (788)$$

Since $(\theta_0 \kappa)^{-2} \gg 1$ in the small-defect regime, this requires either:

1. a strong symmetry-breaking enhancement of the longitudinal projection;
2. an accidental cancellation in the transverse projections;
3. or a breakdown of the $O(1)$ projection hypothesis.

Thus $|\rho| \sim 1$ is fine-tuned relative to the natural Branch B expectation.

CDVII. HONEST SCOPE

This theorem does *not* prove that the full microscopic TECT action must realize the unified-coupling structure (781). What it proves is the following:

$$\boxed{\text{If the mediator couplings come from one symmetry multiplet with } O(1) \text{ projections, then } |\rho| \ll 1 \text{ is natural.}} \quad (789)$$

So the remaining microscopic task is to identify whether the actual full-action truncation realizes this unified-coupling structure.

CDVIII. FINAL SUMMARY

The leading anisotropy ratio is

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} (\theta_0 \kappa)^2.$$

If the three couplings arise from one unified mediator interaction,

$$\alpha_X = g_X c_{\parallel}, \quad \gamma_E = g_X c_E, \quad \gamma_O = g_X c_O,$$

with

$$c_{\parallel}, c_E, c_O = O(1),$$

then

$$\frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} = \frac{c_{\parallel}}{\sqrt{c_E^2 + c_O^2}} = O(1).$$

Therefore

$$|\rho_{\text{LO}}| = O((\theta_0 \kappa)^2) \ll 1$$

in the natural small-defect regime.

So the strongly anisotropic Branch B cone does not require tuning. It is the natural generic prediction whenever the longitudinal and transverse mediator couplings come from one common symmetry multiplet with $O(1)$ projection coefficients.

By contrast, a balanced cone $|\rho| \sim 1$ requires a large enhancement of the longitudinal projection or an accidental suppression of the transverse ones, and is therefore fine-tuned.

CDIX. EXACT DISTRIBUTION OF THE ANISOTROPY PREFACTOR UNDER ISOTROPIC UNIFIED COUPLING

CDX. STARTING POINT

Under the unified-coupling hypothesis,

$$\alpha_X = g_X c_{\parallel}, \quad \gamma_E = g_X c_E, \quad \gamma_O = g_X c_O,$$

the leading anisotropy ratio is

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{c_{\parallel}}{\sqrt{c_E^2 + c_O^2}} (\theta_0 \kappa)^2. \quad (790)$$

Define the dimensionless coupling ratio

$$r := \frac{|c_{\parallel}|}{\sqrt{c_E^2 + c_O^2}}. \quad (791)$$

Then

$$|\rho_{\text{LO}}| = \frac{3}{32} r (\theta_0 \kappa)^2. \quad (792)$$

CDXI. ISOTROPIC MULTIPLY HYPOTHESIS

Assume the projection vector

$$(c_{\parallel}, c_E, c_O)$$

is distributed isotropically on the unit sphere

$$c_{\parallel}^2 + c_E^2 + c_O^2 = 1.$$

Then we may parametrize

$$c_{\parallel} = \cos \vartheta, \quad c_E = \sin \vartheta \cos \varphi, \quad c_O = \sin \vartheta \sin \varphi,$$

with the standard rotationally invariant measure

$$d\mu = \frac{1}{4\pi} \sin \vartheta \, d\vartheta \, d\varphi.$$

Hence

$$\boxed{r = \frac{|\cos \vartheta|}{\sin \vartheta} = |\cot \vartheta|.} \quad (793)$$

CDXII. DISTRIBUTION OF r

For an isotropic unit vector in three dimensions, the variable

$$z := c_{\parallel} = \cos \vartheta$$

is uniformly distributed on $[-1, 1]$. Therefore $|z|$ is uniformly distributed on $[0, 1]$.

Since

$$r = \frac{|z|}{\sqrt{1 - z^2}},$$

the cumulative distribution function of r is

$$\begin{aligned} F_r(R) &= \mathbb{P}(r \leq R) \\ &= \mathbb{P}\left(\frac{|z|}{\sqrt{1 - z^2}} \leq R\right) \\ &= \mathbb{P}\left(|z|^2 \leq \frac{R^2}{1 + R^2}\right) \\ &= \mathbb{P}\left(|z| \leq \frac{R}{\sqrt{1 + R^2}}\right). \end{aligned} \quad (794)$$

Since $|z| \sim \text{Unif}[0, 1]$,

$$\boxed{F_r(R) = \frac{R}{\sqrt{1 + R^2}}, \quad R \geq 0.} \quad (795)$$

Differentiating gives the density:

$$\boxed{f_r(R) = \frac{1}{(1 + R^2)^{3/2}}, \quad R \geq 0.} \quad (796)$$

CDXIII. TYPICAL VALUES

A. Median

The median r_{med} satisfies

$$F_r(r_{\text{med}}) = \frac{1}{2}.$$

Thus

$$\frac{r_{\text{med}}}{\sqrt{1 + r_{\text{med}}^2}} = \frac{1}{2}.$$

Squaring gives

$$4r_{\text{med}}^2 = 1 + r_{\text{med}}^2,$$

hence

$$r_{\text{med}} = \frac{1}{\sqrt{3}} \approx 0.577. \quad (797)$$

Therefore the median anisotropy ratio is

$$|\rho_{\text{LO}}|_{\text{med}} = \frac{3}{32\sqrt{3}} (\theta_0 \kappa)^2 \approx 0.0541 (\theta_0 \kappa)^2. \quad (798)$$

B. Probability of a balanced cone

A balanced cone would require $r \gtrsim (\theta_0 \kappa)^{-2}$. Using (795),

$$\mathbb{P}(r \geq R) = 1 - \frac{R}{\sqrt{1 + R^2}}. \quad (799)$$

For large R ,

$$\mathbb{P}(r \geq R) = \frac{1}{2R^2} + O(R^{-4}). \quad (800)$$

Therefore

$$\mathbb{P}(|\rho_{\text{LO}}| \sim O(1)) \sim O((\theta_0 \kappa)^4) \quad (801)$$

in the natural small-defect regime.

So balanced cones are strongly statistically suppressed.

C. Probability that $r < 1$

As a simple benchmark,

$$F_r(1) = \frac{1}{\sqrt{2}} \approx 0.707.$$

Hence

$$\mathbb{P}(r < 1) = \frac{1}{\sqrt{2}} \approx 70.7\%. \quad (802)$$

So even before multiplying by the small geometric factor $(\theta_0 \kappa)^2$, the coupling ratio is typically not large.

CDXIV. EXPECTATION VALUES

The mean of r diverges:

$$\mathbb{E}[r] = \infty. \quad (803)$$

Indeed,

$$\int_0^\infty R f_r(R) dR = \int_0^\infty \frac{R}{(1 + R^2)^{3/2}} dR$$

has a logarithmic divergence at infinity.

Thus the mean is not a good notion of “typical size”. The median and tail bounds are the correct robust descriptors.

CDXV. MICROSCOPIC CONSEQUENCE FOR TECT

Combining (792) with (797) gives the typical estimate

$$|\rho_{\text{LO}}|_{\text{typ}} \sim O((\theta_0 \kappa)^2), \quad (804)$$

with a concrete median value

$$|\rho_{\text{LO}}|_{\text{med}} \approx 0.0541 (\theta_0 \kappa)^2.$$

Therefore, under isotropic unified coupling, the strongly anisotropic Branch B cone is not merely allowed: it is the statistically typical outcome.

CDXVI. FINAL THEOREM

a. Isotropic unified-coupling theorem. Assume the mediator coupling projections $(c_{\parallel}, c_E, c_O)$ form an isotropically distributed unit vector in the unified coupling space. Then the prefactor

$$r = \frac{|c_{\parallel}|}{\sqrt{c_E^2 + c_O^2}}$$

has exact distribution

$$f_r(R) = \frac{1}{(1 + R^2)^{3/2}}, \quad F_r(R) = \frac{R}{\sqrt{1 + R^2}}.$$

Consequently,

$$|\rho_{\text{LO}}| = \frac{3}{32} r (\theta_0 \kappa)^2 \quad (805)$$

is typically of order $(\theta_0 \kappa)^2$, with median

$$|\rho_{\text{LO}}|_{\text{med}} = \frac{3}{32\sqrt{3}} (\theta_0 \kappa)^2. \quad (806)$$

Thus the generic Branch B prediction is a strongly anisotropic cone, while balanced cones are statistically suppressed.

CDXVII. FINAL SUMMARY

If the unified coupling projections $(c_{\parallel}, c_E, c_O)$ are isotropic on the unit sphere, then

$$r := \frac{|c_{\parallel}|}{\sqrt{c_E^2 + c_O^2}}$$

has exact CDF and density

$$F_r(R) = \frac{R}{\sqrt{1 + R^2}}, \quad f_r(R) = \frac{1}{(1 + R^2)^{3/2}}.$$

Therefore

$$|\rho_{\text{LO}}| = \frac{3}{32} r (\theta_0 \kappa)^2$$

has typical size

$$|\rho_{\text{LO}}|_{\text{typ}} \sim O((\theta_0 \kappa)^2).$$

More concretely, the median is

$$|\rho_{\text{LO}}|_{\text{med}} = \frac{3}{32\sqrt{3}}(\theta_0\kappa)^2 \approx 0.0541 (\theta_0\kappa)^2.$$

Also,

$$\mathbb{P}(r < 1) = \frac{1}{\sqrt{2}} \approx 70.7\%,$$

and the probability of a balanced cone is parametrically suppressed:

$$\mathbb{P}(|\rho_{\text{LO}}| \sim O(1)) \sim O((\theta_0\kappa)^4).$$

So under isotropic unified coupling, the strongly anisotropic Branch B cone is the statistically typical TECT prediction.

CDXVIII. NUMERICAL EXAMPLES FOR THE ANISOTROPY RATIO

Using the isotropic unified-coupling distribution, the leading anisotropy ratio is

$$|\rho_{\text{LO}}| = \frac{3}{32}r(\theta_0\kappa)^2.$$

The median corresponds to

$$r_{\text{med}} = \frac{1}{\sqrt{3}},$$

so

$$|\rho_{\text{LO}}|_{\text{med}} = \frac{3}{32\sqrt{3}}(\theta_0\kappa)^2 \approx 0.0541(\theta_0\kappa)^2.$$

The 90th percentile corresponds to

$$r_{0.9} = \frac{0.9}{\sqrt{1-0.9^2}} \approx 2.0647,$$

so

$$|\rho_{\text{LO}}|_{90\%} \approx 0.1936(\theta_0\kappa)^2.$$

Therefore:

$$\theta_0\kappa = 0.1 \implies |\rho|_{\text{med}} \approx 5.41 \times 10^{-4}, \quad |\rho|_{90\%} \approx 1.94 \times 10^{-3}.$$

$$\theta_0\kappa = 0.2 \implies |\rho|_{\text{med}} \approx 2.16 \times 10^{-3}, \quad |\rho|_{90\%} \approx 7.74 \times 10^{-3}.$$

$$\theta_0\kappa = 0.3 \implies |\rho|_{\text{med}} \approx 4.87 \times 10^{-3}, \quad |\rho|_{90\%} \approx 1.74 \times 10^{-2}.$$

$$\theta_0\kappa = 0.5 \implies |\rho|_{\text{med}} \approx 1.35 \times 10^{-2}, \quad |\rho|_{90\%} \approx 4.84 \times 10^{-2}.$$

$$\theta_0\kappa = 1.0 \implies |\rho|_{\text{med}} \approx 5.41 \times 10^{-2}, \quad |\rho|_{90\%} \approx 1.94 \times 10^{-1}.$$

CDXIX. EXACT PROBABILITY FORMULA FOR $|\rho| < \rho_*$

Let

$$a := \frac{3}{32}, \quad x := \theta_0 \kappa.$$

Then

$$|\rho_{\text{LO}}| = a r x^2.$$

Hence

$$\mathbb{P}(|\rho_{\text{LO}}| < \rho_*) = \mathbb{P}\left(r < \frac{\rho_*}{ax^2}\right) = F_r\left(\frac{\rho_*}{ax^2}\right).$$

Using

$$F_r(R) = \frac{R}{\sqrt{1+R^2}},$$

we obtain

$$\boxed{\mathbb{P}(|\rho_{\text{LO}}| < \rho_*) = \frac{u}{\sqrt{1+u^2}}, \quad u := \frac{32 \rho_*}{3(\theta_0 \kappa)^2}.} \quad (807)$$

CDXX. NUMERICAL EXAMPLES

For $\rho_* = 0.1$:

$$\theta_0 \kappa = 0.1 \implies \mathbb{P}(|\rho| < 0.1) \approx 0.999956.$$

$$\theta_0 \kappa = 0.2 \implies \mathbb{P}(|\rho| < 0.1) \approx 0.999297.$$

$$\theta_0 \kappa = 0.3 \implies \mathbb{P}(|\rho| < 0.1) \approx 0.996490.$$

$$\theta_0 \kappa = 0.5 \implies \mathbb{P}(|\rho| < 0.1) \approx 0.973646.$$

$$\theta_0 \kappa = 1.0 \implies \mathbb{P}(|\rho| < 0.1) \approx 0.729537.$$

For $\rho_* = 0.01$:

$$\theta_0 \kappa = 0.1 \implies \mathbb{P}(|\rho| < 0.01) \approx 0.995327.$$

$$\theta_0 \kappa = 0.2 \implies \mathbb{P}(|\rho| < 0.01) \approx 0.936329.$$

$$\theta_0 \kappa = 0.3 \implies \mathbb{P}(|\rho| < 0.01) \approx 0.764600.$$

$$\theta_0 \kappa = 0.5 \implies \mathbb{P}(|\rho| < 0.01) \approx 0.392480.$$

$$\theta_0 \kappa = 1.0 \implies \mathbb{P}(|\rho| < 0.01) \approx 0.106070.$$

CDXXI. GROUP-VELOCITY ANISOTROPY AND CONE ASPECT RATIO

CDXXII. DEFINITIONS

For the exact Branch B reduced PDE,

$$\omega(\mathbf{p})^2 = v_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 (p_1^2 + p_2^2), \quad \rho := \frac{v_{\parallel}}{v_{\perp}}.$$

Define the cone aspect ratio

$$A := \frac{p_{\parallel}^{\max}}{p_{\perp}^{\max}} = \frac{v_{\perp}}{|v_{\parallel}|} = \frac{1}{|\rho|}. \quad (808)$$

The characteristic group-velocity anisotropy is likewise

$$\frac{|v_{g,\perp}|}{|v_{g,\parallel}|} \sim \frac{v_{\perp}}{|v_{\parallel}|} = \frac{1}{|\rho|} = A. \quad (809)$$

Thus the cone aspect ratio and the velocity anisotropy ratio are the same invariant quantity.

CDXXIII. MEDIAN ESTIMATE UNDER ISOTROPIC UNIFIED COUPLING

Under isotropic unified coupling,

$$r_{\text{med}} = \frac{1}{\sqrt{3}},$$

hence

$$|\rho|_{\text{med}} = \frac{3}{32\sqrt{3}} (\theta_0 \kappa)^2 \approx 0.0541266 (\theta_0 \kappa)^2. \quad (810)$$

Therefore

$$A_{\text{med}} = \frac{1}{|\rho|_{\text{med}}} = \frac{32\sqrt{3}}{3} (\theta_0 \kappa)^{-2} \approx 18.475 (\theta_0 \kappa)^{-2}. \quad (811)$$

CDXXIV. NUMERICAL TABLE

The median numerical predictions are:

$\theta_0 \kappa$	$ \rho _{\text{med}}$	$A_{\text{med}} = \frac{1}{ \rho _{\text{med}}}$	$\frac{ v_{g,\perp} }{ v_{g,\parallel} } \sim A_{\text{med}}$
0.1	5.41×10^{-4}	1.85×10^3	1.85×10^3
0.2	2.17×10^{-3}	4.62×10^2	4.62×10^2
0.3	4.87×10^{-3}	2.05×10^2	2.05×10^2
0.5	1.35×10^{-2}	7.39×10^1	7.39×10^1
1.0	5.41×10^{-2}	1.85×10^1	1.85×10^1

CDXXV. 90TH-PERCENTILE TABLE

Using

$$r_{0.9} \approx 2.0647,$$

we get

$$\boxed{|\rho|_{90\%} \approx 0.1936 (\theta_0 \kappa)^2, \quad A_{90\%} \approx 5.165 (\theta_0 \kappa)^{-2}.} \quad (812)$$

Hence:

$\theta_0 \kappa$	$ \rho _{90\%}$	$A_{90\%} = \frac{1}{ \rho _{90\%}}$
0.1	1.94×10^{-3}	5.16×10^2
0.2	7.74×10^{-3}	1.29×10^2
0.3	1.74×10^{-2}	5.74×10^1
0.5	4.84×10^{-2}	2.07×10^1
1.0	1.94×10^{-1}	5.16

CDXXVI. INTERPRETATION

These numbers imply:

1. Even at $\theta_0 \kappa = 1$, the median cone remains strongly anisotropic:

$$A_{\text{med}} \approx 18.5.$$

2. In the genuinely small-defect regime $\theta_0 \kappa \leq 0.3$, the typical anisotropy is enormous:

$$A_{\text{med}} \gtrsim 2 \times 10^2.$$

3. Therefore the generic Branch B prediction is not merely anisotropy, but an extremely transverse-dominated Dirac cone.

CDXXVII. COMPRESSED CONSEQUENCE

Thus the Branch B phase predicts

$$\boxed{A \sim O((\theta_0 \kappa)^{-2}), \quad \frac{|v_{g,\perp}|}{|v_{g,\parallel}|} \sim O((\theta_0 \kappa)^{-2}),} \quad (813)$$

with a large numerical prefactor already at the median level.

CDXXVIII. FINAL SUMMARY

Under isotropic unified coupling, the median anisotropy ratio is

$$|\rho|_{\text{med}} = \frac{3}{32\sqrt{3}} (\theta_0 \kappa)^2 \approx 0.0541266 (\theta_0 \kappa)^2.$$

Therefore the cone aspect ratio and the group-velocity anisotropy are both

$$A_{\text{med}} = \frac{1}{|\rho|_{\text{med}}} = \frac{32\sqrt{3}}{3} (\theta_0 \kappa)^{-2} \approx 18.475 (\theta_0 \kappa)^{-2}.$$

Numerically:

$$\theta_0 \kappa = 0.1 \Rightarrow A_{\text{med}} \approx 1.85 \times 10^3,$$

$$\theta_0 \kappa = 0.2 \Rightarrow A_{\text{med}} \approx 4.62 \times 10^2,$$

$$\theta_0 \kappa = 0.3 \Rightarrow A_{\text{med}} \approx 2.05 \times 10^2,$$

$$\theta_0 \kappa = 0.5 \Rightarrow A_{\text{med}} \approx 7.39 \times 10^1,$$

$$\theta_0 \kappa = 1.0 \Rightarrow A_{\text{med}} \approx 1.85 \times 10^1.$$

Thus the generic Branch B prediction is an extremely transverse-dominated Dirac cone:

$$A \sim O((\theta_0 \kappa)^{-2}), \quad \frac{|v_{g,\perp}|}{|v_{g,\parallel}|} \sim O((\theta_0 \kappa)^{-2}).$$

CDXXIX. INVERSE ESTIMATION OF THE DEFECT PARAMETER FROM THE CONE ASPECT RATIO

CDXXX. STARTING POINT

At leading order, the anisotropy ratio is

$$|\rho_{\text{LO}}| = \frac{3}{32} r (\theta_0 \kappa)^2, \quad r := \frac{|c_{\parallel}|}{\sqrt{c_E^2 + c_O^2}}. \quad (814)$$

The cone aspect ratio is

$$A := \frac{1}{|\rho|}. \quad (815)$$

Therefore, at leading order,

$$A_{\text{LO}} = \frac{32}{3r} (\theta_0 \kappa)^{-2}. \quad (816)$$

CDXXXI. DIRECT INVERSE FORMULA

Solving (816) for the defect parameter gives

$$\theta_0 \kappa = \sqrt{\frac{32}{3rA}}. \quad (817)$$

Thus a measurement of the aspect ratio A determines the combination $\theta_0 \kappa$ up to the coupling-projection factor r .

CDXXXII. VELOCITY-RATIO FORM

Since

$$A = \frac{v_{\perp}}{|v_{\parallel}|},$$

we also have

$$\theta_0 \kappa = \sqrt{\frac{32}{3r} \frac{|v_{\parallel}|}{v_{\perp}}}. \quad (818)$$

Hence the same inverse estimate may be extracted either from the cone geometry or from the group-velocity anisotropy.

CDXXXIII. MEDIAN INVERSE ESTIMATE UNDER ISOTROPIC UNIFIED COUPLING

Under isotropic unified coupling, the median value is

$$r_{\text{med}} = \frac{1}{\sqrt{3}}.$$

Therefore

$$(\theta_0 \kappa)_{\text{med}} = \sqrt{\frac{32\sqrt{3}}{3A}} \approx \frac{4.298}{\sqrt{A}}. \quad (819)$$

Equivalently,

$$A_{\text{med}} = \frac{32\sqrt{3}}{3} (\theta_0 \kappa)^{-2} \approx 18.475 (\theta_0 \kappa)^{-2}. \quad (820)$$

CDXXXIV. NUMERICAL INVERSE EXAMPLES

Using the median inverse formula (819):

$$A = 10 \implies (\theta_0 \kappa)_{\text{med}} \approx 1.36.$$

$$A = 20 \implies (\theta_0 \kappa)_{\text{med}} \approx 0.961.$$

$$A = 50 \implies (\theta_0 \kappa)_{\text{med}} \approx 0.608.$$

$$A = 100 \implies (\theta_0 \kappa)_{\text{med}} \approx 0.430.$$

$$A = 200 \implies (\theta_0 \kappa)_{\text{med}} \approx 0.304.$$

$$A = 500 \implies (\theta_0 \kappa)_{\text{med}} \approx 0.192.$$

$$A = 1000 \implies (\theta_0 \kappa)_{\text{med}} \approx 0.136.$$

CDXXXV. CREDIBLE INTERVAL FROM COUPLING UNCERTAINTY

If the coupling factor r is not known exactly but lies in an interval

$$r \in [r_{\min}, r_{\max}], \quad 0 < r_{\min} \leq r_{\max},$$

then the inferred defect parameter lies in the interval

$$\sqrt{\frac{32}{3r_{\max}A}} \leq \theta_0 \kappa \leq \sqrt{\frac{32}{3r_{\min}A}}. \quad (821)$$

Thus the uncertainty in $\theta_0 \kappa$ is only square-root sensitive to the uncertainty in r .

CDXXXVI. TAIL-BASED INFERENCE

From the exact isotropic unified-coupling distribution

$$F_r(R) = \frac{R}{\sqrt{1+R^2}},$$

we may derive confidence-style inversion rules.

Given a confidence level $p \in (0, 1)$, define

$$r_p = \frac{p}{\sqrt{1-p^2}}. \quad (822)$$

Then

$$\mathbb{P}(r \leq r_p) = p,$$

and therefore

$$\mathbb{P}\left(\theta_0\kappa \geq \sqrt{\frac{32}{3r_p A}}\right) = p. \quad (823)$$

Equivalently, one may use r_p to construct percentile-based estimates of $\theta_0\kappa$.

CDXXXVII. COMPRESSED PRACTICAL FORMULA

For practical purposes, the cleanest inversion rule is:

$$\theta_0\kappa \approx \frac{4.298}{\sqrt{A}} \quad (\text{median estimate under isotropic unified coupling}). \quad (824)$$

Thus measuring the cone aspect ratio immediately gives an estimate of the effective defect strength parameter.

CDXXXVIII. INTERPRETATION

This inversion formula has a clear meaning:

1. very large aspect ratio A implies a weak/small defect parameter $\theta_0\kappa$;
2. moderate aspect ratio $A \sim 10-20$ corresponds to $\theta_0\kappa \sim O(1)$;
3. the small-defect regime $\theta_0\kappa \ll 1$ corresponds to very large aspect ratios $A \gg 10$.

Therefore the observed cone anisotropy provides a direct low-energy probe of the microscopic defect geometry.

CDXXXIX. FINAL SUMMARY

The leading relation between the cone aspect ratio and the defect parameter is

$$A = \frac{1}{|\rho_{\text{LO}}|} = \frac{32}{3r}(\theta_0\kappa)^{-2}, \quad r = \frac{|c_{\parallel}|}{\sqrt{c_E^2 + c_O^2}}.$$

Hence the inverse estimate is

$$\theta_0\kappa = \sqrt{\frac{32}{3rA}}.$$

Under isotropic unified coupling, the median value $r_{\text{med}} = 1/\sqrt{3}$ gives

$$(\theta_0 \kappa)_{\text{med}} = \sqrt{\frac{32\sqrt{3}}{3A}} \approx \frac{4.298}{\sqrt{A}}.$$

Examples:

$$A = 100 \Rightarrow (\theta_0 \kappa)_{\text{med}} \approx 0.430, \quad A = 200 \Rightarrow (\theta_0 \kappa)_{\text{med}} \approx 0.304, \quad A = 1000 \Rightarrow (\theta_0 \kappa)_{\text{med}} \approx 0.136.$$

Thus measuring the cone aspect ratio directly probes the microscopic defect-strength combination $\theta_0 \kappa$. Large aspect ratio means small defect strength; small-defect regimes correspond to very large anisotropy.

CDXL. ERROR PROPAGATION FROM THE CONE ASPECT RATIO TO THE DEFECT PARAMETER

CDXLI. STARTING POINT

The leading inverse relation is

$$\theta_0 \kappa = \sqrt{\frac{32}{3rA}}, \quad (825)$$

where

$$A = \frac{1}{|\rho|}, \quad r = \frac{|c_{\parallel}|}{\sqrt{c_E^2 + c_O^2}}.$$

For the median isotropic unified-coupling estimate,

$$\theta_0 \kappa \approx \frac{4.298}{\sqrt{A}}. \quad (826)$$

CDXLII. DIFFERENTIAL ERROR PROPAGATION AT FIXED r

Treat r as fixed first. Let

$$x := \theta_0 \kappa.$$

From (825),

$$x = C A^{-1/2}, \quad C := \sqrt{\frac{32}{3r}}.$$

Differentiate:

$$dx = -\frac{1}{2} C A^{-3/2} dA.$$

Since $x = C A^{-1/2}$, this becomes

$$\frac{dx}{x} = -\frac{1}{2} \frac{dA}{A}. \quad (827)$$

Hence, for small observational uncertainty,

$$\frac{\delta(\theta_0 \kappa)}{\theta_0 \kappa} = \frac{1}{2} \frac{\delta A}{A}. \quad (828)$$

Therefore the inferred defect parameter is square-root stable with respect to aspect-ratio uncertainty.

CDXLIII. ABSOLUTE ERROR FORMULA

From (825),

$$\frac{\partial(\theta_0\kappa)}{\partial A} = -\frac{1}{2}\sqrt{\frac{32}{3r}}A^{-3/2}.$$

Hence the absolute propagated error is

$$\delta(\theta_0\kappa) \approx \frac{1}{2}\sqrt{\frac{32}{3r}}A^{-3/2}\delta A. \quad (829)$$

Using $\theta_0\kappa = \sqrt{32/(3rA)}$, this may also be written as

$$\delta(\theta_0\kappa) \approx \frac{\theta_0\kappa}{2A}\delta A. \quad (830)$$

For the median isotropic estimate,

$$\delta(\theta_0\kappa) \approx \frac{4.298}{2}A^{-3/2}\delta A = 2.149A^{-3/2}\delta A. \quad (831)$$

CDXLIV. EXAMPLES

a. Example 1: $A = 100$. Then

$$(\theta_0\kappa)_{\text{med}} \approx \frac{4.298}{10} = 0.430.$$

If $\delta A = 10$ (a 10% relative uncertainty), then

$$\frac{\delta(\theta_0\kappa)}{\theta_0\kappa} = \frac{1}{2}\frac{10}{100} = 0.05,$$

so

$$\delta(\theta_0\kappa) \approx 0.0215.$$

Thus

$$\theta_0\kappa \approx 0.430 \pm 0.022.$$

b. Example 2: $A = 200$. Then

$$(\theta_0\kappa)_{\text{med}} \approx \frac{4.298}{\sqrt{200}} \approx 0.304.$$

If $\delta A = 20$ (again 10%), then

$$\frac{\delta(\theta_0\kappa)}{\theta_0\kappa} = 0.05,$$

hence

$$\delta(\theta_0\kappa) \approx 0.0152.$$

Thus

$$\theta_0\kappa \approx 0.304 \pm 0.015.$$

c. *Example 3:* $A = 1000$. Then

$$(\theta_0 \kappa)_{\text{med}} \approx \frac{4.298}{\sqrt{1000}} \approx 0.136.$$

If $\delta A = 100$ (again 10%), then

$$\frac{\delta(\theta_0 \kappa)}{\theta_0 \kappa} = 0.05,$$

so

$$\boxed{\delta(\theta_0 \kappa) \approx 0.0068.}$$

Thus

$$\theta_0 \kappa \approx 0.136 \pm 0.0068.$$

CDXLV. INCLUDING UNCERTAINTY IN r

If the coupling-projection factor r is also uncertain, then

$$\theta_0 \kappa = \left(\frac{32}{3}\right)^{1/2} r^{-1/2} A^{-1/2}.$$

Taking logarithmic differentials gives

$$\boxed{\frac{d(\theta_0 \kappa)}{\theta_0 \kappa} = -\frac{1}{2} \frac{dA}{A} - \frac{1}{2} \frac{dr}{r}.} \quad (832)$$

Hence, for independent small uncertainties,

$$\boxed{\left(\frac{\delta(\theta_0 \kappa)}{\theta_0 \kappa}\right)^2 \approx \frac{1}{4} \left(\frac{\delta A}{A}\right)^2 + \frac{1}{4} \left(\frac{\delta r}{r}\right)^2.} \quad (833)$$

So the relative uncertainty in r enters on the same square-root footing as the relative uncertainty in A .

CDXLVI. CONFIDENCE INTERVAL FORM

If one uses the median estimator for r , then the dominant observational uncertainty comes from A , and the practical propagated interval is

$$\boxed{\theta_0 \kappa \approx \frac{4.298}{\sqrt{A}} \left(1 \pm \frac{1}{2} \frac{\delta A}{A}\right)} \quad (834)$$

to leading order in $\delta A/A$.

CDXLVII. INTERPRETATION

The error propagation has a favorable structure:

1. the inferred defect parameter depends only on $A^{-1/2}$, not A^{-1} ;
2. therefore relative errors are reduced by a factor of 2;
3. large aspect-ratio measurements still allow reasonably stable estimation of $\theta_0 \kappa$.

Thus cone anisotropy is not only a theoretically meaningful observable but also a quantitatively robust probe of the defect-strength scale.

CDXLVIII. FINAL SUMMARY

From the inverse relation

$$\theta_0 \kappa = \sqrt{\frac{32}{3rA}},$$

the leading small-error propagation at fixed r is

$$\frac{\delta(\theta_0 \kappa)}{\theta_0 \kappa} = \frac{1}{2} \frac{\delta A}{A}.$$

So the defect parameter is square-root stable with respect to aspect-ratio uncertainty. Equivalently,

$$\delta(\theta_0 \kappa) \approx \frac{\theta_0 \kappa}{2A} \delta A.$$

Under the median isotropic unified-coupling estimate,

$$\theta_0 \kappa \approx \frac{4.298}{\sqrt{A}},$$

so

$$\delta(\theta_0 \kappa) \approx 2.149 A^{-3/2} \delta A.$$

Examples for a 10% relative uncertainty in A :

$$A = 100 \Rightarrow \theta_0 \kappa \approx 0.430 \pm 0.022,$$

$$A = 200 \Rightarrow \theta_0 \kappa \approx 0.304 \pm 0.015,$$

$$A = 1000 \Rightarrow \theta_0 \kappa \approx 0.136 \pm 0.0068.$$

If r is also uncertain, then

$$\left(\frac{\delta(\theta_0 \kappa)}{\theta_0 \kappa} \right)^2 \approx \frac{1}{4} \left(\frac{\delta A}{A} \right)^2 + \frac{1}{4} \left(\frac{\delta r}{r} \right)^2.$$

Thus aspect-ratio measurements provide a quantitatively robust probe of the microscopic defect-strength combination $\theta_0 \kappa$.

CDXLIX. FINAL ANISOTROPY-DEFECT THEOREM AND OBSERVATIONAL COROLLARY

[Anisotropy-defect theorem for the exact Branch B phase] Assume the exact Branch B phase theorem holds, so that the reduced infrared defect PDE is

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0, \quad (835)$$

with

$$v_{\perp} := \sqrt{v_E^2 + v_O^2}, \quad \rho := \frac{v_{\parallel}}{v_{\perp}}. \quad (836)$$

Then the cone aspect ratio is

$$A := \frac{p_{\parallel}^{\max}}{p_{\perp}^{\max}} = \frac{1}{|\rho|} = \frac{v_{\perp}}{|v_{\parallel}|}. \quad (837)$$

For the concrete one-mediator Branch B truncation with unified coupling projections

$$\alpha_X = g_X c_{\parallel}, \quad \gamma_E = g_X c_E, \quad \gamma_O = g_X c_O,$$

define

$$r := \frac{|c_{\parallel}|}{\sqrt{c_E^2 + c_O^2}}. \quad (838)$$

Then the leading anisotropy ratio satisfies

$$|\rho_{\text{LO}}| = \frac{3}{32} r (\theta_0 \kappa)^2, \quad (839)$$

equivalently

$$A_{\text{LO}} = \frac{32}{3r} (\theta_0 \kappa)^{-2}. \quad (840)$$

Therefore the effective defect-strength parameter is determined by the observable anisotropy through

$$\theta_0 \kappa = \sqrt{\frac{32}{3rA}}. \quad (841)$$

Hence the exact Branch B phase has the following interpretation:

$$\boxed{\text{the low-energy anisotropy is not merely qualitative but quantitatively encodes the microscopic defect-strength combination } \theta_0 \kappa} \quad (842)$$

[Median observational estimate under isotropic unified coupling] Assume further that the projection vector

$$(c_{\parallel}, c_E, c_O)$$

is isotropically distributed on the unit sphere in coupling space. Then the median value of the coupling factor is

$$r_{\text{med}} = \frac{1}{\sqrt{3}}. \quad (843)$$

Consequently,

$$|\rho_{\text{LO}}|_{\text{med}} = \frac{3}{32\sqrt{3}} (\theta_0 \kappa)^2 \approx 0.0541266 (\theta_0 \kappa)^2, \quad (844)$$

$$A_{\text{med}} = \frac{32\sqrt{3}}{3} (\theta_0 \kappa)^{-2} \approx 18.475 (\theta_0 \kappa)^{-2}, \quad (845)$$

and the median inverse estimate is

$$(\theta_0 \kappa)_{\text{med}} = \sqrt{\frac{32\sqrt{3}}{3A}} \approx \frac{4.298}{\sqrt{A}}. \quad (846)$$

[Error-propagation corollary] At fixed r , the leading inverse relation

$$\theta_0 \kappa = \sqrt{\frac{32}{3rA}}$$

implies

$$\boxed{\frac{\delta(\theta_0 \kappa)}{\theta_0 \kappa} = \frac{1}{2} \frac{\delta A}{A}}. \quad (847)$$

Thus the inferred defect parameter is square-root stable with respect to aspect-ratio uncertainty. Equivalently,

$$\boxed{\delta(\theta_0 \kappa) \approx \frac{\theta_0 \kappa}{2A} \delta A}. \quad (848)$$

[Natural-regime corollary] If the small-defect regime holds,

$$0 < \theta_0 \kappa \ll 1,$$

and the unified coupling projections are $O(1)$, then

$$\boxed{|\rho_{\text{LO}}| \ll 1, \quad A_{\text{LO}} \gg 1}. \quad (849)$$

Therefore the generic explicit Branch B prediction is

$$\boxed{\text{a strongly transverse-dominated anisotropic Dirac cone whose measured aspect ratio directly probes the underlying defect strength}} \quad (850)$$

CDL. FINAL SUMMARY

For the exact Branch B phase,

$$\rho = \frac{v_{\parallel}}{v_{\perp}}, \quad A = \frac{1}{|\rho|} = \frac{v_{\perp}}{|v_{\parallel}|}.$$

In the concrete one-mediator unified-coupling truncation,

$$|\rho_{\text{LO}}| = \frac{3}{32} r (\theta_0 \kappa)^2, \quad r = \frac{|c_{\parallel}|}{\sqrt{c_E^2 + c_O^2}},$$

so

$$A_{\text{LO}} = \frac{32}{3r} (\theta_0 \kappa)^{-2}, \quad \theta_0 \kappa = \sqrt{\frac{32}{3rA}}.$$

Thus the observable anisotropy and the microscopic defect strength are directly linked.

Under isotropic unified coupling,

$$r_{\text{med}} = \frac{1}{\sqrt{3}},$$

hence

$$|\rho_{\text{LO}}|_{\text{med}} = \frac{3}{32\sqrt{3}} (\theta_0 \kappa)^2, \quad A_{\text{med}} = \frac{32\sqrt{3}}{3} (\theta_0 \kappa)^{-2},$$

and

$$(\theta_0 \kappa)_{\text{med}} \approx \frac{4.298}{\sqrt{A}}.$$

Moreover, at fixed r ,

$$\frac{\delta(\theta_0 \kappa)}{\theta_0 \kappa} = \frac{1}{2} \frac{\delta A}{A}.$$

Therefore the exact Branch B phase predicts that large cone anisotropy is the natural signature of weak defects, and the measured aspect ratio provides a quantitatively robust probe of the microscopic defect-strength combination $\theta_0 \kappa$.

CDLI. PROJECT: TECT — MASTER CONTENTS FOR CONTINUATION (UPDATED SNAPSHOT)

CDLII. PROJECT GOAL

The long-term goal of TECT (Topological Energy Condensate Theory) is to derive, from a microscopic condensate/defect framework:

1. emergent spacetime-like defect kinematics,
2. emergent protected fermionic infrared sectors,
3. eventually gauge structure and realistic matter phenomenology.

At the present stage, the most rigorously developed part is the emergent defect-fermion Dirac sector.

CDLIII. WORKING RULES

1. Conversation language: Korean.
2. Mathematical derivations: English LaTeX.
3. Always provide:
 - (a) short Korean explanation,
 - (b) full detailed LaTeX derivation,
 - (c) final summary (LaTeX),
 - (d) next step.
4. Avoid philosophical exposition.
5. Focus on rigorous mathematical derivations and honest separation between what is proved, conditionally proved, and open.

CDLIV. GLOBAL STATUS

The old scalar Bragg-pair route is no longer the correct mainline. Within the currently known safe microscopic operator library, the scalar identity-channel Bragg block remains nonzero and cannot be canceled by the known safe scalar/alignment/Class II sectors. Therefore the correct mainline is the graded $2 + 2$ reduced route.

The reduced Dirac theorem is already closed. The actual local CP^2 projector-defect geometry is compatible with the S2 branch. The branch ambiguity has now been pushed much further and, within the existing local isotropic TECT algebra, Branch B is effectively exact.

CDLV. WHAT IS FULLY CLOSED

A. Reduced graded Dirac theorem

The reduced first-order symbol must take the S2 form

$$\boxed{A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).} \quad (851)$$

The exact nondegeneracy condition is

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (852)$$

B. Actual local defect geometry

For the local projector-defect background

$$P_B(\xi) = z_B(\xi)z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1,$$

the tangent bundle is

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp.$$

Using the explicit real representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

the tangent Pauli operators are

$$\Sigma_1, \Sigma_2, \Sigma_3.$$

C. Scalar-channel no-go

Inside the existing safe local isotropic operator library, the scalar identity-channel Bragg block cannot be removed. So the old scalar mainline is ruled out.

D. Natural S2 branch and transverse finite test

The reduced symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

and the transverse sector is reduced to the finite nonzero test

$$(\Lambda_E, \Lambda_O) \neq (0, 0).$$

CDLVI. BRANCH A / BRANCH B STATUS

A. Original branch question

The old unresolved longitudinal question was:

$$\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0, \quad \text{Branch B: } \Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.$$

B. Minimal complete LO realization

For the minimal complete lowest-order TECT realization already explicitly constructed,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.$$

Therefore the explicit concrete currently constructed branch is Branch B.

C. Why Branch A fails in the existing local isotropic algebra

For the real defect representative,

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0, \quad (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3.$$

Therefore no linear longitudinal tangent selector of the form

$$\theta'(\xi) \Sigma_3$$

can be generated from the existing local isotropic TECT algebra.

This remains true not only at lowest order but to all local orders within the same isotropic local variable algebra. Higher local corrections generate only the odd primitive tower:

$$\theta'^3 \Sigma_3, \quad \theta'^5 \Sigma_3, \quad \theta'^7 \Sigma_3, \quad \dots$$

Thus

$$\boxed{\Lambda_{\parallel}^{(0)} = 0} \tag{853}$$

is exact inside the existing local isotropic TECT algebra.

D. Exact Branch B conclusion

Hence, within the existing local isotropic TECT framework, Branch B is exact.

CDLVII. EXACT BRANCH B REDUCED PDE

The exact reduced Branch B PDE is

$$\boxed{\left[i\partial_t - v_{\parallel} \sigma_3 (-i\partial_{\parallel}) - v_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - v_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0,} \tag{854}$$

with

$$v_{\perp} := \sqrt{v_E^2 + v_O^2}, \quad \rho := \frac{v_{\parallel}}{v_{\perp}}.$$

Up to transverse frame rotation and overall rescaling, the infrared phenomenology is controlled by the single anisotropy ratio ρ .

CDLVIII. EXACT BRANCH B PHASE THEOREM

The current main theorem proved/assembled in conditional-exact form is:

a. Exact Branch B phase theorem. Assume:

1. the theory stays within the existing local isotropic TECT variable algebra,
2. the defect belongs to an admissible monotone class,
3. the leading microscopic coefficients satisfy

$$\lambda_c^{(1)} \neq 0, \quad (\tilde{\Lambda}_E^{(1)}, \tilde{\Lambda}_O^{(1)}) \neq (0, 0),$$

4. the defect lies in the natural small-defect regime

$$0 < \theta_0 \kappa \ll 1,$$

and higher coefficient towers have finite relative growth.

Then:

1. Branch B is exact,
2. $I_3 \neq 0$ and $J_1 \neq 0$,
3. higher-order corrections are subleading,
4. hence

$$v_{\parallel} \neq 0, \quad v_{\perp} \neq 0.$$

Therefore the reduced defect PDE defines a stable anisotropic Dirac phase.

CDLIX. OVERLAP NONVANISHING THEOREM

For admissible monotone defects with

$$\theta'(\xi) \text{ of fixed sign,} \quad w(\xi) := f_+(\xi)f_-(\xi) \geq 0, \quad w \not\equiv 0, \quad \cos \theta(\xi) \geq c_0 > 0$$

on the effective support, one has

$$I_3 = \int d\xi w(\xi) \theta'(\xi)^3 \neq 0, \quad J_1 = \int d\xi w(\xi) \theta'(\xi) \cos \theta(\xi) \neq 0.$$

CDLX. HIGHER-ORDER DOMINANCE

The exact coefficients are

$$v_{\parallel} = \sum_{n \geq 1} \frac{\lambda_c^{(n)}}{2} I_{2n+1}, \quad v_E = \sum_{n \geq 1} \tilde{\Lambda}_E^{(n)} J_{2n-1}, \quad v_O = \sum_{n \geq 1} \tilde{\Lambda}_O^{(n)} J_{2n-1}.$$

In the natural small-defect regime

$$\theta(\xi) = \theta_0 \Theta(\kappa \xi), \quad 0 < \theta_0 \kappa \ll 1,$$

the overlap towers scale as

$$I_{2n+1} = O((\theta_0 \kappa)^{2n+1}), \quad J_{2n-1} = O((\theta_0 \kappa)^{2n-1}),$$

so the higher-order remainders satisfy

$$\frac{|\Delta v_{\parallel}|}{|v_{\parallel}^{\text{LO}}|} = O((\theta_0 \kappa)^2), \quad \frac{\Delta v_{\perp}}{v_{\perp}^{\text{LO}}} = O((\theta_0 \kappa)^2).$$

Thus the Branch B phase is generically stable in an open natural small-defect regime.

CDLXI. CONCRETE ONE-MEDIATOR TRUNCATION

A concrete predictive truncation was introduced:

$$\mathcal{L}_{\text{trunc}} = \mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_E X_i^a \mathcal{E}_i^a[P, \psi] + \gamma_O X_i^a \mathcal{O}_i^a[P, \psi].$$

Integrating out X_i^a gives

$$\lambda_c^{(1)} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}, \quad \tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2} C_E, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} C_O.$$

In canonical explicit normalization,

$$C_{\parallel} = C_E = C_O = 1.$$

CDLXII. LEADING ANISOTROPY RATIO

In the one-mediator truncation,

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X C_{\parallel}}{\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} (\theta_0 \kappa)^2.$$

In canonical normalization,

$$\boxed{\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} (\theta_0 \kappa)^2.} \quad (855)$$

Important consequence:

$$\beta_X, M_X$$

drop out of the ratio ρ_{LO} . So the anisotropy is controlled by the relative coupling hierarchy

$$\alpha_X : \gamma_E : \gamma_O$$

and the geometric factor $(\theta_0 \kappa)^2$.

CDLXIII. UNIFIED-COUPLING NATURALNESS

If the couplings arise from a single mediator multiplet:

$$\alpha_X = g_X c_{\parallel}, \quad \gamma_E = g_X c_E, \quad \gamma_O = g_X c_O,$$

with

$$c_{\parallel}, c_E, c_O = O(1),$$

then

$$\frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} = \frac{c_{\parallel}}{\sqrt{c_E^2 + c_O^2}} = O(1).$$

Therefore

$$\boxed{|\rho_{\text{LO}}| = O((\theta_0 \kappa)^2) \ll 1} \quad (856)$$

in the natural small-defect regime. So the strongly anisotropic Branch B cone is the natural generic prediction.

Balanced cones $|\rho| \sim 1$ require strong enhancement of the longitudinal projection or accidental suppression of the transverse ones, and are therefore fine-tuned.

CDLXIV. ISOTROPIC UNIFIED-COUPLING DISTRIBUTION

Under the isotropic unified-coupling hypothesis, the projection vector

$$(c_{\parallel}, c_E, c_O)$$

is isotropically distributed on the unit sphere. Define

$$r := \frac{|c_{\parallel}|}{\sqrt{c_E^2 + c_O^2}}.$$

Then

$$F_r(R) = \frac{R}{\sqrt{1+R^2}}, \quad f_r(R) = \frac{1}{(1+R^2)^{3/2}}.$$

The median is

$$r_{\text{med}} = \frac{1}{\sqrt{3}}.$$

Hence

$$|\rho_{\text{LO}}|_{\text{med}} = \frac{3}{32\sqrt{3}}(\theta_0\kappa)^2 \approx 0.0541266 (\theta_0\kappa)^2,$$

and

$$A_{\text{med}} = \frac{1}{|\rho|_{\text{med}}} = \frac{32\sqrt{3}}{3}(\theta_0\kappa)^{-2} \approx 18.475 (\theta_0\kappa)^{-2}.$$

Balanced cones are statistically suppressed.

CDLXV. NUMERICAL ANISOTROPY EXAMPLES

Median examples:

$$\theta_0\kappa = 0.1 \Rightarrow A_{\text{med}} \approx 1.85 \times 10^3,$$

$$\theta_0\kappa = 0.2 \Rightarrow A_{\text{med}} \approx 4.62 \times 10^2,$$

$$\theta_0\kappa = 0.3 \Rightarrow A_{\text{med}} \approx 2.05 \times 10^2,$$

$$\theta_0\kappa = 0.5 \Rightarrow A_{\text{med}} \approx 7.39 \times 10^1,$$

$$\theta_0\kappa = 1.0 \Rightarrow A_{\text{med}} \approx 1.85 \times 10^1.$$

Thus the natural Branch B prediction is not merely mild anisotropy but an extremely transverse-dominated Dirac cone.

CDLXVI. ANISOTROPY-DEFECT THEOREM

For the exact Branch B phase,

$$\rho = \frac{v_{\parallel}}{v_{\perp}}, \quad A = \frac{1}{|\rho|} = \frac{v_{\perp}}{|v_{\parallel}|}.$$

In the concrete unified-coupling truncation,

$$|\rho_{\text{LO}}| = \frac{3}{32} r (\theta_0\kappa)^2, \quad A_{\text{LO}} = \frac{32}{3r} (\theta_0\kappa)^{-2},$$

so

$$\boxed{\theta_0\kappa = \sqrt{\frac{32}{3rA}}.} \tag{857}$$

Thus the observable low-energy anisotropy directly encodes the microscopic defect-strength combination $\theta_0\kappa$.

CDLXVII. MEDIAN INVERSE ESTIMATE

Under isotropic unified coupling,

$$r_{\text{med}} = \frac{1}{\sqrt{3}},$$

so the practical median inversion rule is

$$\boxed{\theta_0 \kappa \approx \frac{4.298}{\sqrt{A}}.} \quad (858)$$

Examples:

$$A = 100 \Rightarrow \theta_0 \kappa \approx 0.430,$$

$$A = 200 \Rightarrow \theta_0 \kappa \approx 0.304,$$

$$A = 1000 \Rightarrow \theta_0 \kappa \approx 0.136.$$

CDLXVIII. ERROR PROPAGATION

At fixed r ,

$$\theta_0 \kappa = \sqrt{\frac{32}{3rA}},$$

so

$$\boxed{\frac{\delta(\theta_0 \kappa)}{\theta_0 \kappa} = \frac{1}{2} \frac{\delta A}{A}.} \quad (859)$$

Hence a 10% relative uncertainty in A leads to only a 5% relative uncertainty in the inferred $\theta_0 \kappa$.

Thus cone anisotropy is not only a qualitative signature but a quantitatively robust observable probe of defect strength.

CDLXIX. WHAT IS NOT YET CLOSED

The following are *not* yet unconditionally closed:

1. Full microscopic derivation of a realistic Standard Model gauge sector with full RG data.
2. Proton decay prediction.
3. Neutrino mass prediction.
4. Coupling-constant unification prediction.
5. Full matter embedding for realistic baryon/lepton sector phenomenology.

Current honest status: TECT presently predicts the defect-fermion infrared Branch B Dirac phase much more strongly than it predicts standard GUT observables.

CDLXX. GUT-STYLE PREDICTION ASSESSMENT

For the standard GUT-type observables:

- a. Proton decay.* Not yet predictable at the present stage. Requires explicit baryon-number-violating operators and their suppression scale.
- b. Neutrino mass.* Closest possible future target, but not yet predicted. Requires explicit tiny mass-generation mechanism and neutrino-sector identification.
- c. Coupling unification.* Not yet predictable. Requires full gauge normalization, matter content, beta functions, and threshold structure.

Thus at the current stage none of these three is yet quantitatively derived.

CDLXXI. STRONGEST HONEST CURRENT CONCLUSION

The strongest justified statement at the current stage is:

Within the existing local isotropic TECT algebra, the defect infrared theory is on an exact Branch B anisotropic Dirac phase. (860)

In the natural small-defect regime, this phase is generically stable and strongly transverse-dominated. (861)

Its cone anisotropy A provides a direct quantitative probe of the microscopic defect-strength combination $\theta_0\kappa$. (862)

CDLXXII. EXACT NEXT FRONTIER

The next real frontier is no longer branch selection. It is quantitative microscopic prediction of the anisotropy ratio:

$$\rho = \frac{v_{\parallel}}{v_{\perp}}.$$

More concretely, the next tasks are:

1. choose richer full-action truncations and compute

$$\lambda_c^{(1)}, \quad \tilde{\Lambda}_E^{(1)}, \quad \tilde{\Lambda}_O^{(1)};$$

2. test whether unified-coupling structure actually emerges from the full microscopic truncation;
3. estimate the numerical range of ρ beyond the minimal one-mediator model;
4. only after that, revisit possible links to realistic neutrino-mass generation or later gauge phenomenology.

CDLXXIII. FINAL SUMMARY

Current TECT status:

1. The reduced defect fermion theory is on an exact Branch B phase inside the existing local isotropic TECT algebra.
2. The exact reduced PDE is an anisotropic Dirac equation with coefficients $v_{\parallel}, v_E, v_O$.
3. In the natural small-defect regime, the phase is generically stable and strongly transverse-dominated.
4. The key infrared observable is the anisotropy ratio

$$\rho = \frac{v_{\parallel}}{v_{\perp}}, \quad v_{\perp} = \sqrt{v_E^2 + v_O^2}, \quad A = \frac{1}{|\rho|}.$$

5. In the minimal unified-coupling truncation,

$$|\rho_{\text{LO}}| = \frac{3}{32}r(\theta_0\kappa)^2, \quad \theta_0\kappa = \sqrt{\frac{32}{3rA}}.$$

6. Under isotropic unified coupling, the median inversion rule is

$$\theta_0\kappa \approx \frac{4.298}{\sqrt{A}}.$$

7. Standard GUT observables such as proton decay, neutrino mass, and coupling unification are not yet quantitatively predicted.

Next frontier: compute $\lambda_c^{(1)}, \tilde{\Lambda}_E^{(1)}, \tilde{\Lambda}_O^{(1)}$ in richer full microscopic truncations and sharpen the prediction for ρ .